

GENIFY

LECTURE 15

27-August-2025

TOPICS COVERED:

RIS

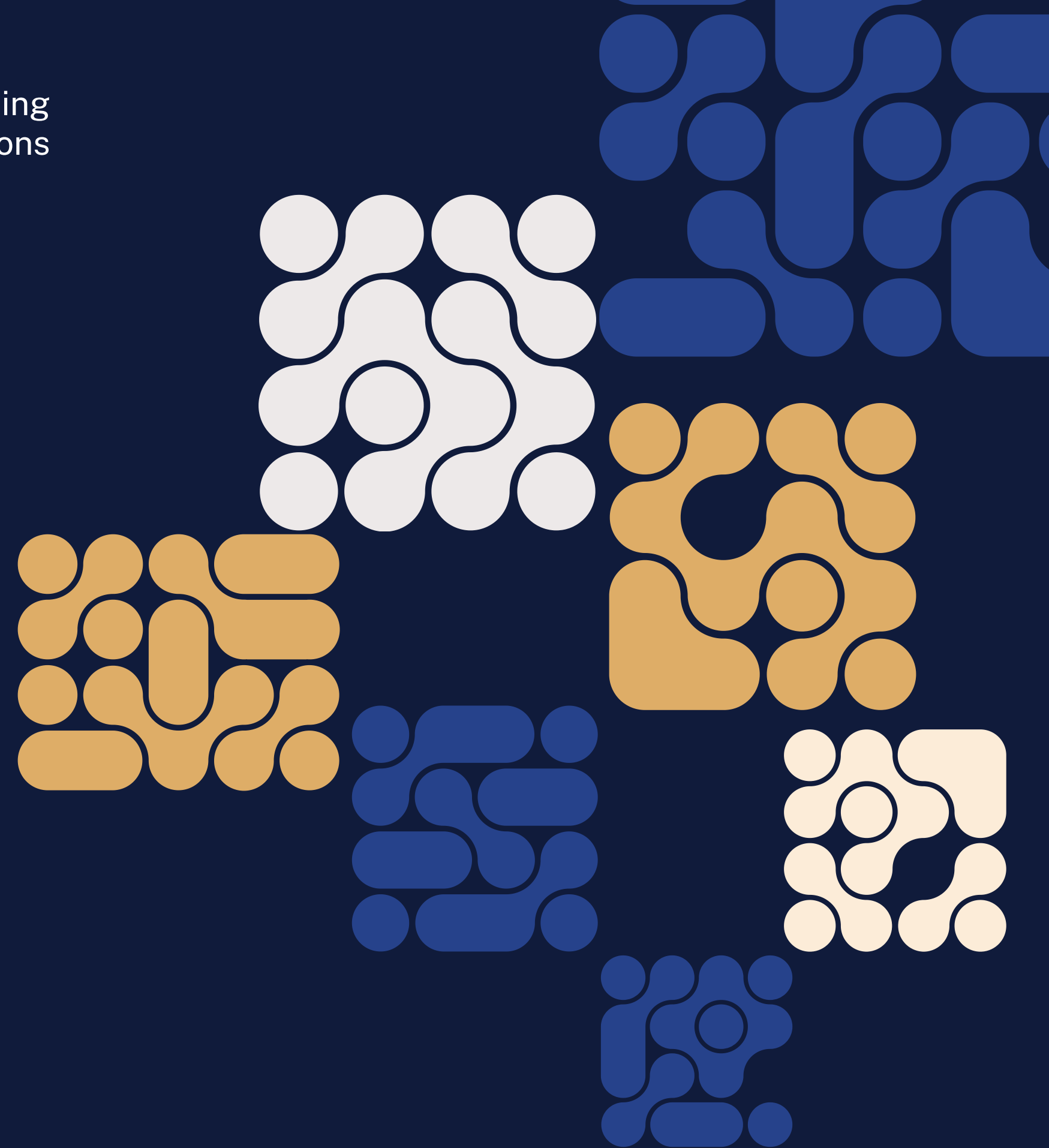
- TRANSFORMERS
 - SELF ATTENTION

Presenter

Shahzaib Kashif

Co-Presenter

Shayan Hassan Baig





SELF
ATTENTION

Contextual Word Embeddings

isTech	0.2
canCall	0.1
isSmart	0.19
.	.
.	.
isFruit	0.3
isFood	0.18

APPLE

Contextual Word Embeddings

isTech	0.2
canCall	0.1
isSmart	0.19
.	.
.	.
isFruit	0.3
isFood	0.18

APPLE

I LOVE APPLE PHONES



Contextual Word Embeddings

isTech	0.2
canCall	0.1
isSmart	0.19
.	.
.	.
isFruit	0.3
isFood	0.18

APPLE

I LOVE APPLE PHONES



I LOVE APPLE JUICE



Contextual Word Embeddings

isTech	0.2
canCall	0.1
isSmart	0.19
.	.
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isFruit	0.3
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APPLE

I LOVE APPLE PHONES



I LOVE APPLE JUICE



APPLE



Tech Company

Fruits

Contextual Word Embeddings

isTech	0.2
canCall	0.1
isSmart	0.19
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APPLE

I LOVE APPLE JUICE



APPLE



Tech Company

Fruits

Contextual Word Embeddings

isTech	0.2
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APPLE

I LOVE APPLE PHONES



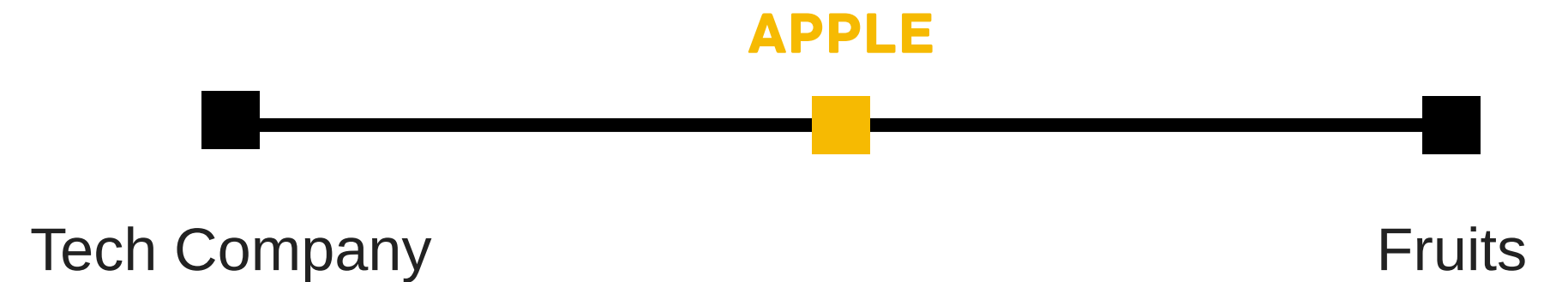
APPLE



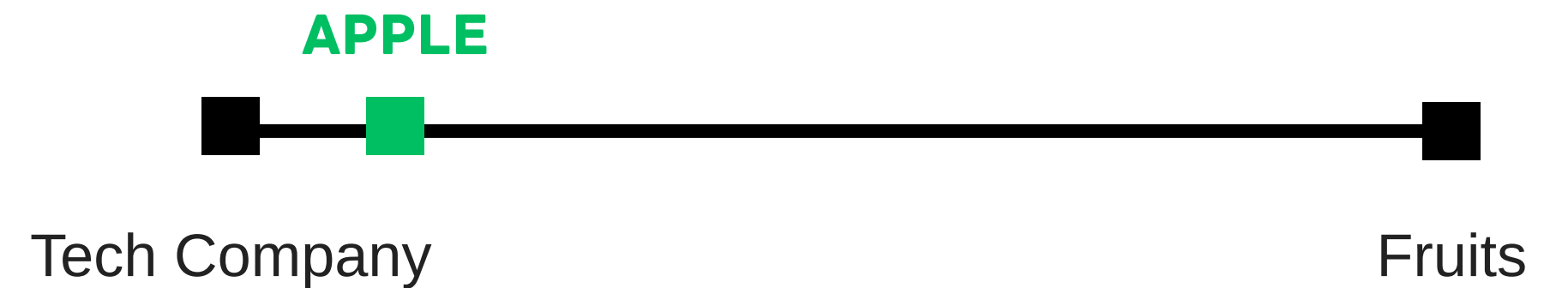
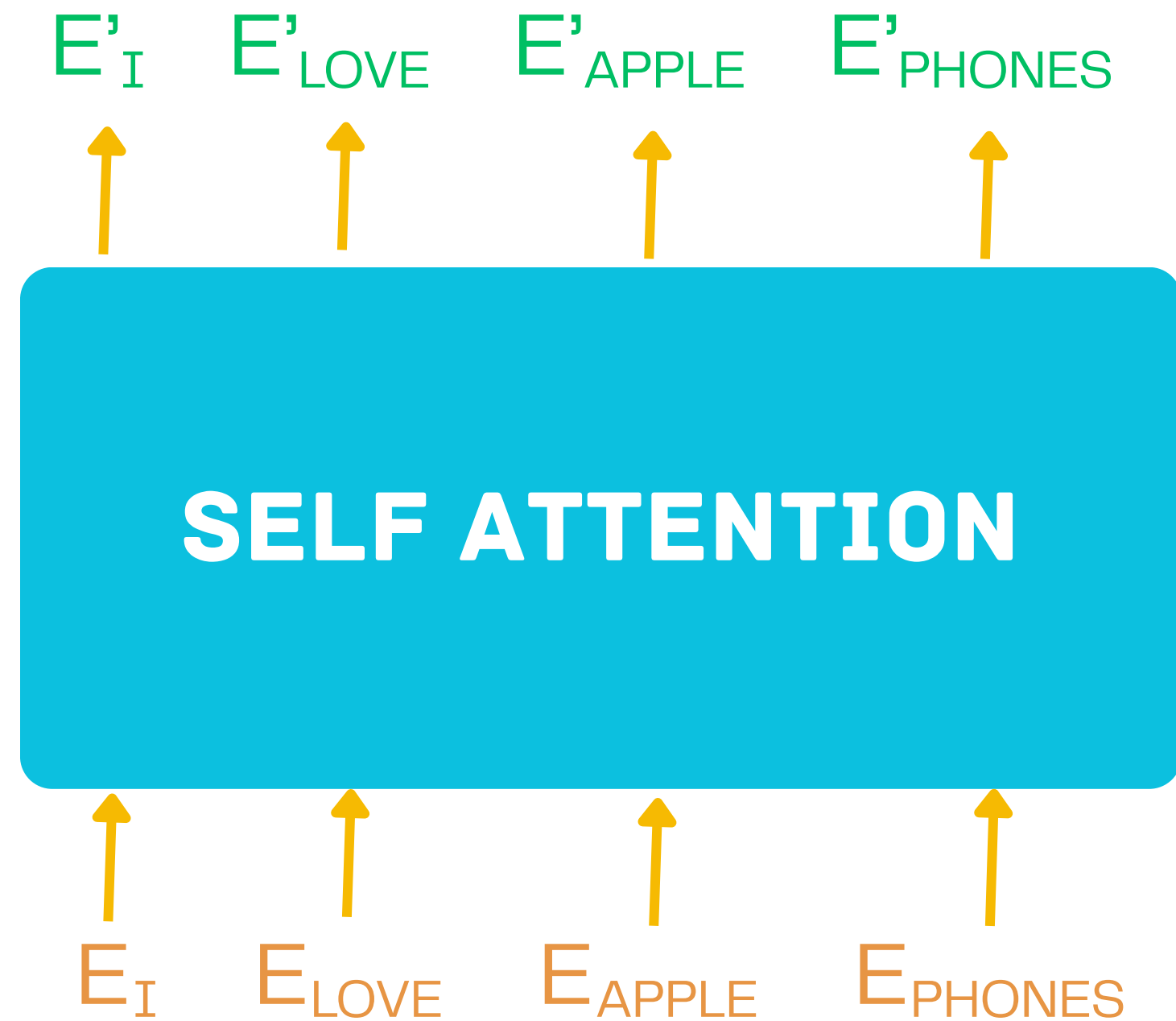
Tech Company

Fruits

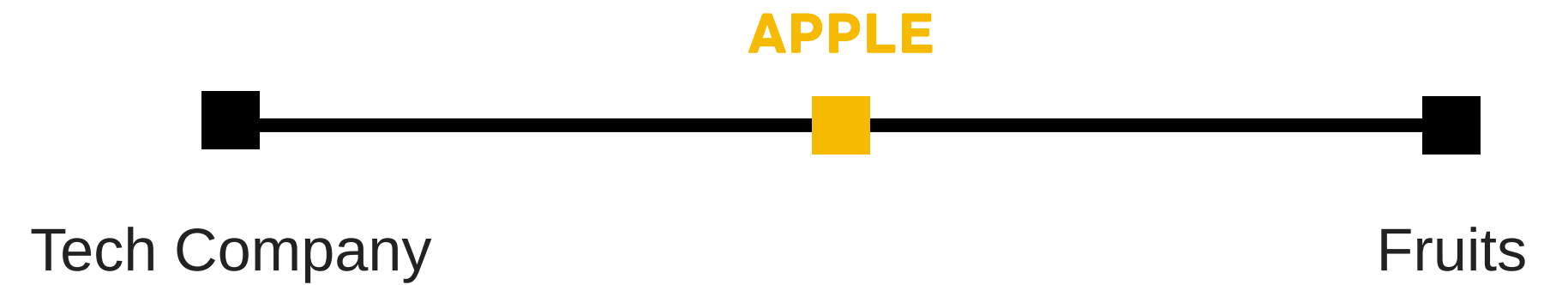
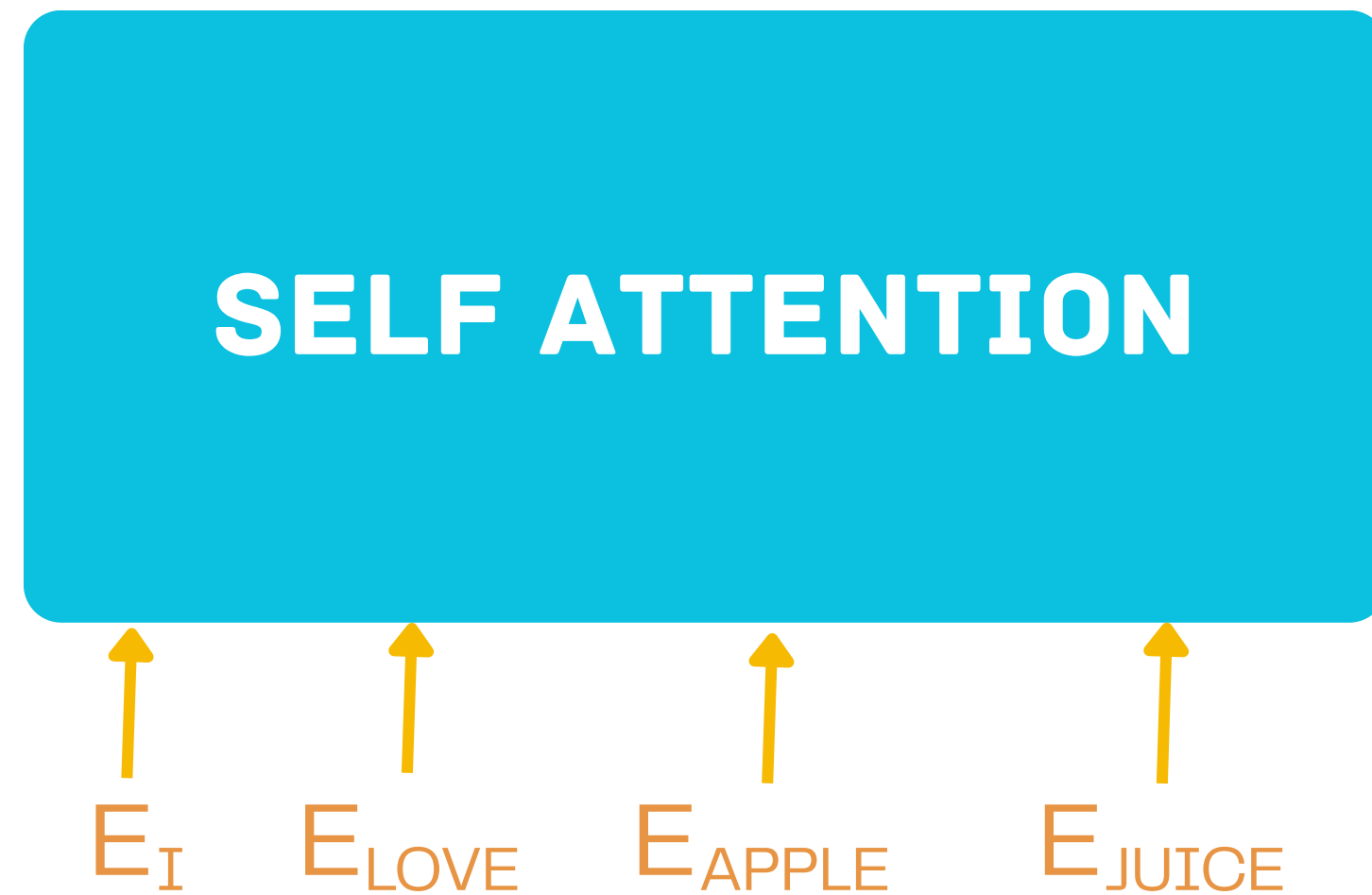
Self Attention - Brief Overview



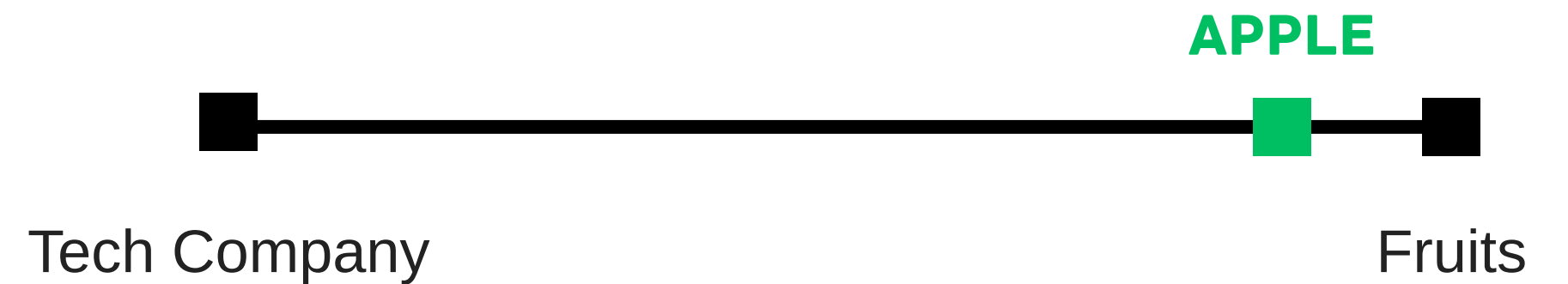
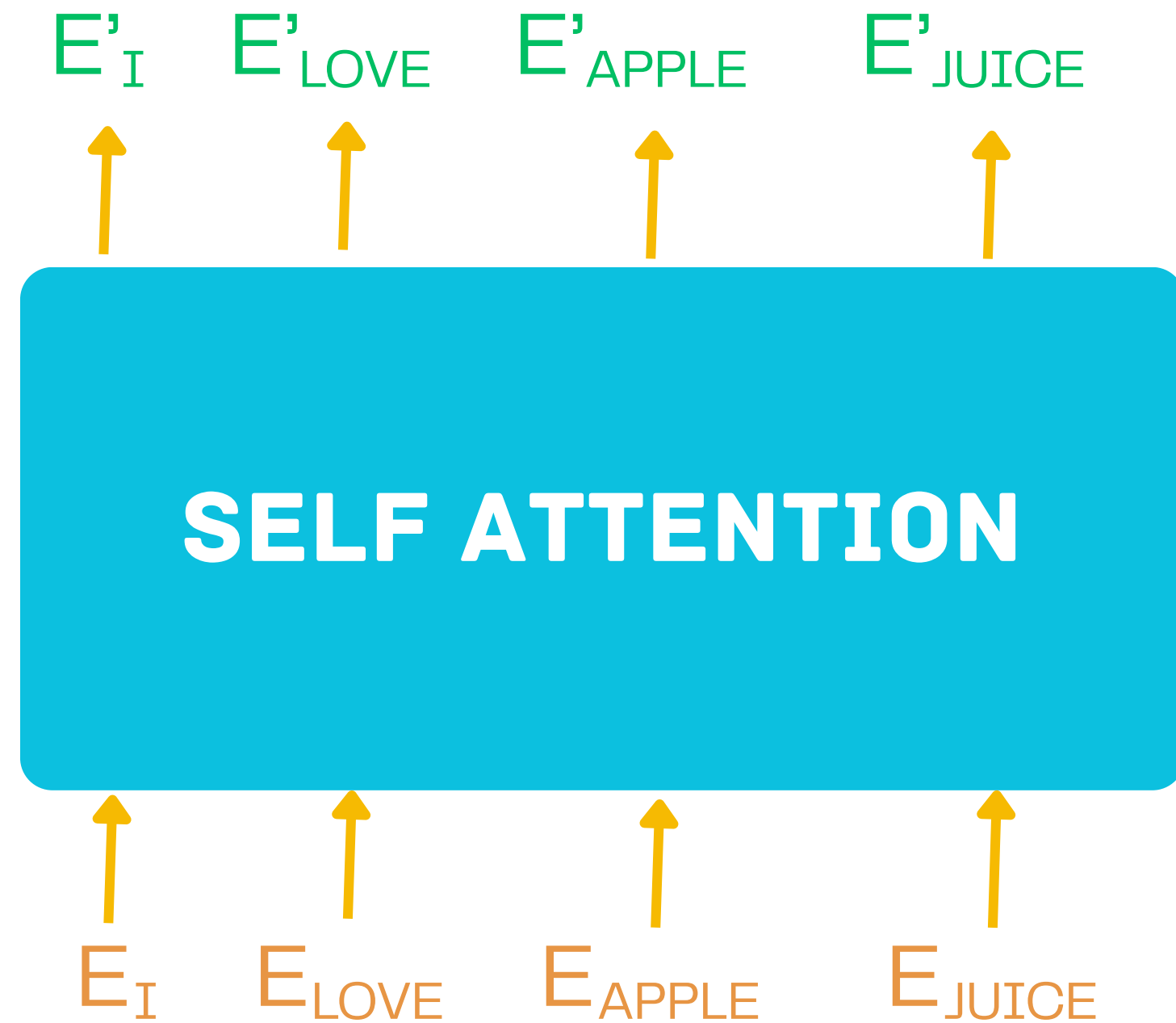
Self Attention - Brief Overview



Self Attention - Brief Overview

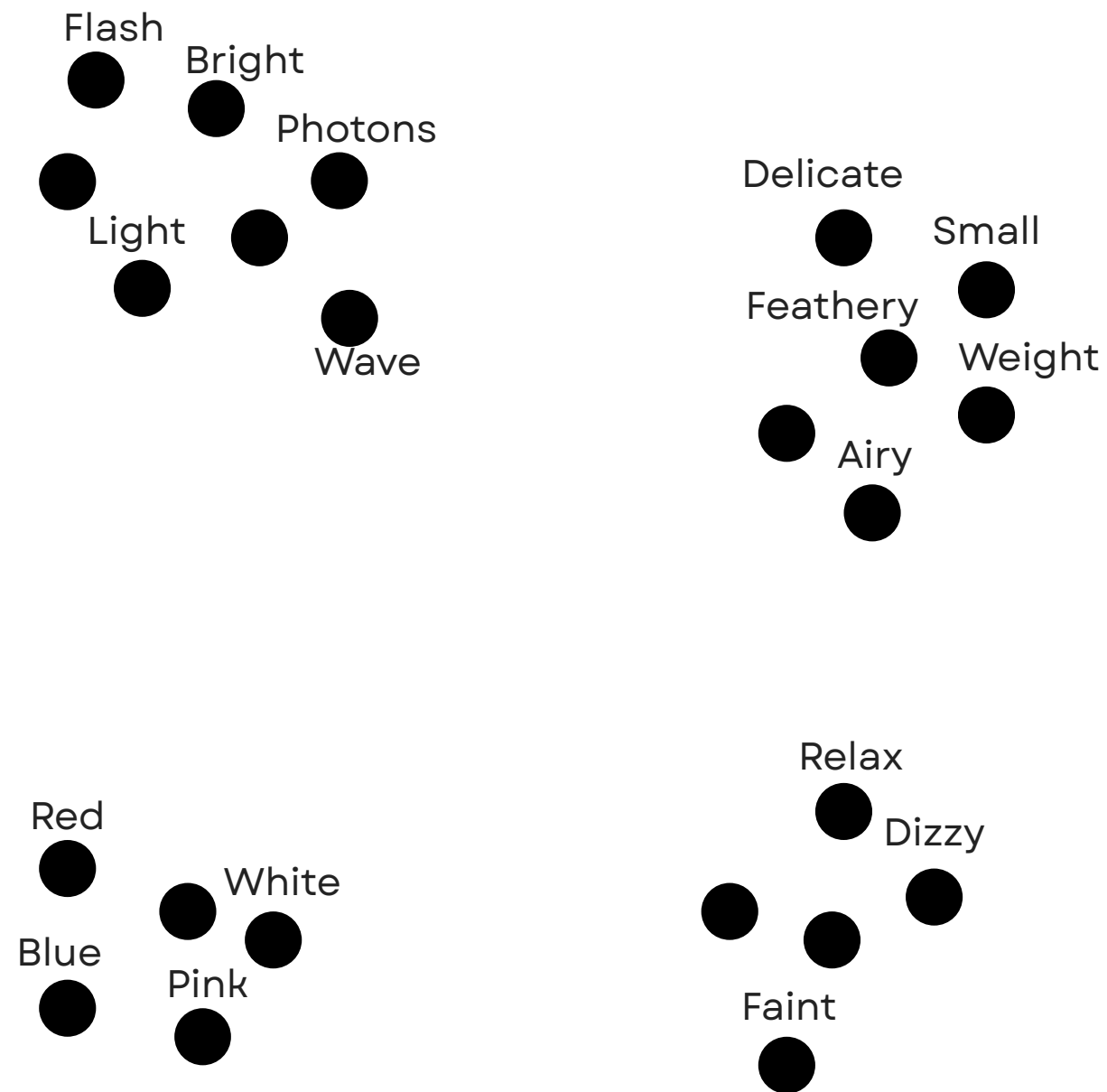


Self Attention - Brief Overview



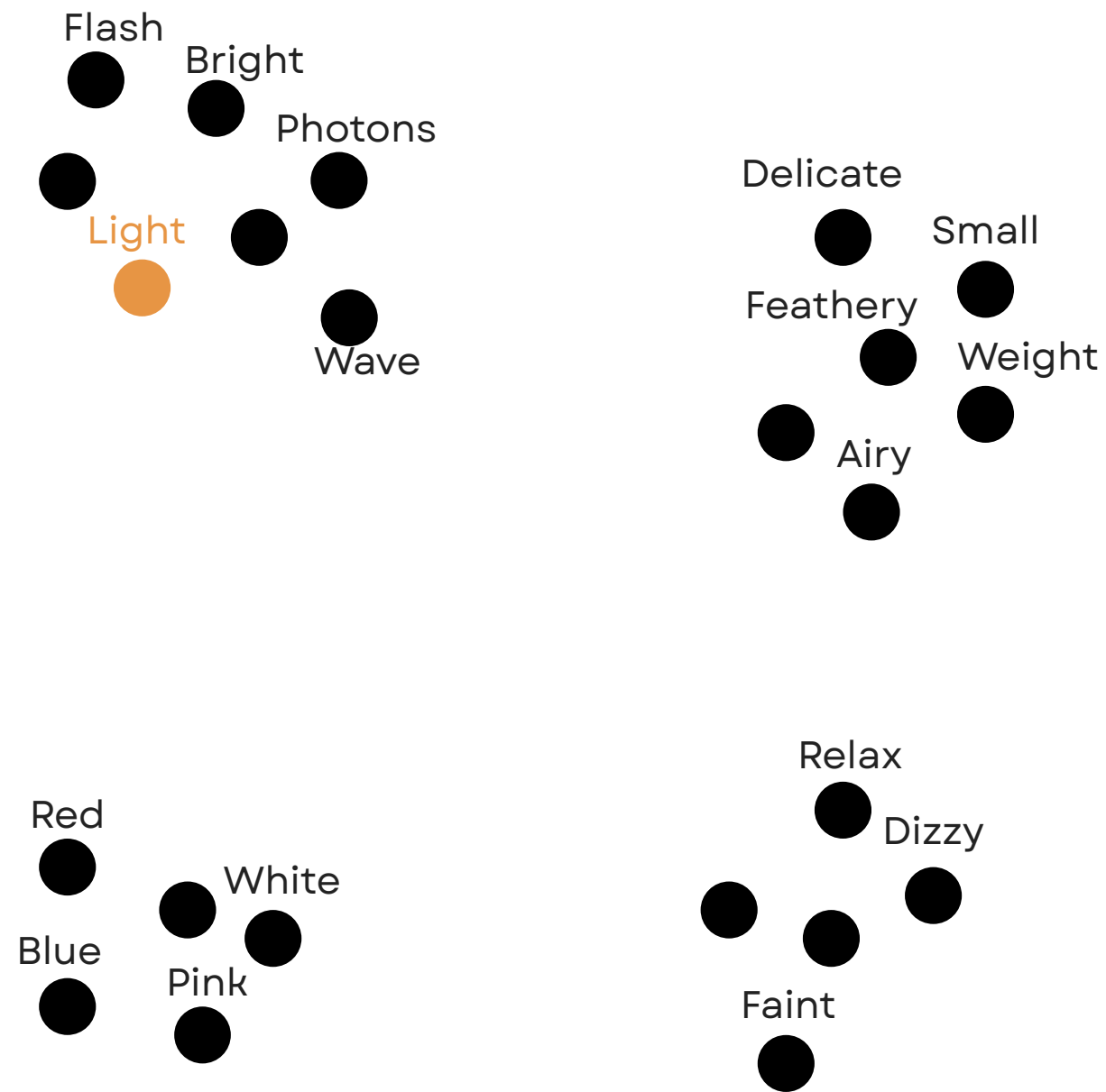
Self Attention - Brief Overview

Word Embeddings



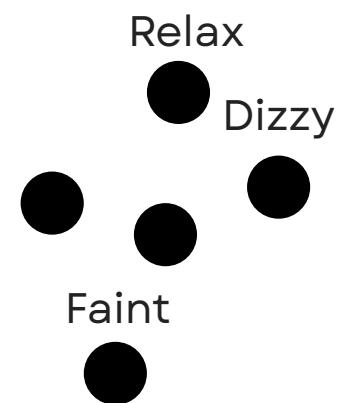
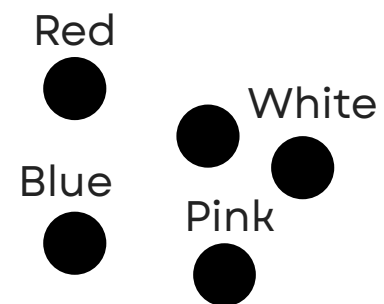
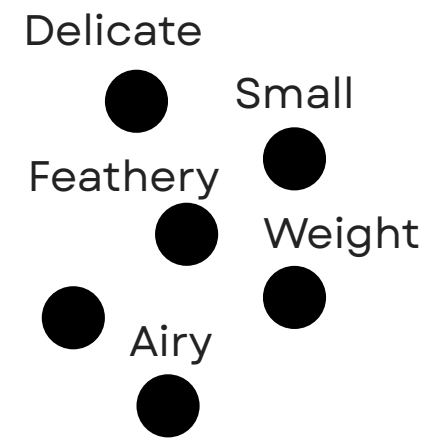
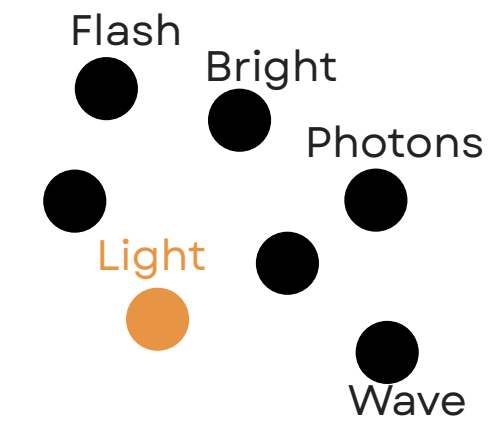
Self Attention - Brief Overview

Word Embeddings



Self Attention - Brief Overview

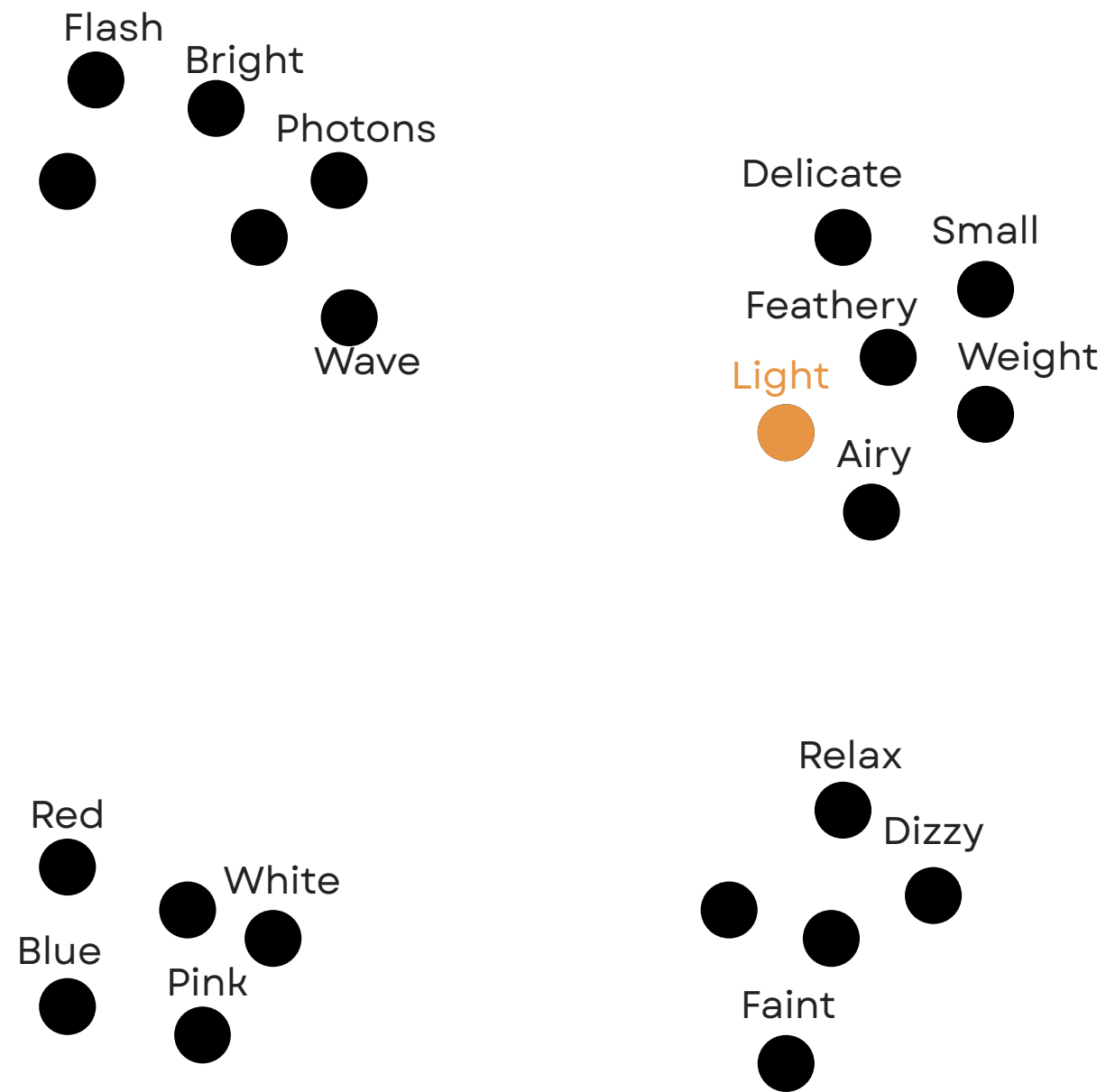
Word Embeddings




LIGHT WEIGHT

Self Attention - Brief Overview

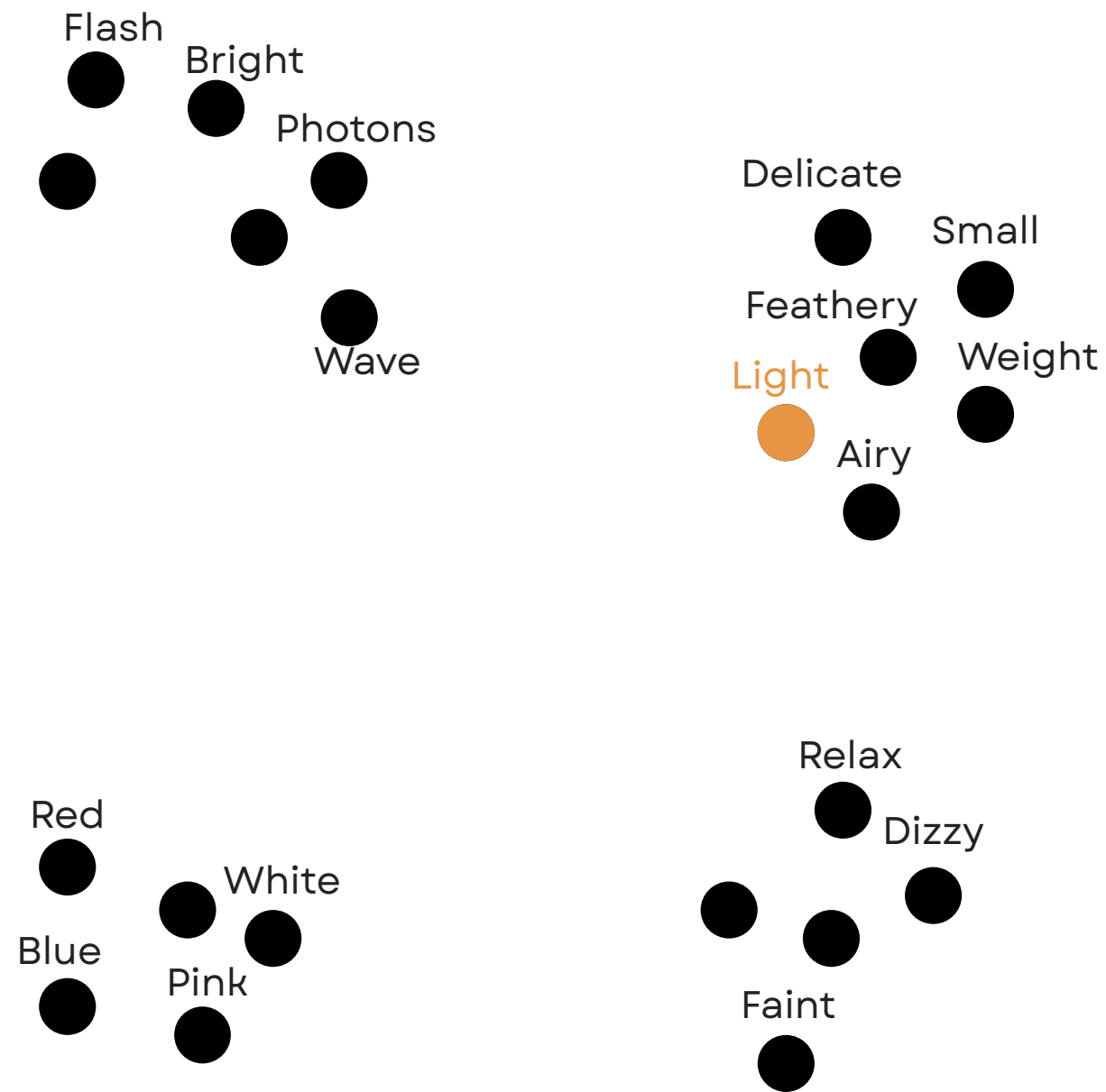
Word Embeddings




LIGHT WEIGHT

Self Attention - Brief Overview

Word Embeddings



LIGHT WEIGHT

LIGHT BLUE

Self Attention - Brief Overview

Word Embeddings

Flash
Bright
Photons
Wave

Delicate
Small
Feathery
Weight
Airy

Red
Blue
Light
White
Pink

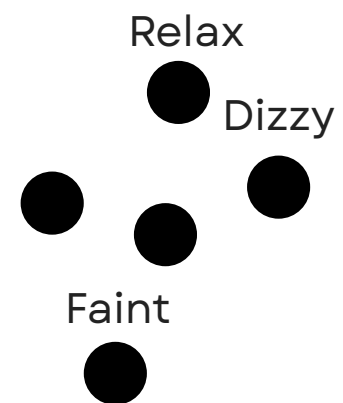
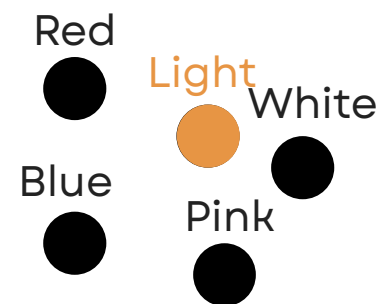
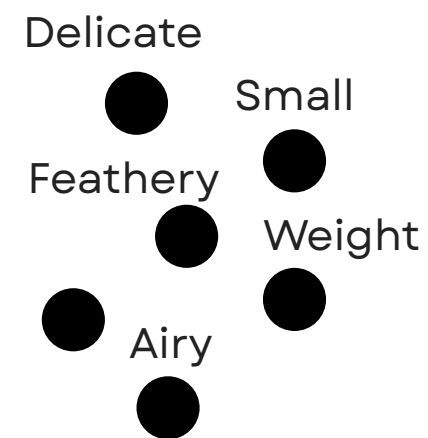
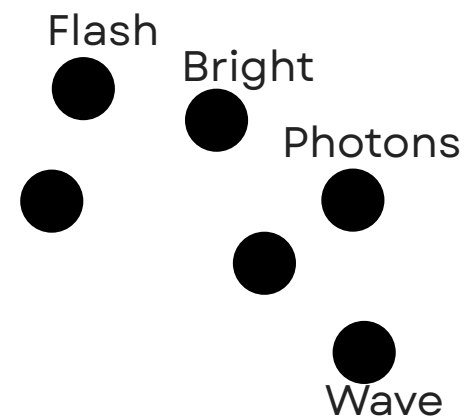
Relax
Dizzy
Faint

LIGHT WEIGHT

LIGHT BLUE

Self Attention - Brief Overview

Word Embeddings



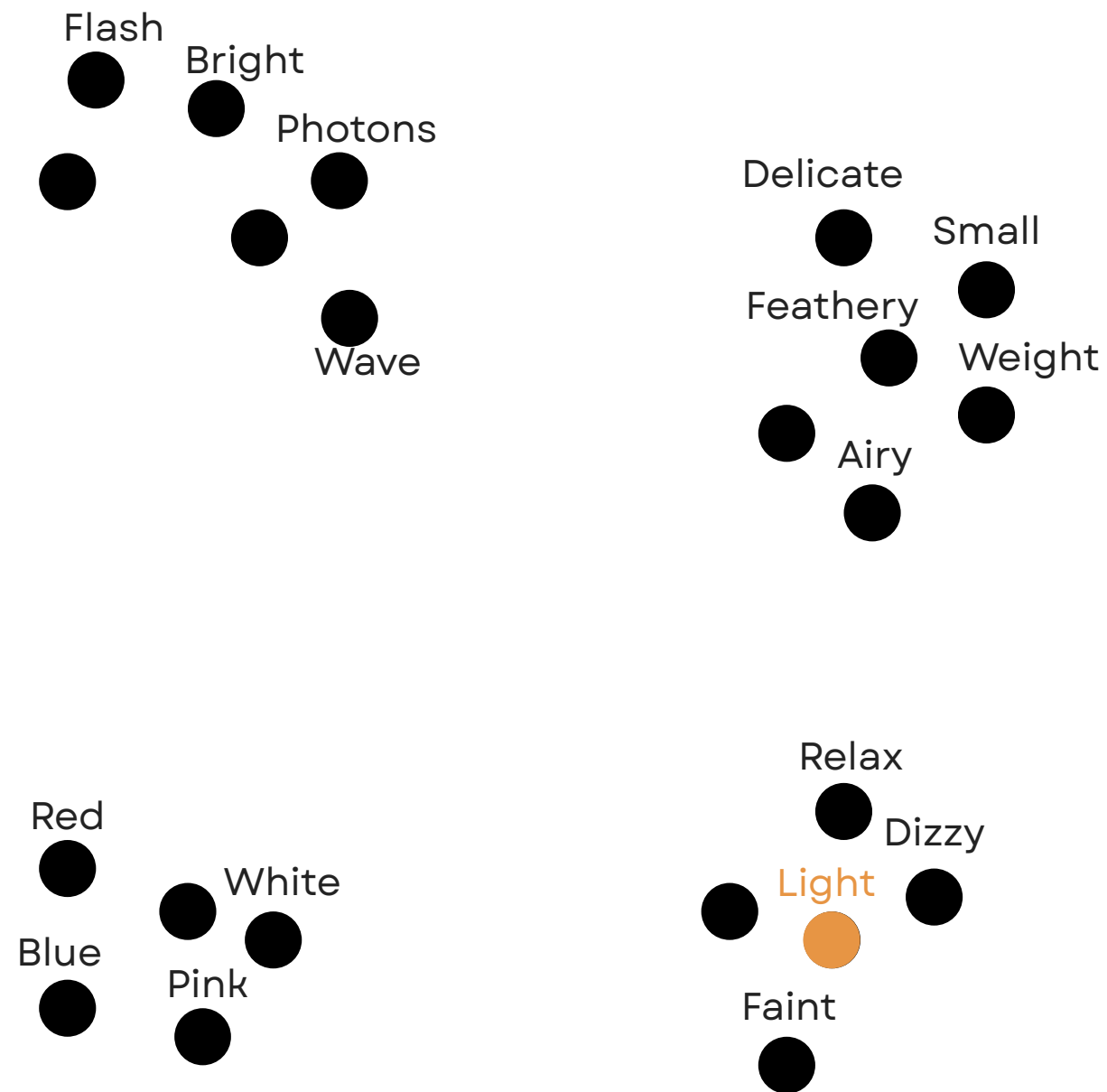

LIGHT WEIGHT


LIGHT BLUE


LIGHT HEADED

Self Attention - Brief Overview

Word Embeddings



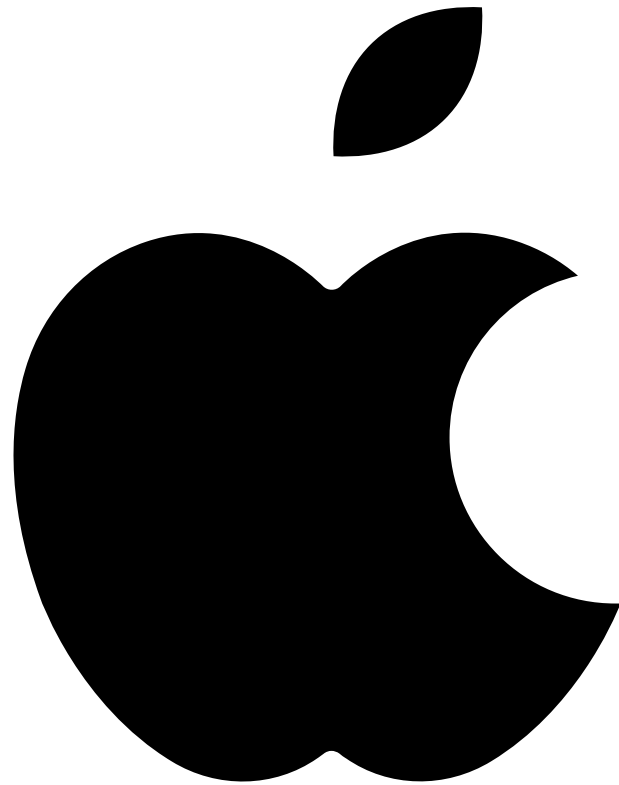

LIGHT WEIGHT


LIGHT BLUE

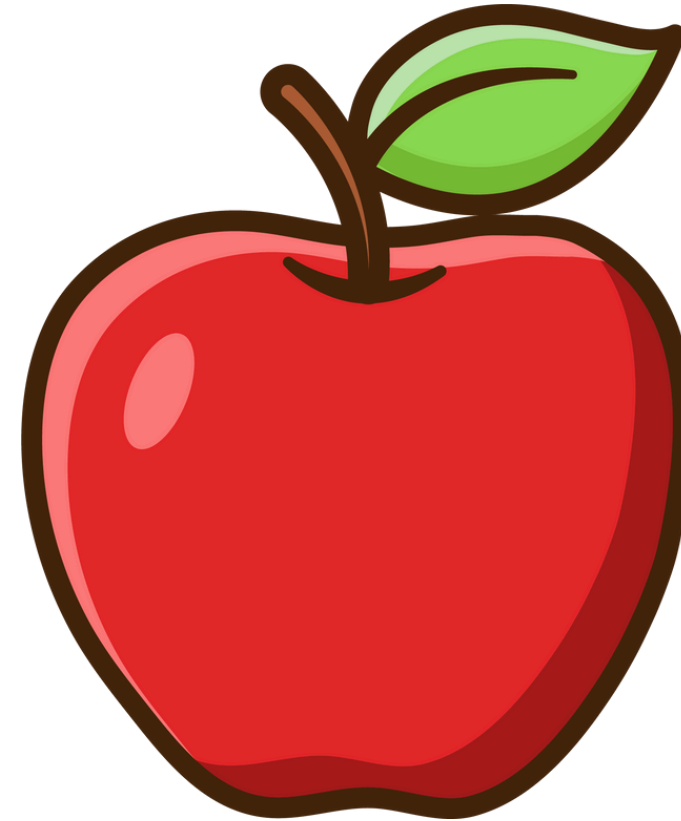

LIGHT HEADED

Self Attention - Mathematics

LOVE APPLE PHONE

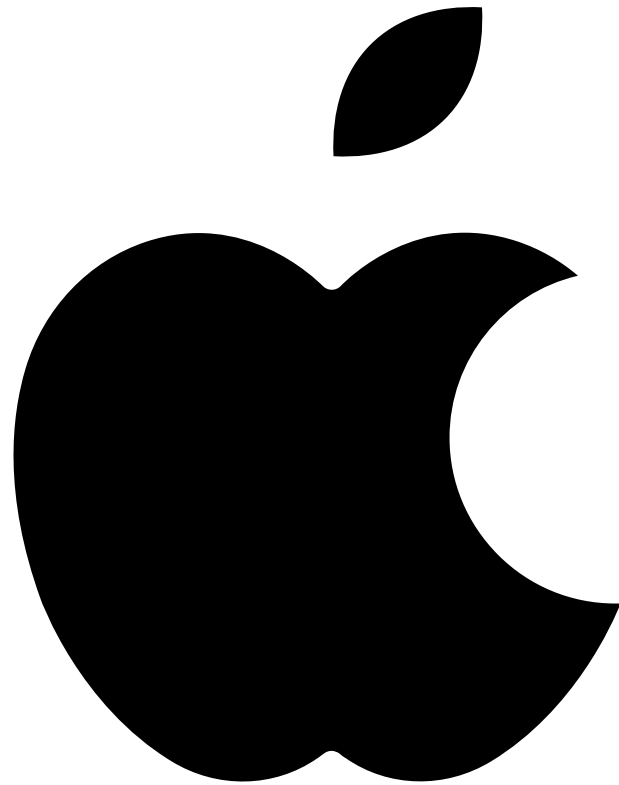


LOVE APPLE JUICE

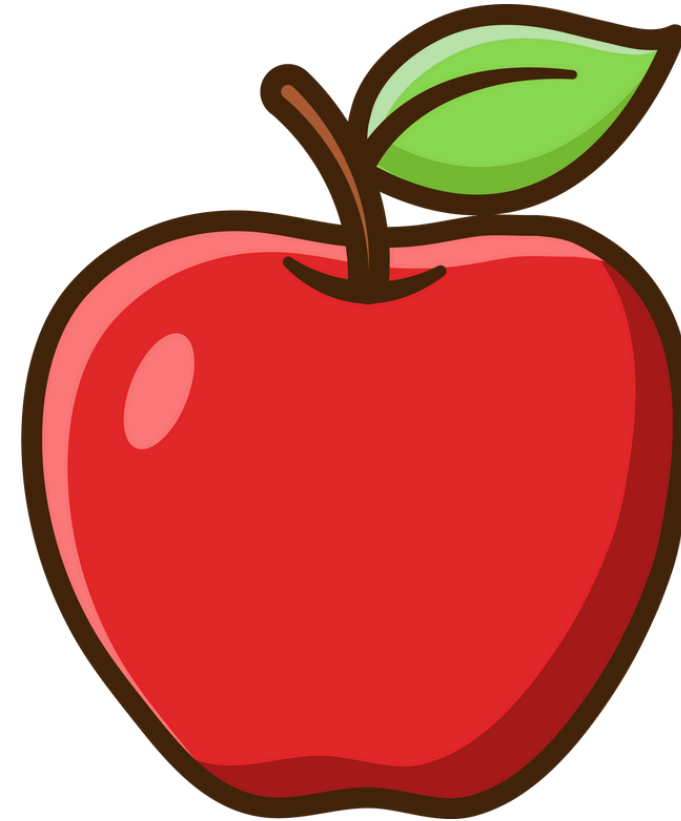


Self Attention - Mathematics

LOVE APPLE PHONE

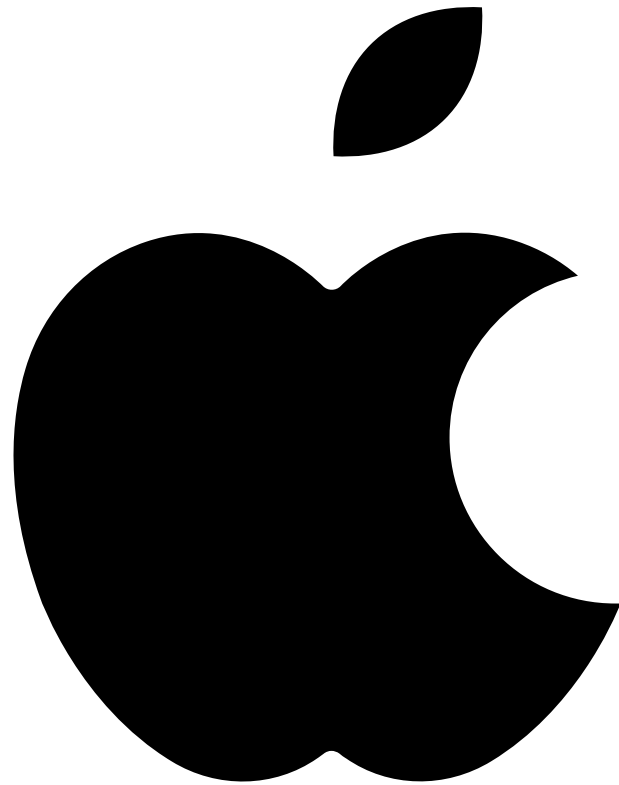


LOVE APPLE JUICE

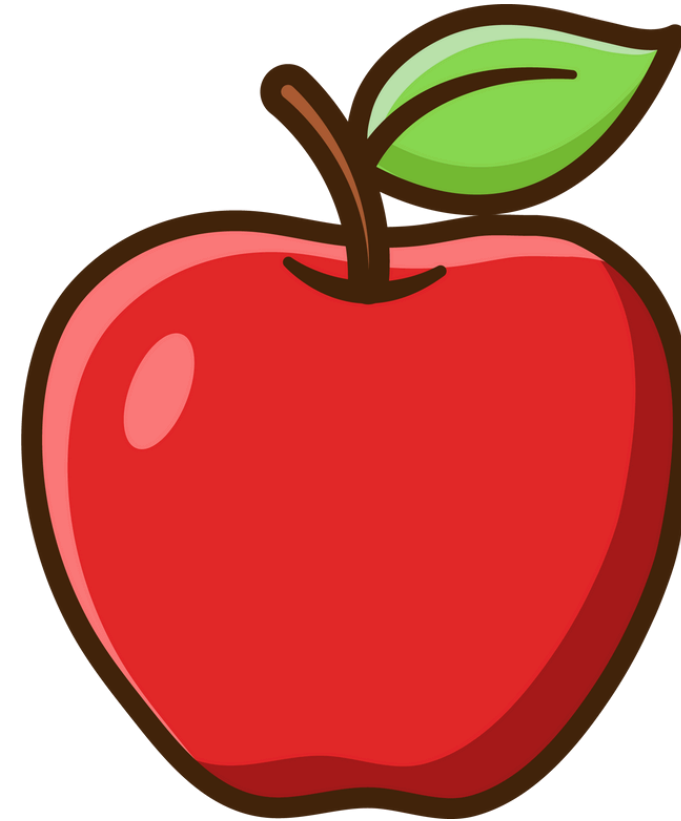


Self Attention - Mathematics


LOVE APPLE PHONE




LOVE APPLE JUICE



LOVE APPLE PHONE

Apple' =

APPLE

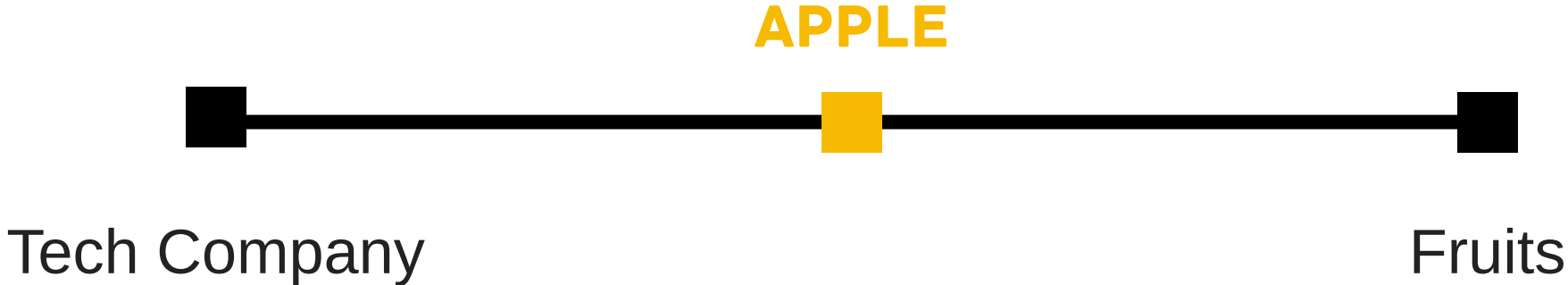


Tech Company

Fruits

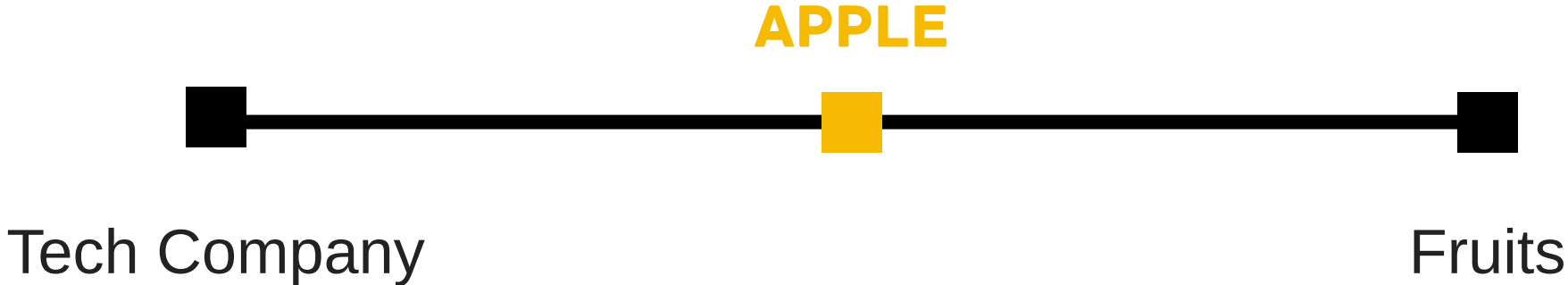
LOVE APPLE PHONE

$Apple' = 0.1 \times Love$



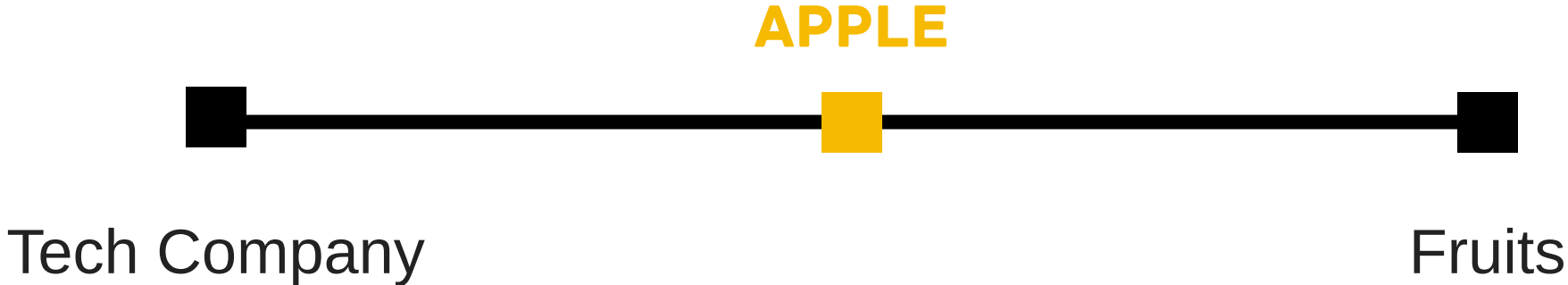
LOVE APPLE PHONE

$Apple' = 0.1 \times Love + 0.5 \times Apple$



LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$



LOVE APPLE PHONE

E_{love}

Apple' = 0.1 x Love + 0.5 x Apple + 0.4 x Phone

APPLE

Tech Company

Fruits

LOVE APPLE PHONE

E_{love}

E_{apple}

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

E_{love}

E_{apple}

E_{phone}

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

APPLE



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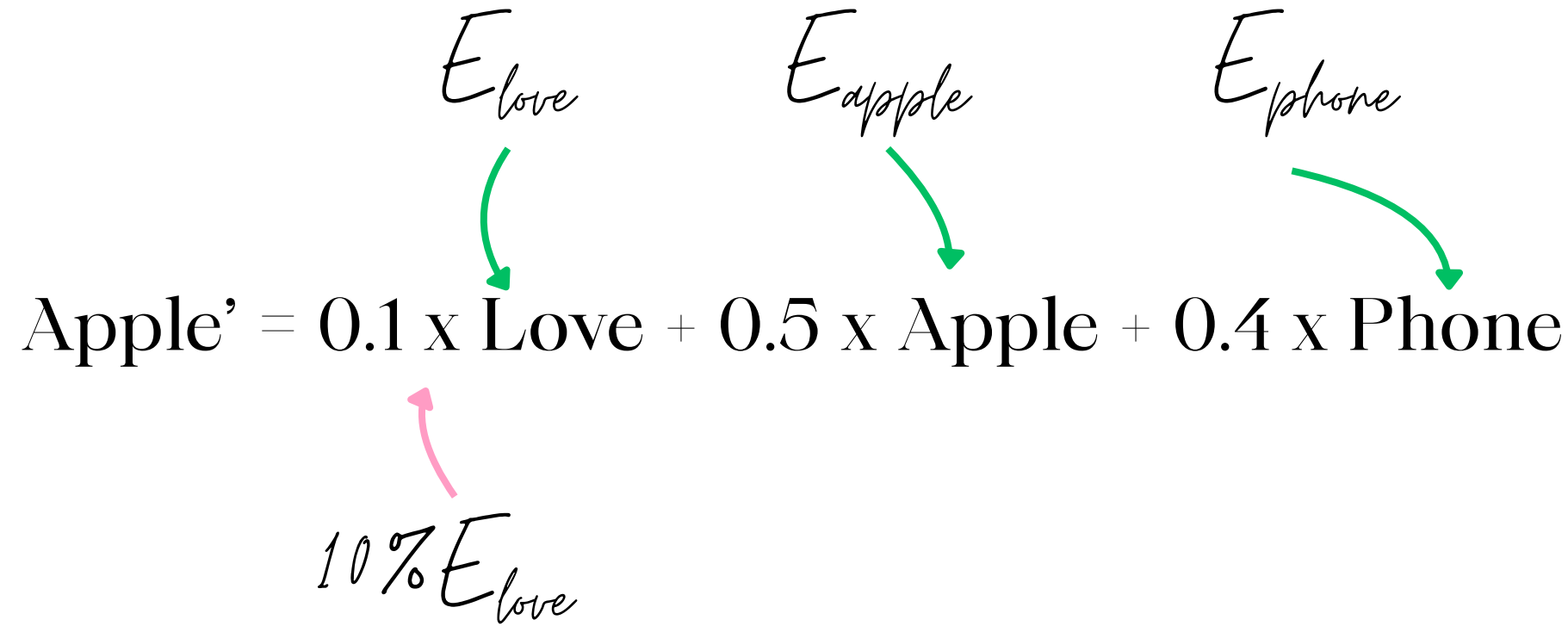
Fruits

LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

E_{love} *E_{apple}* *E_{phone}*

10% E_{love}



APPLE



Tech Company

Fruits

LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

E_{love} E_{apple} E_{phone}

$10\% E_{\text{love}}$ $50\% E_{\text{apple}}$

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

E_{love} *E_{apple}* *E_{phone}*

10% E_{love} *50% E_{apple}* *40% E_{phone}*

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

E_{love} E_{apple} E_{phone}

$10\% E_{\text{love}}$ $50\% E_{\text{apple}}$ $40\% E_{\text{phone}}$

APPLE



Tech Company

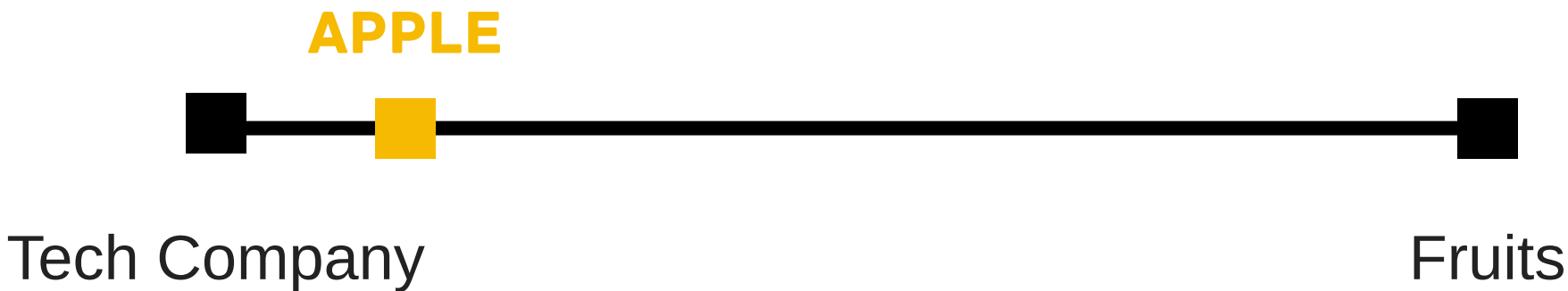
Fruits

LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

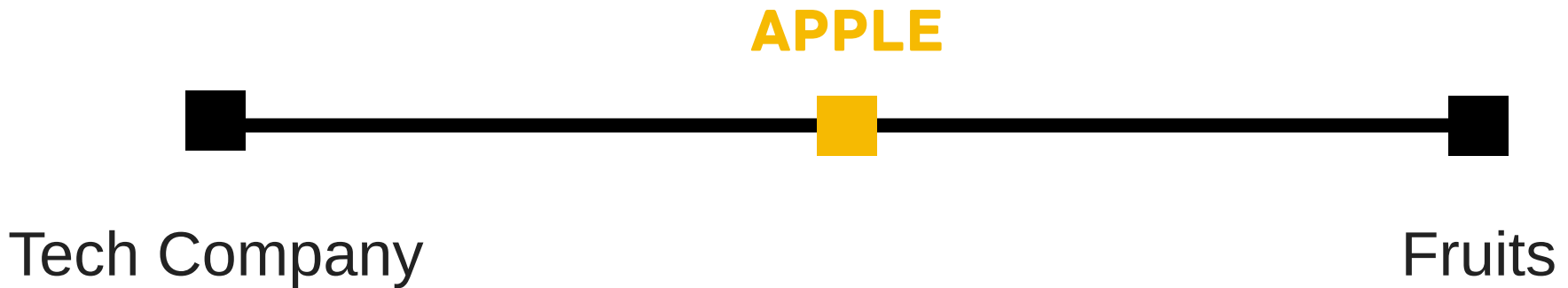
E_{love} *E_{apple}* *E_{phone}*

10% E_{love} *50% E_{apple}* *40% E_{phone}*



LOVE APPLE JUICE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Juice}$$

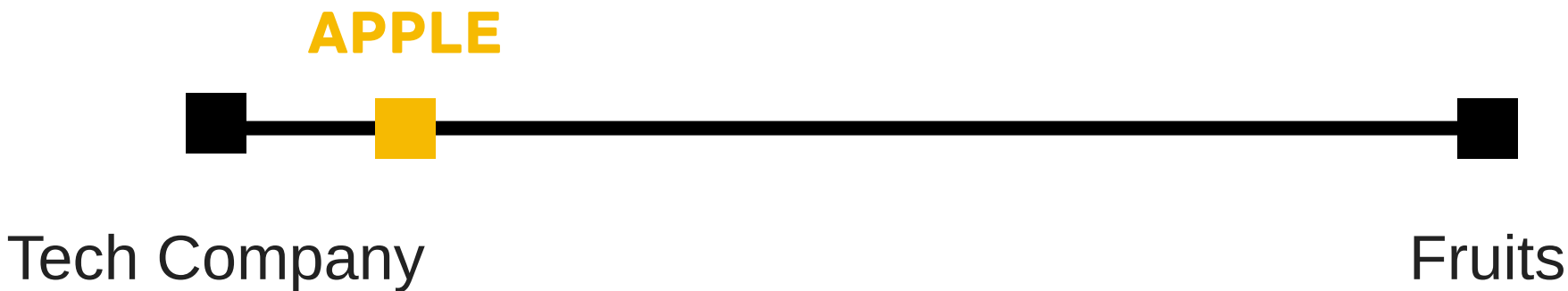


LOVE APPLE PHONE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Phone}$$

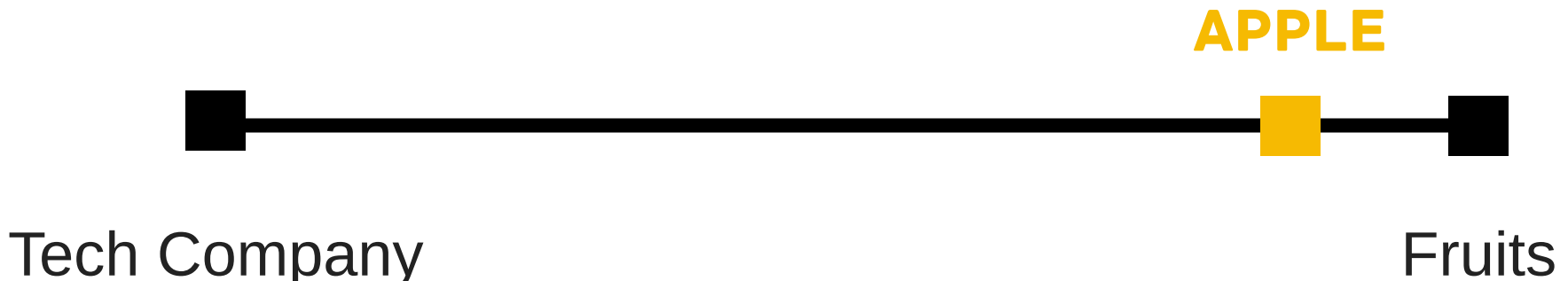
E_{love} E_{apple} E_{phone}

$10\% E_{\text{love}}$ $50\% E_{\text{apple}}$ $40\% E_{\text{phone}}$



LOVE APPLE JUICE

$$\text{Apple}' = 0.1 \times \text{Love} + 0.5 \times \text{Apple} + 0.4 \times \text{Juice}$$



LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

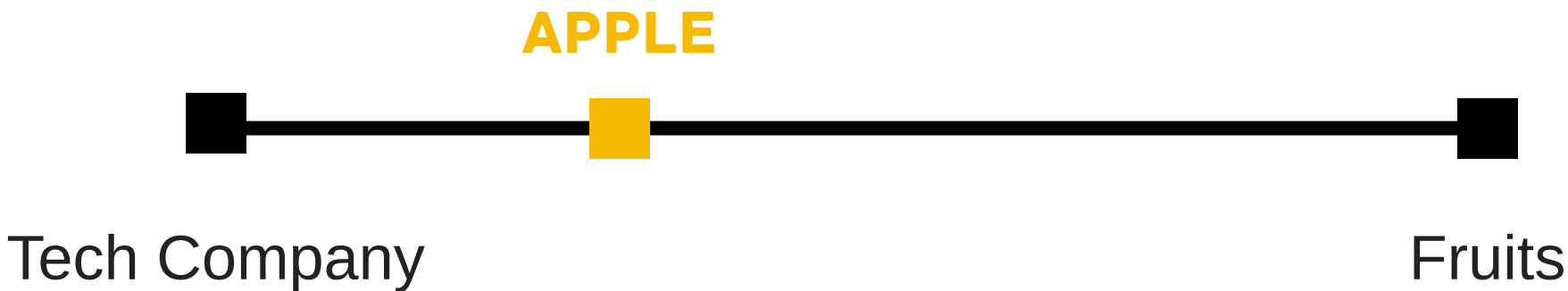
LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

(apple x love)

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

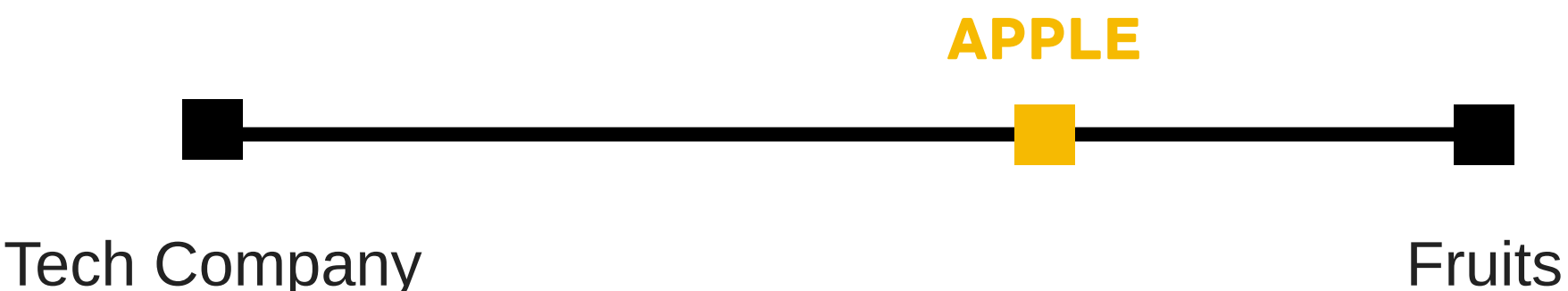


LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$



LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(apple x apple)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

(apple x love)

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

(apple x apple)

(apple x love)

(apple x phone)

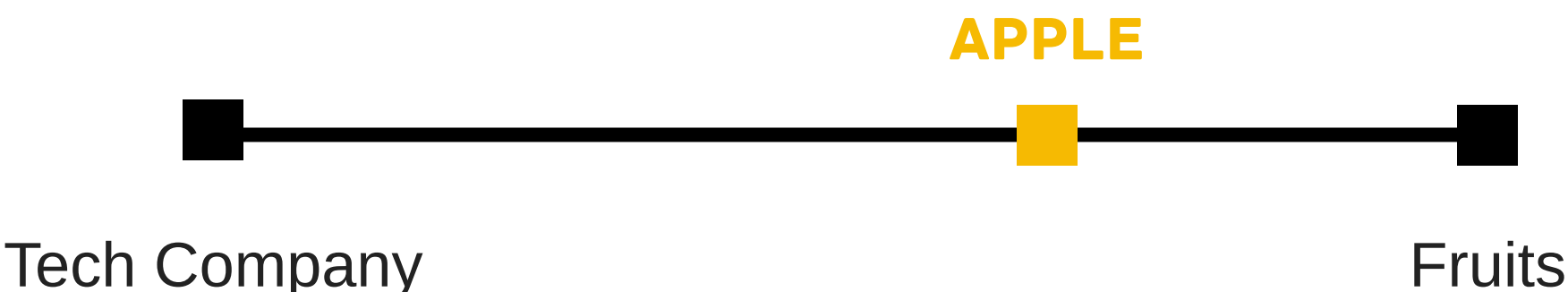
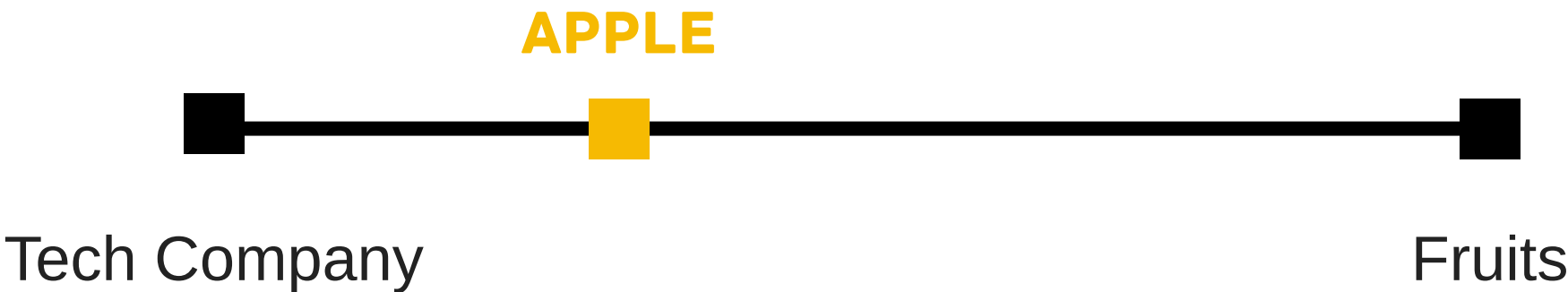
$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$



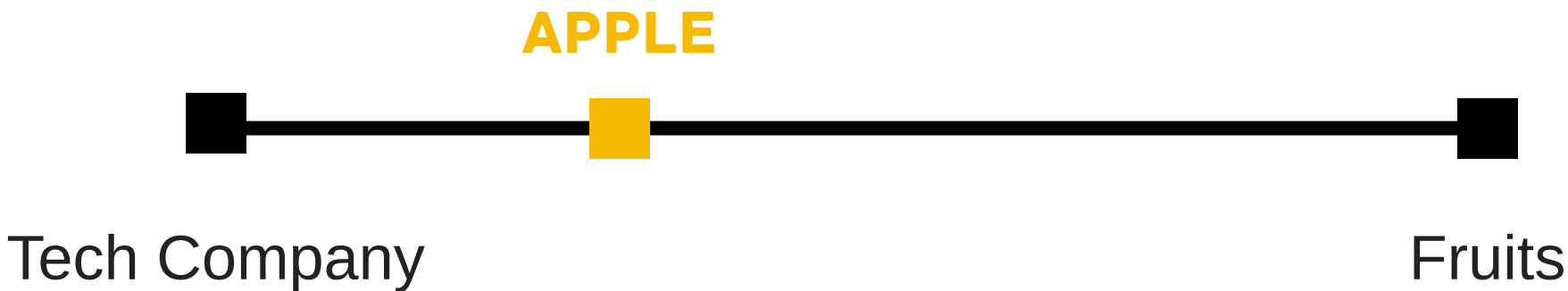
LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

(apple x love) *(apple x apple)* *(apple x phone)*

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$



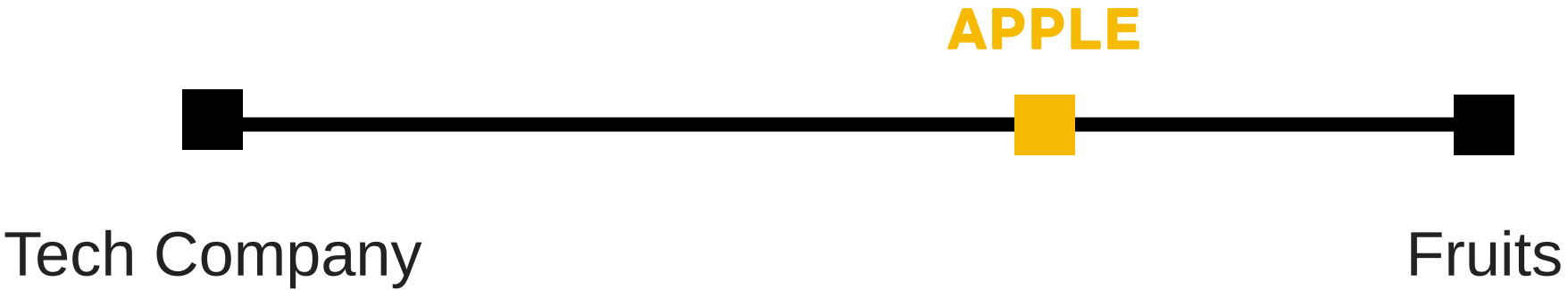
LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

(juice x love) *(juice x apple)* *(juice x juice)*



4

3

2

1

0

-4

-3

-2

-1

0

1

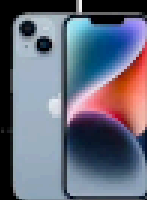
2

3

4



Love



Phones

Apple



Juice

4

3

2

1

0

-4

-3

-2

-1

0

1

2

3

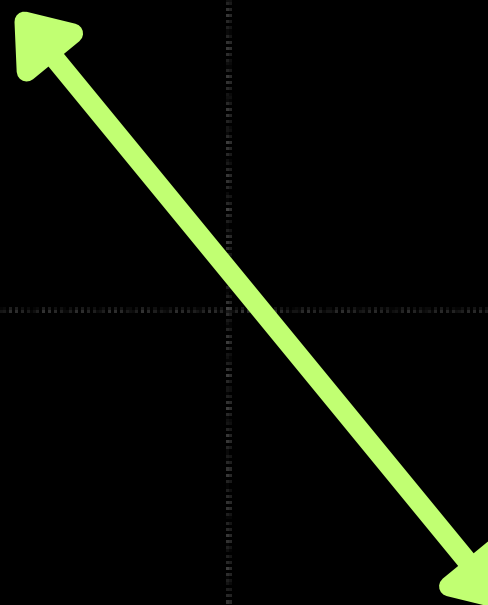
4



Love



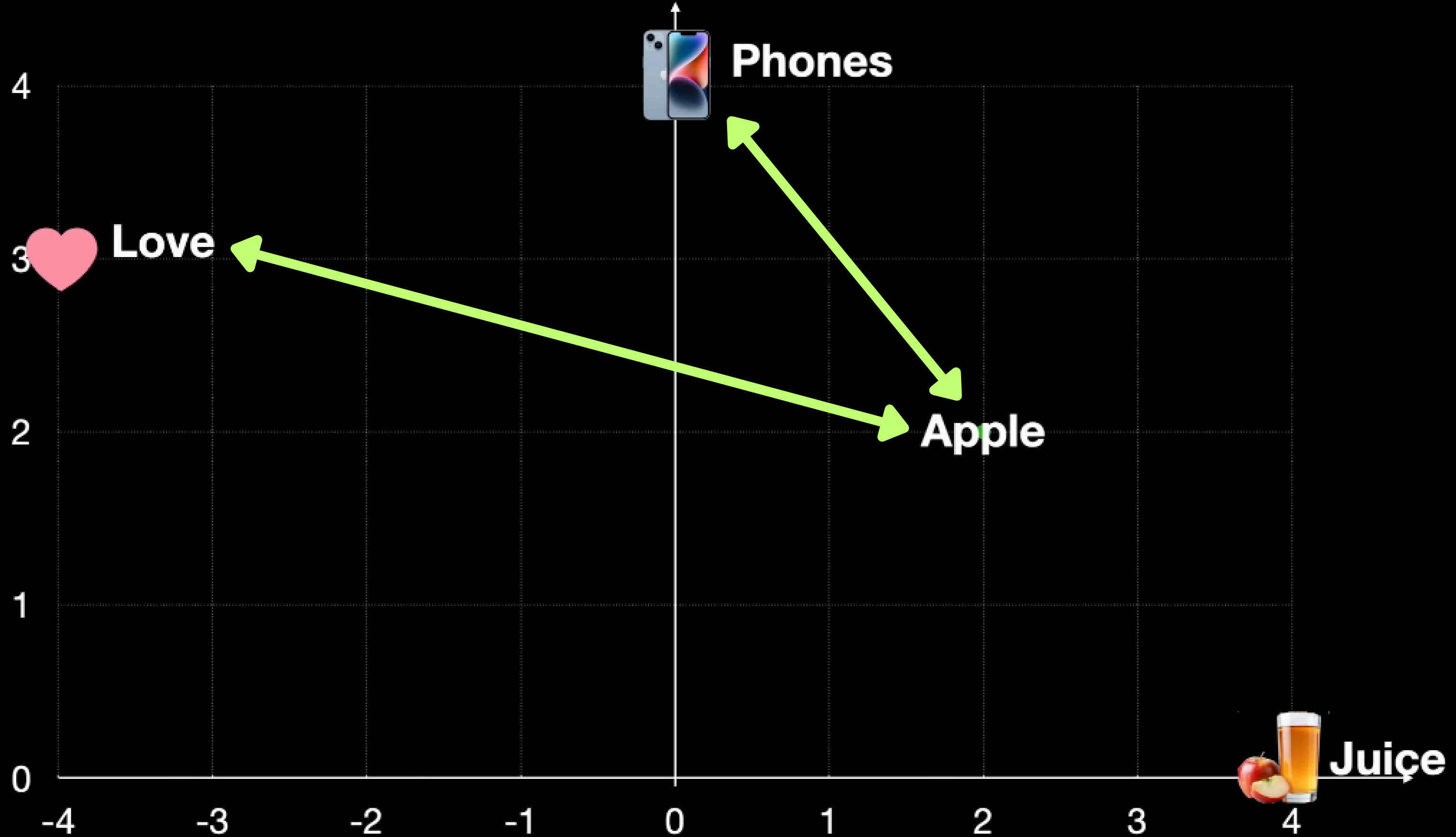
Phones

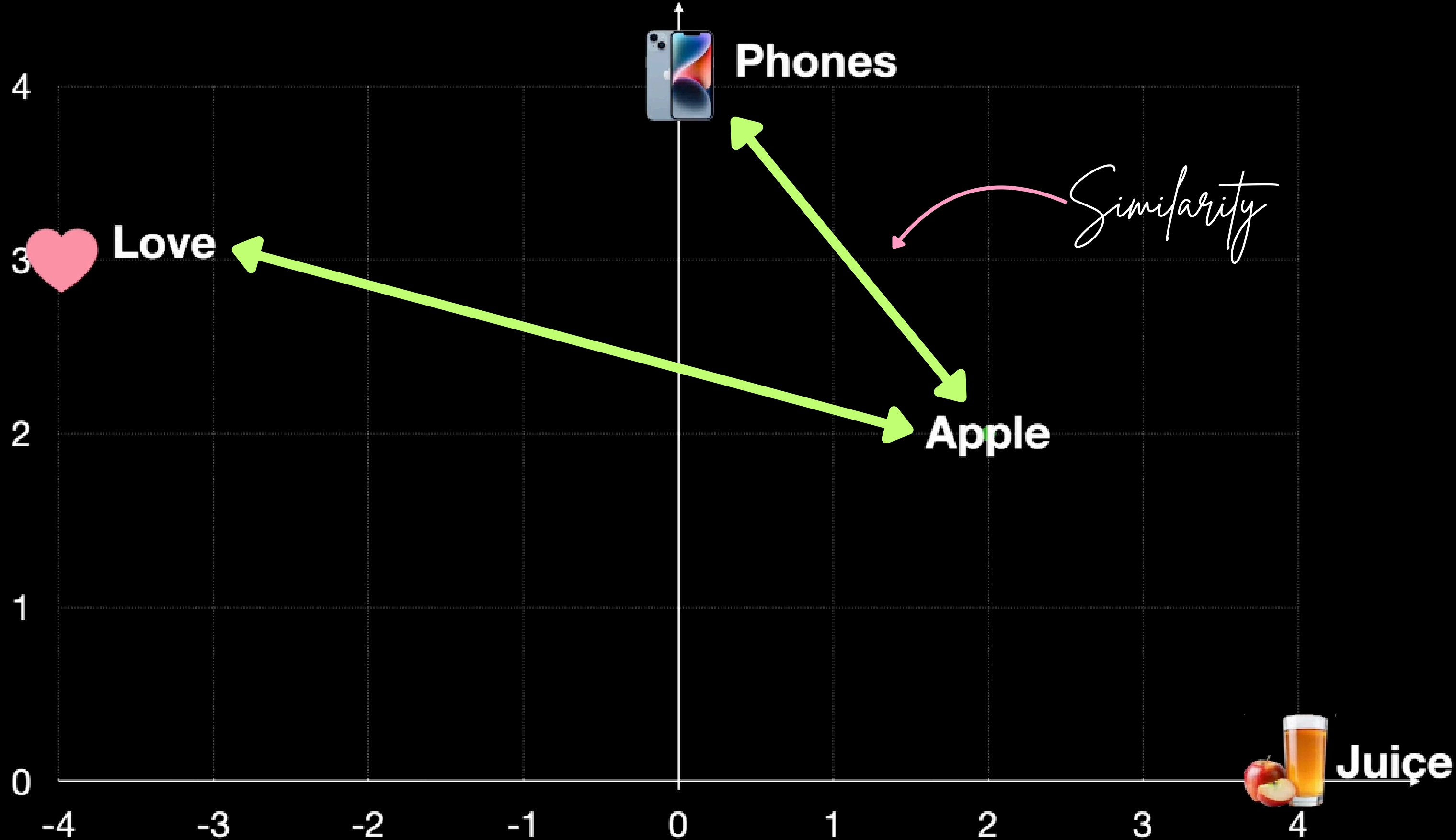


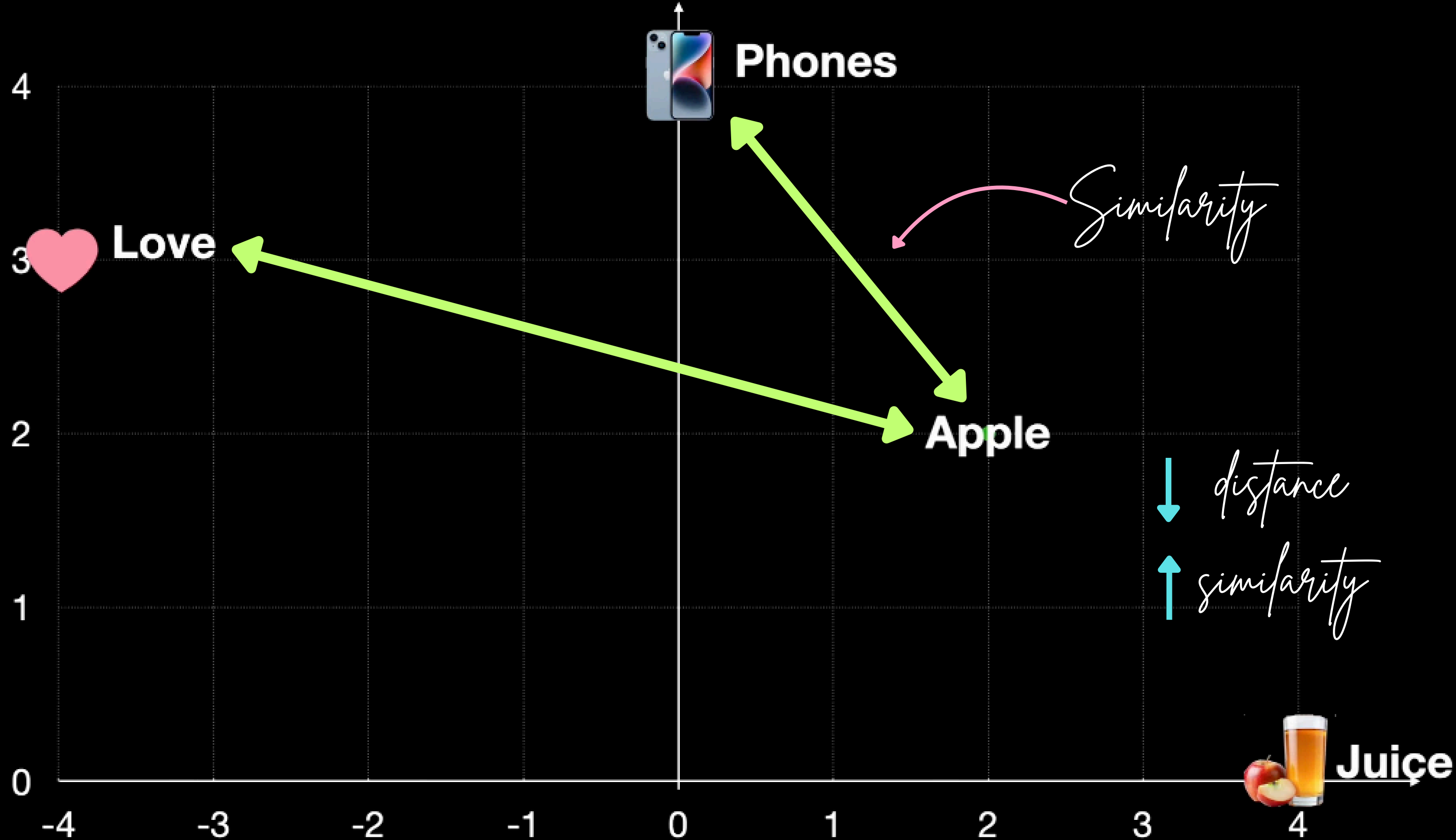
Apple



Juice







LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(apple x apple)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

(apple x love) *(apple x phone)*

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

(juice x apple)

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

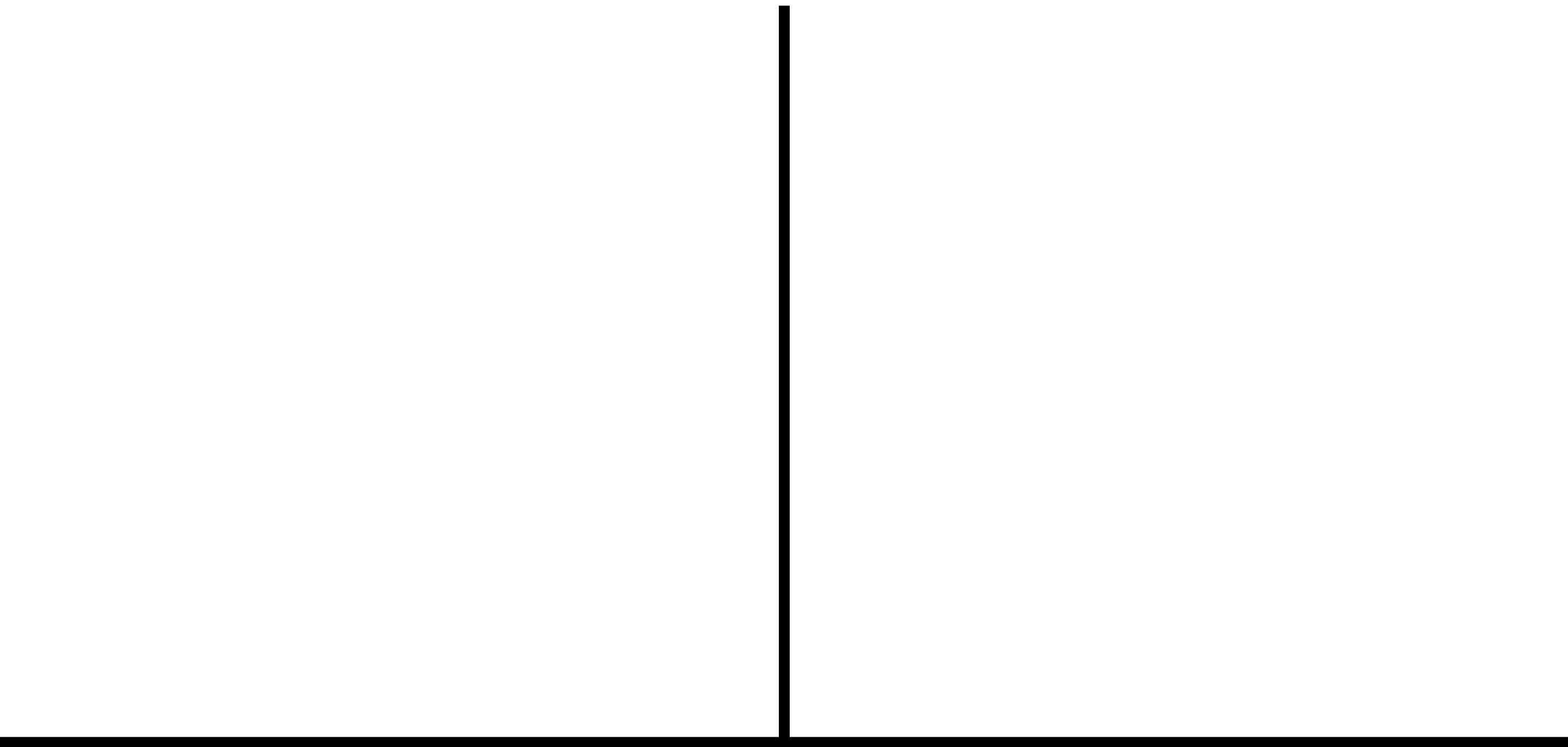
(juice x love) *(juice x juice)*

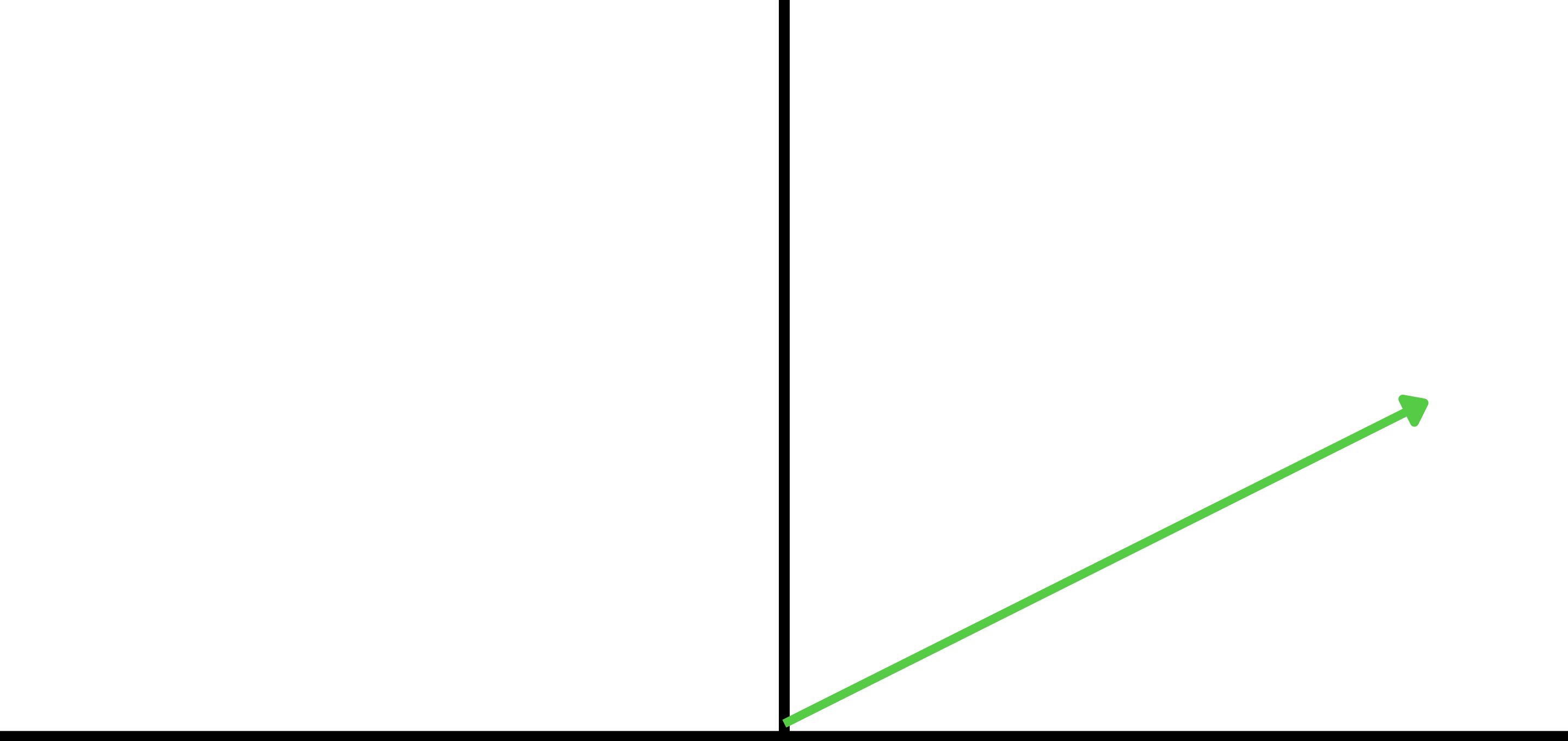
APPLE

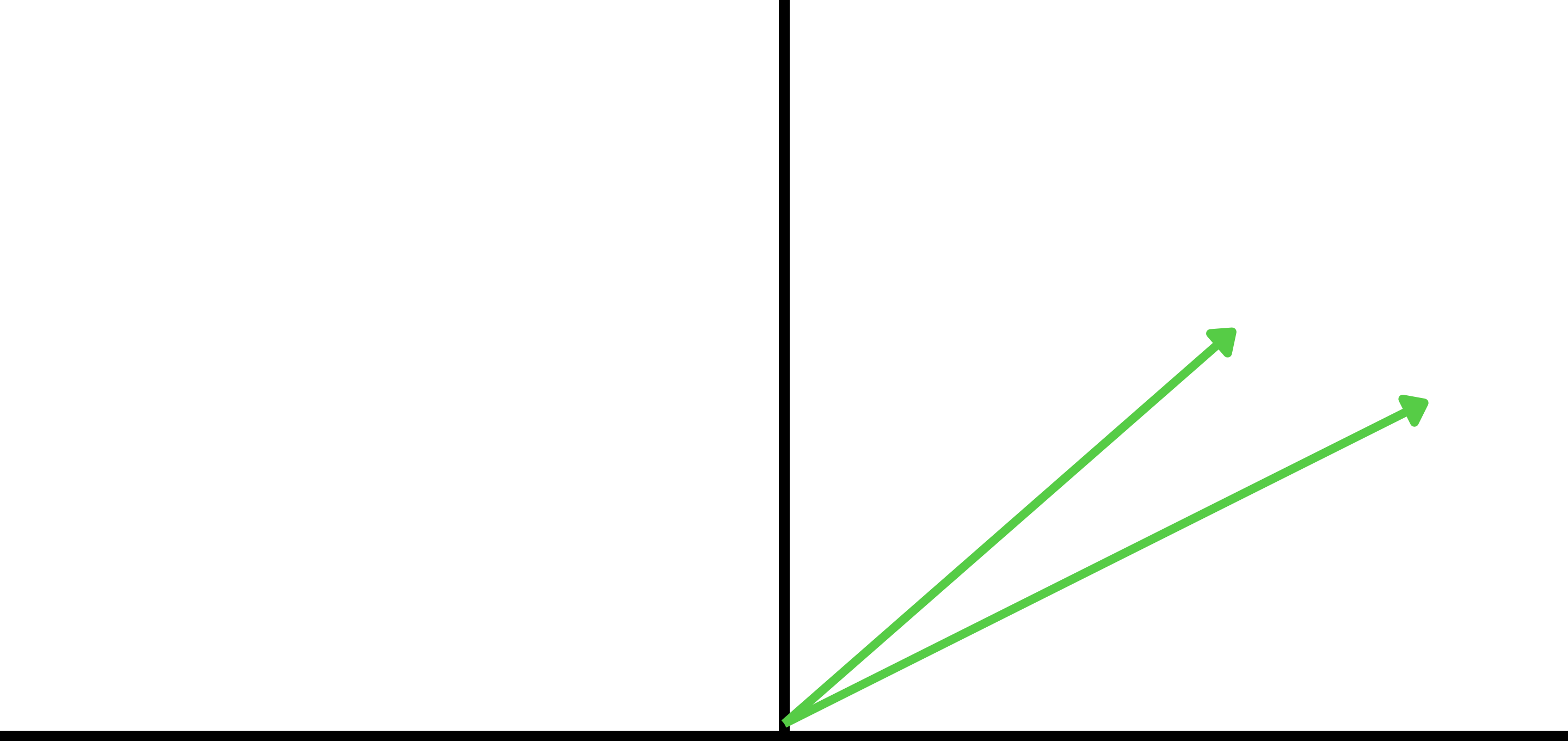


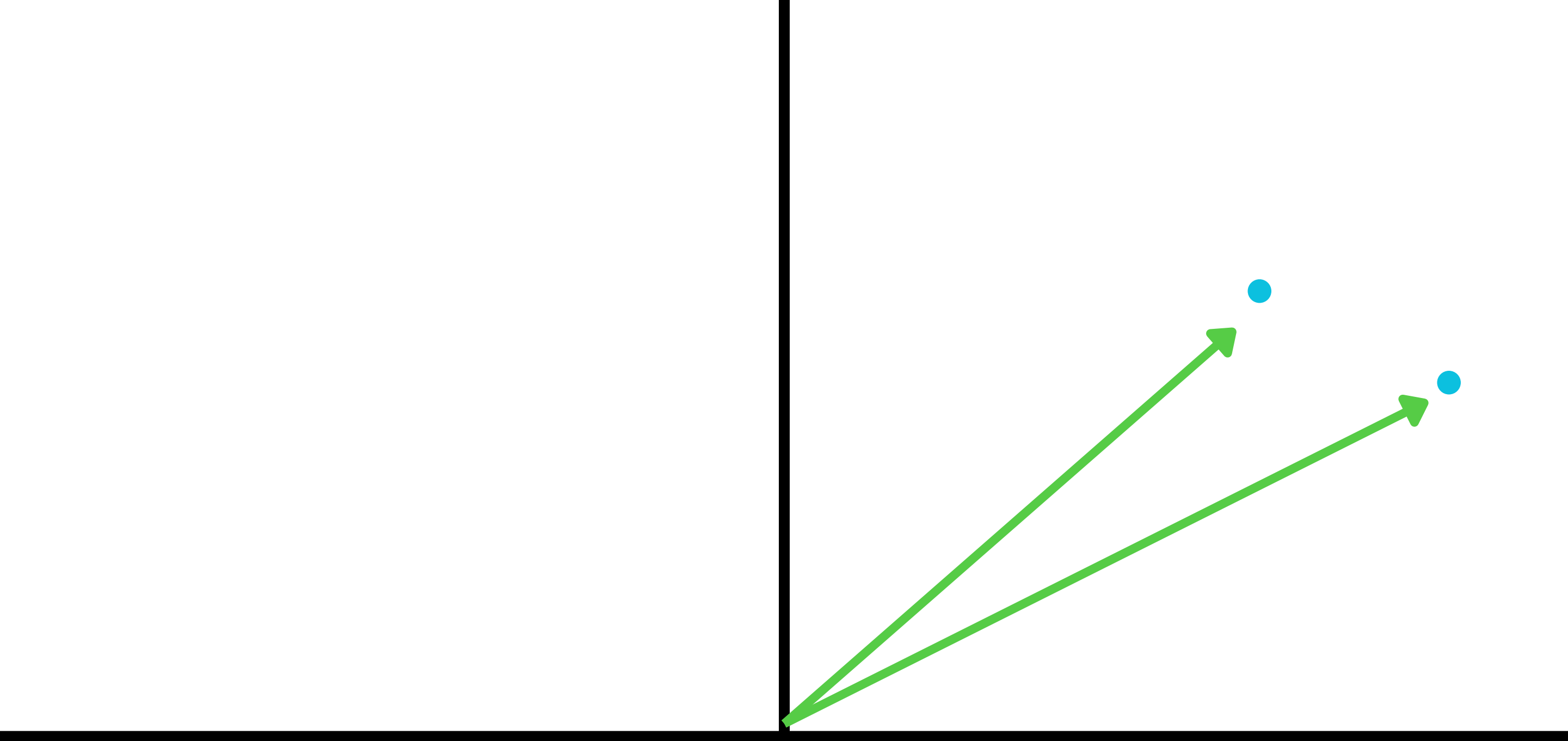
Tech Company

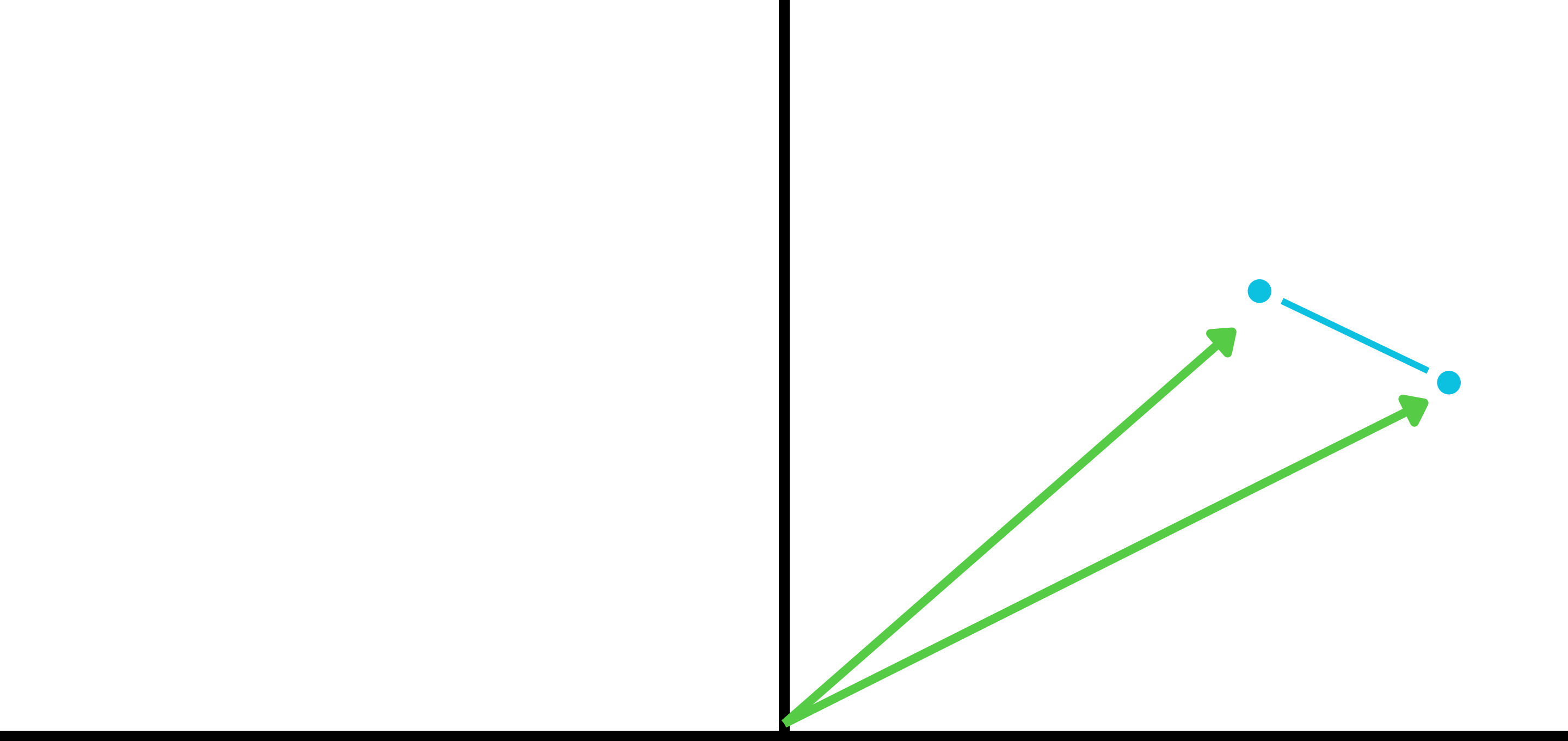
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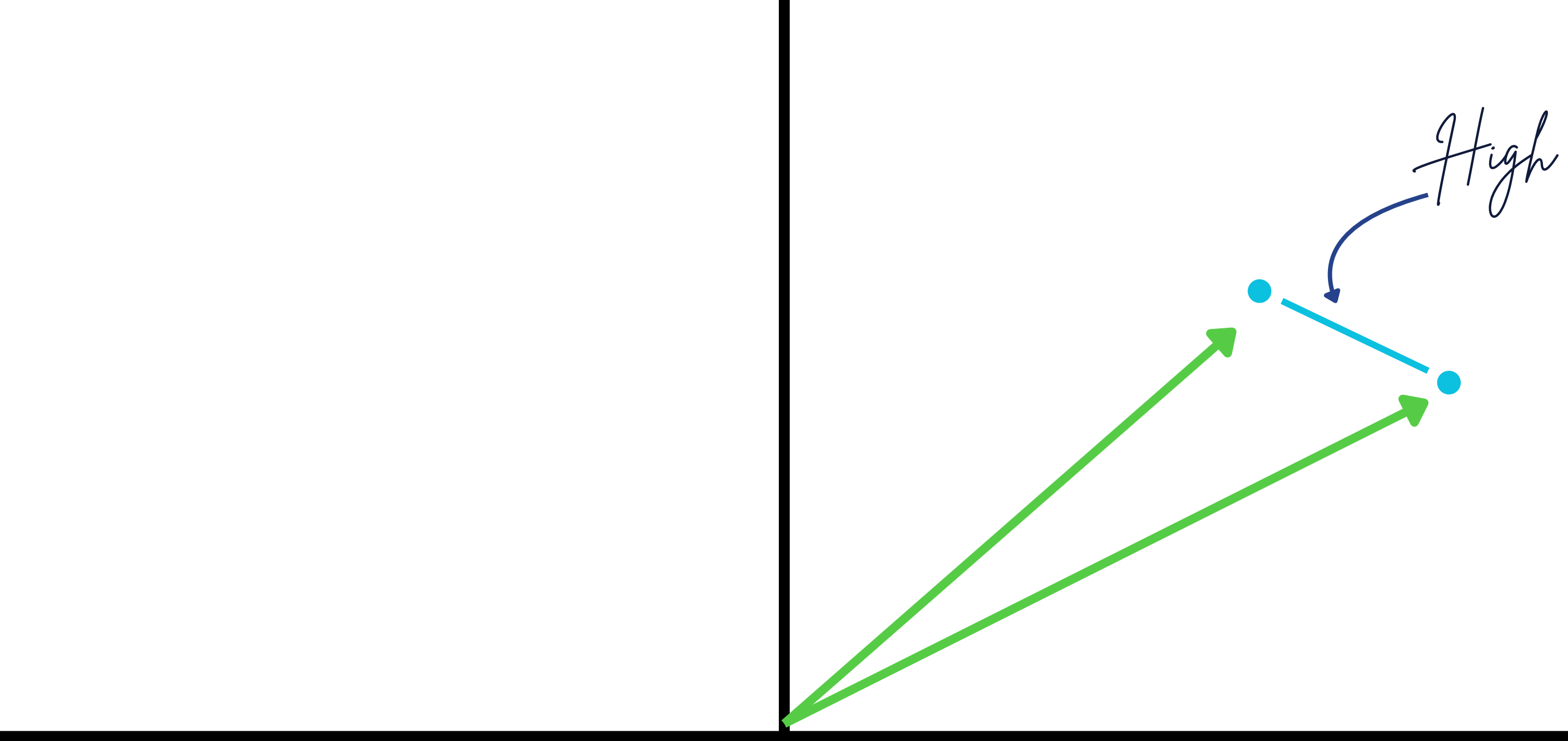


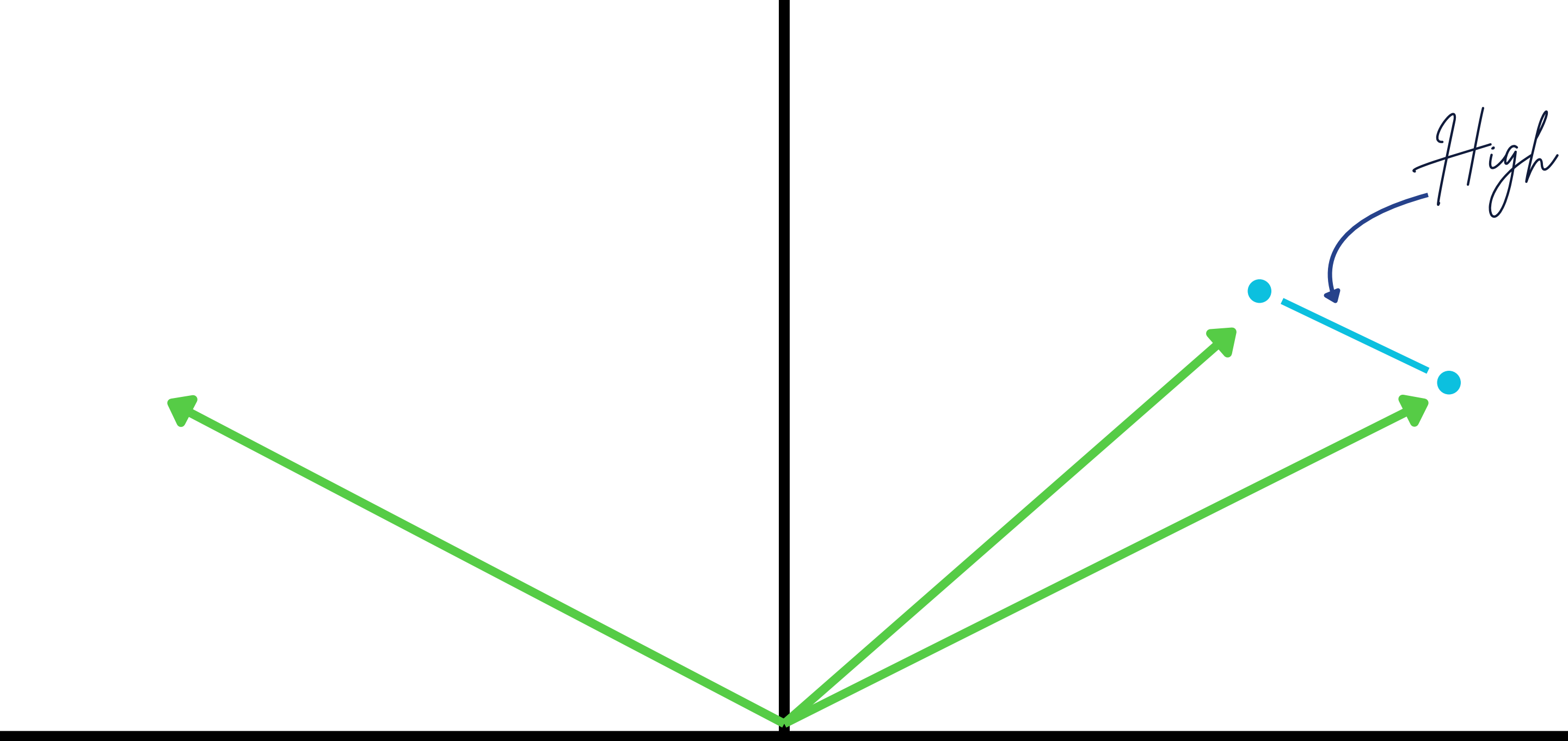




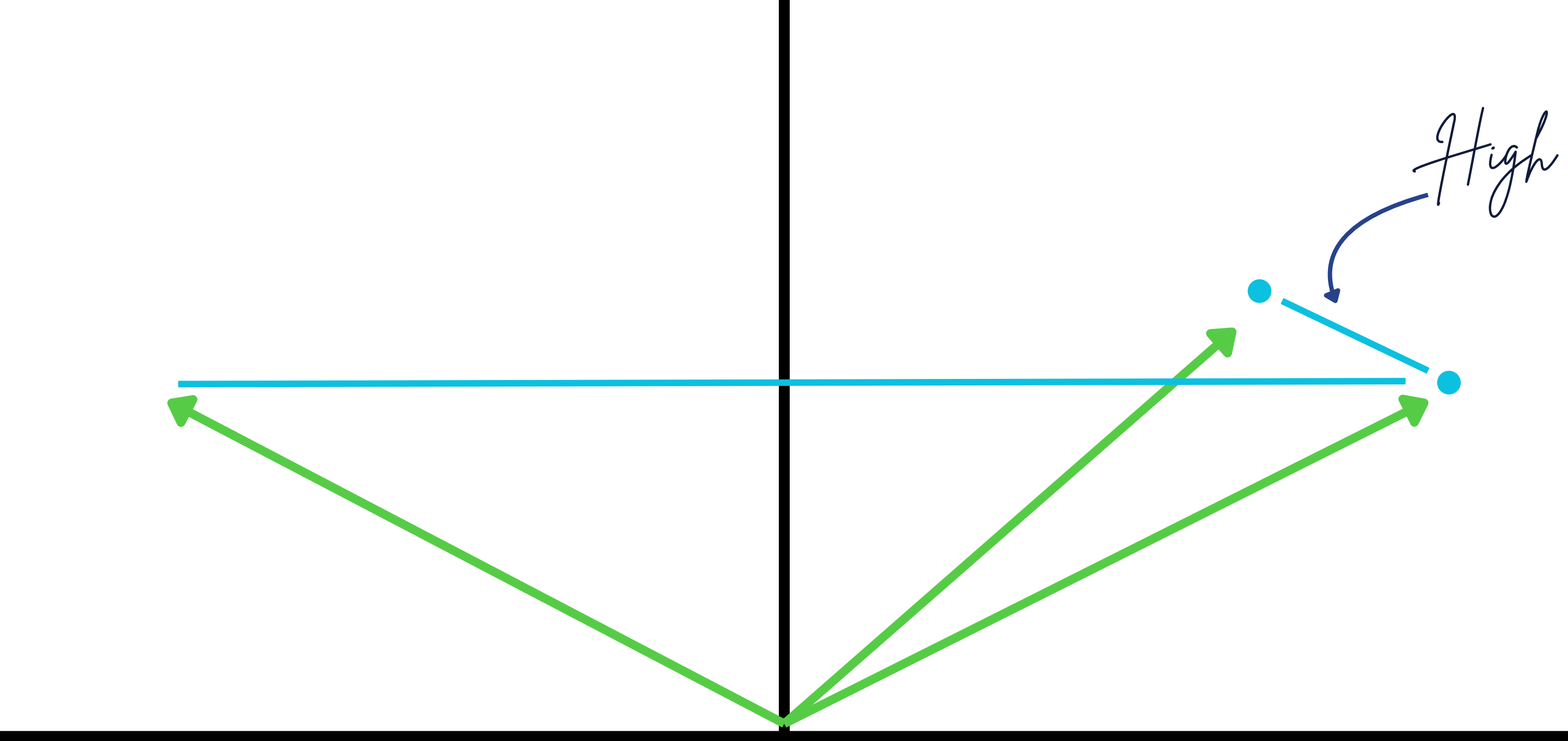




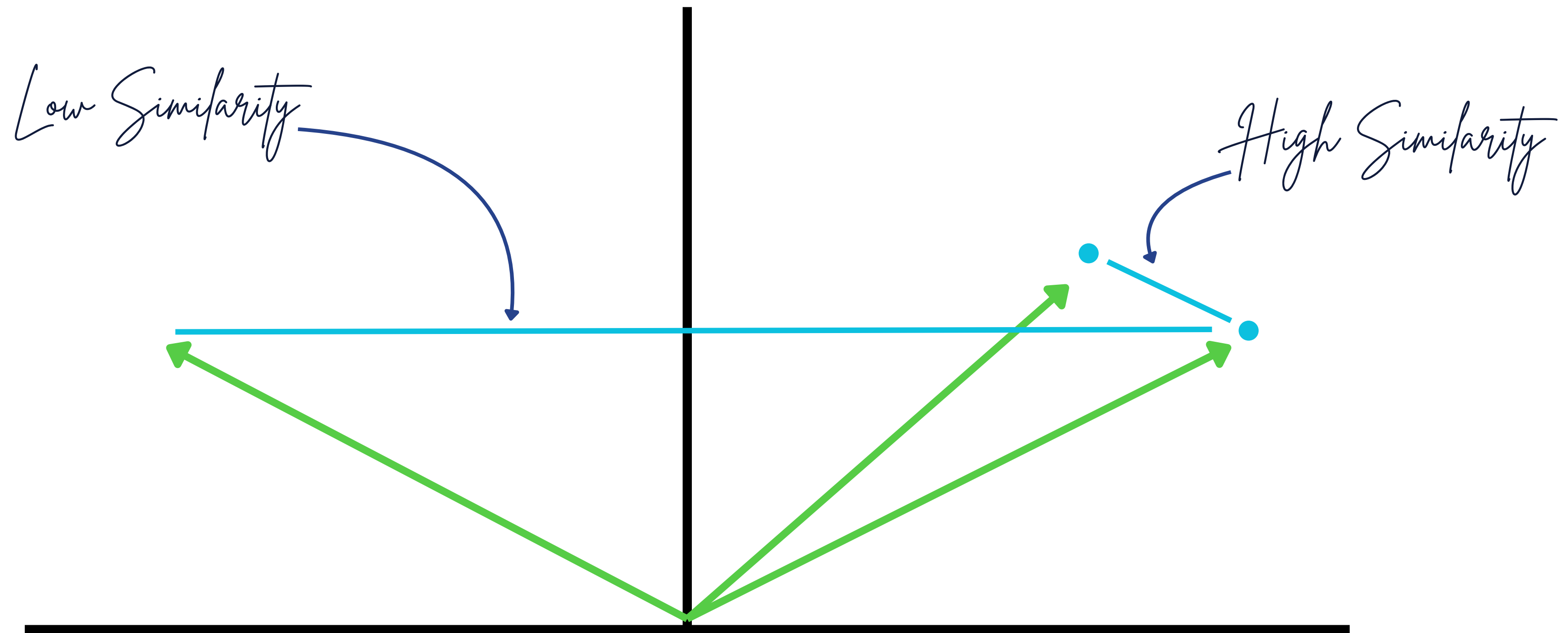




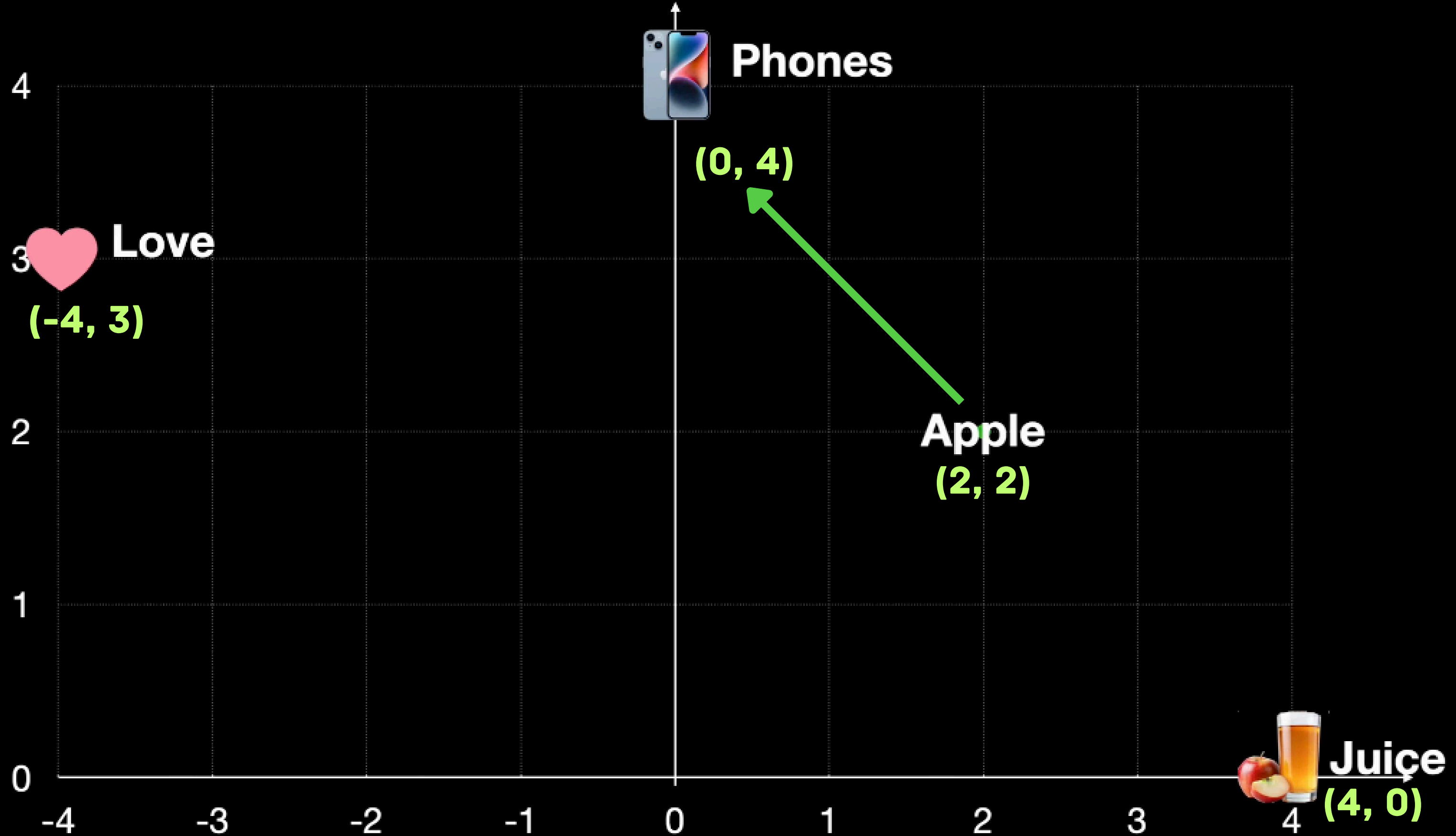
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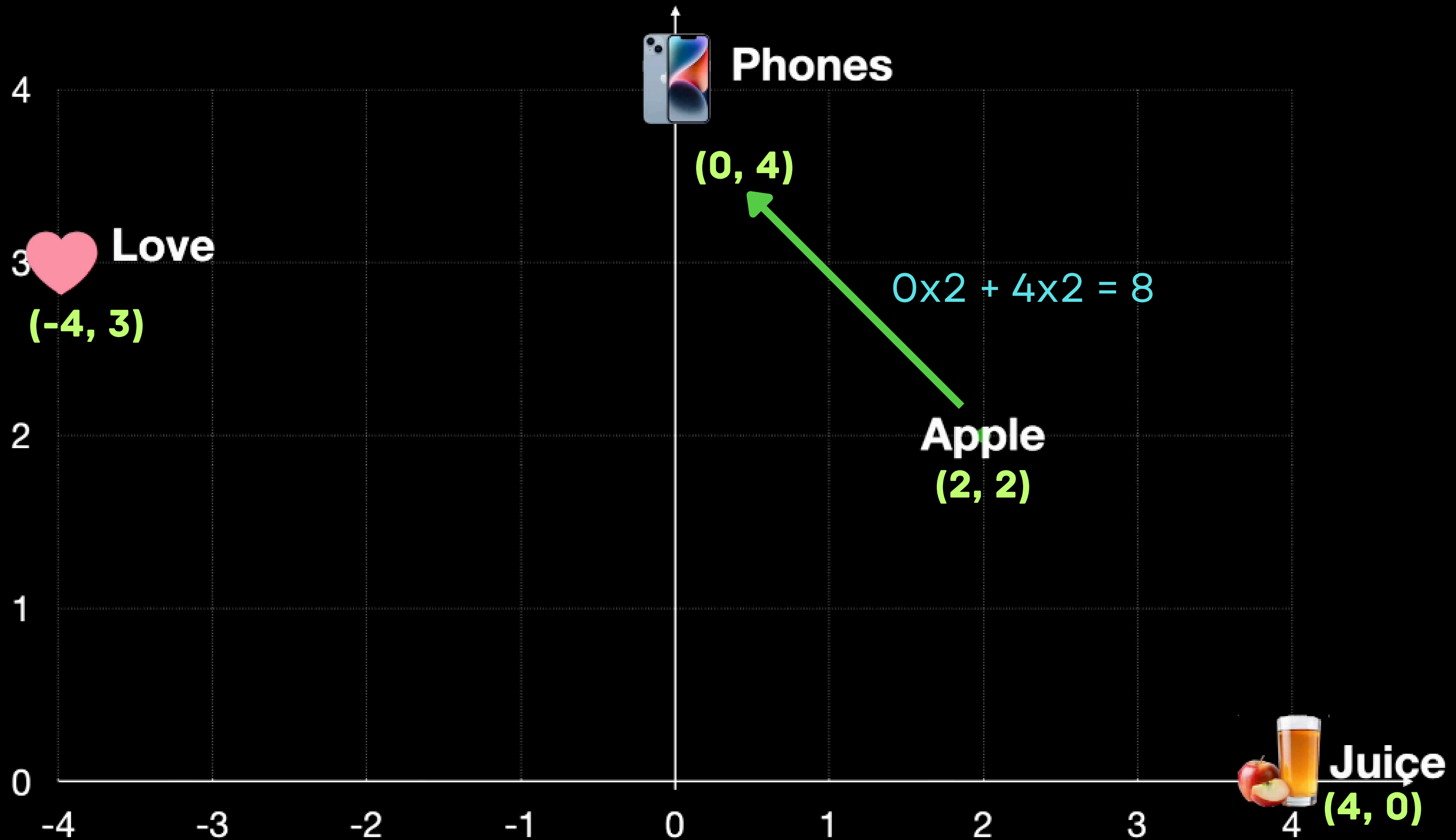


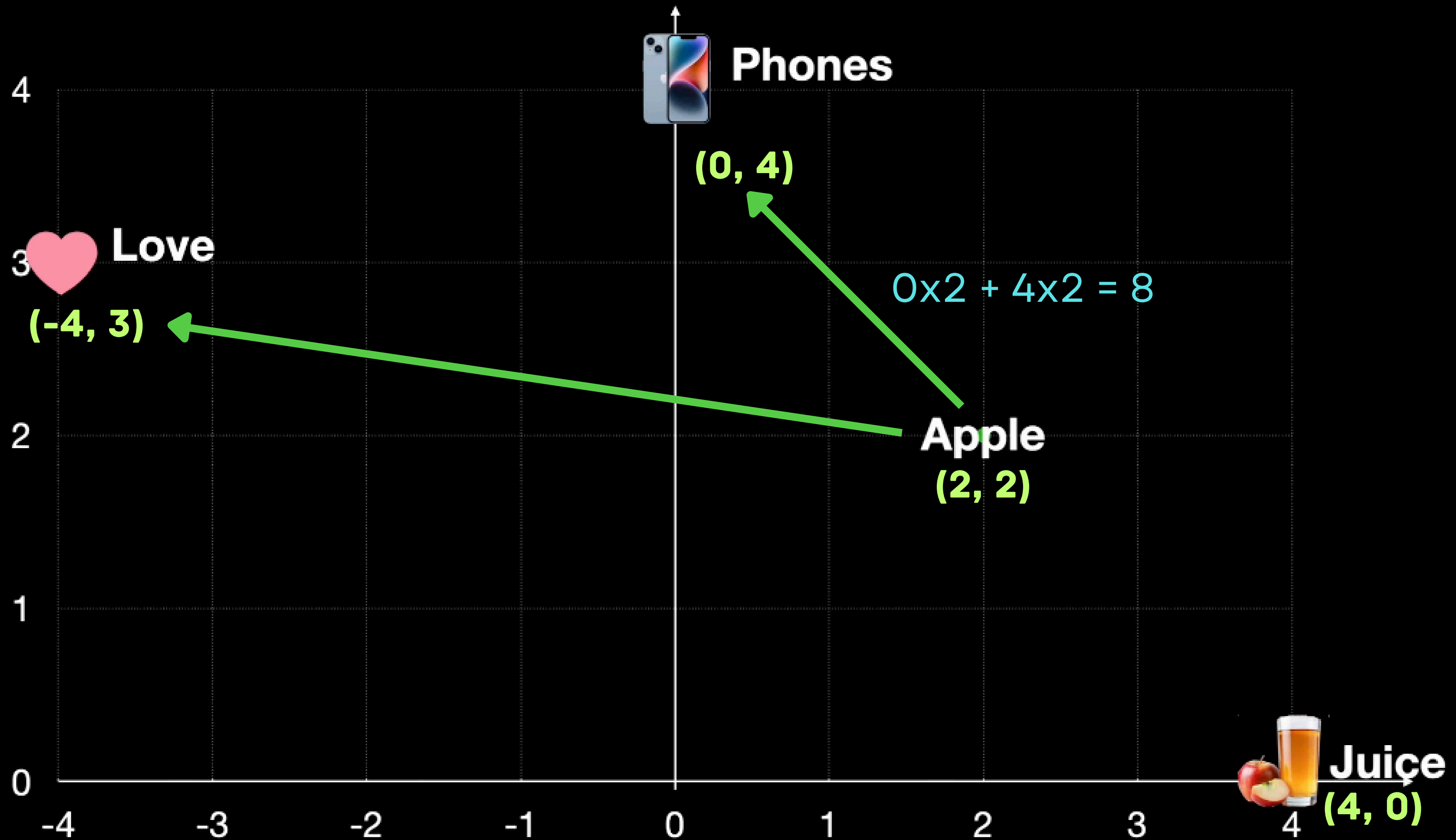
High Similarity

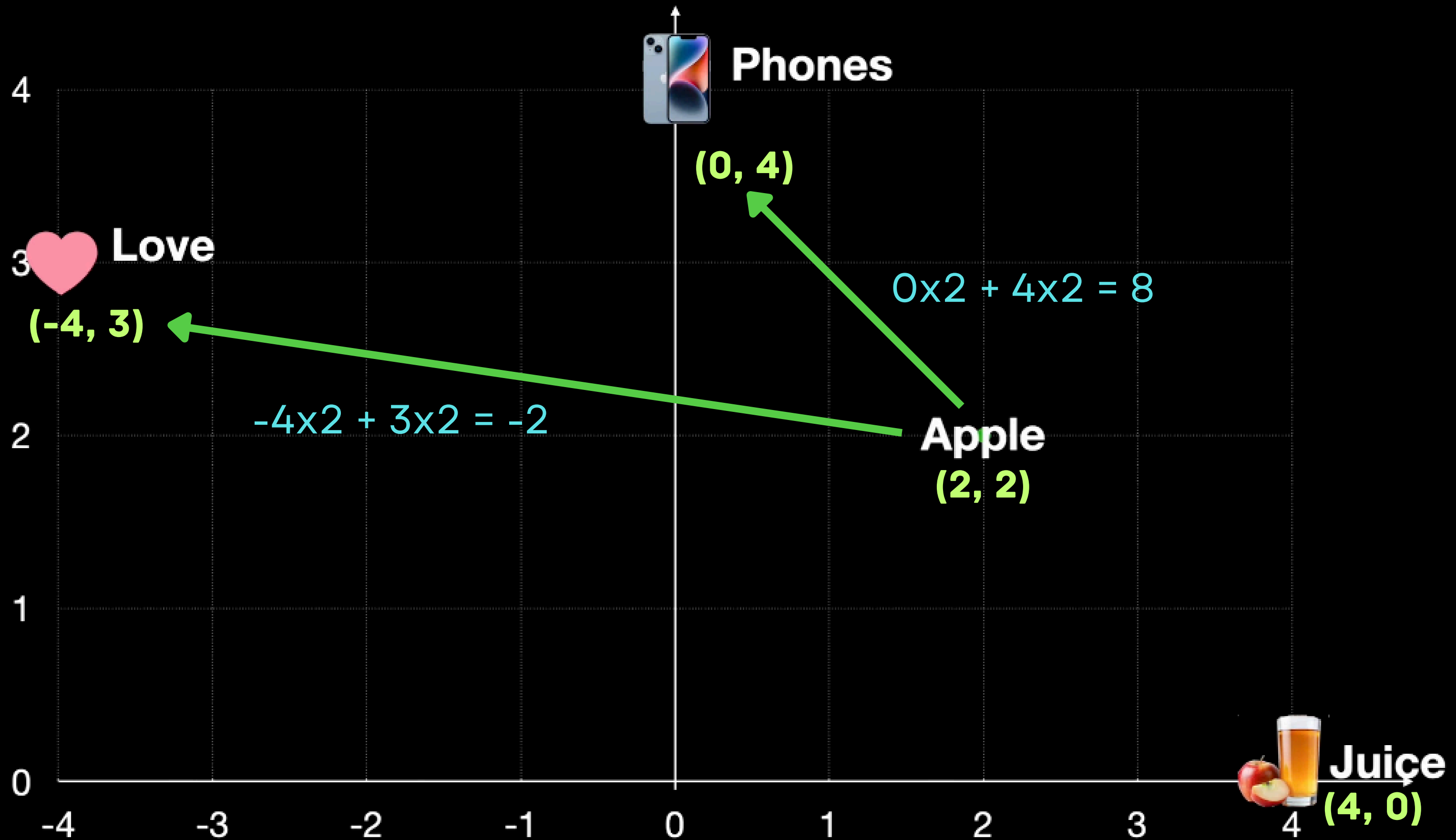


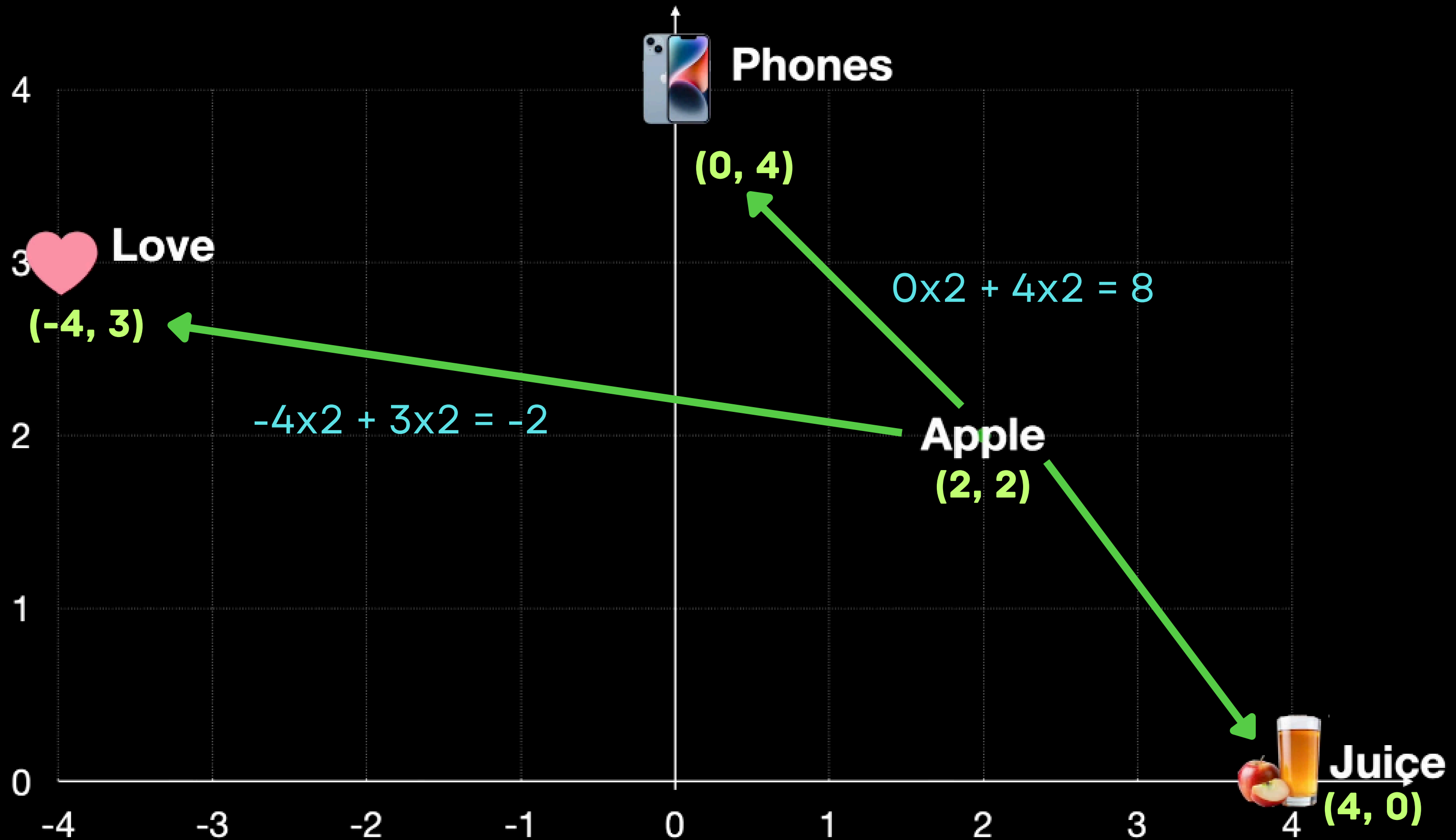


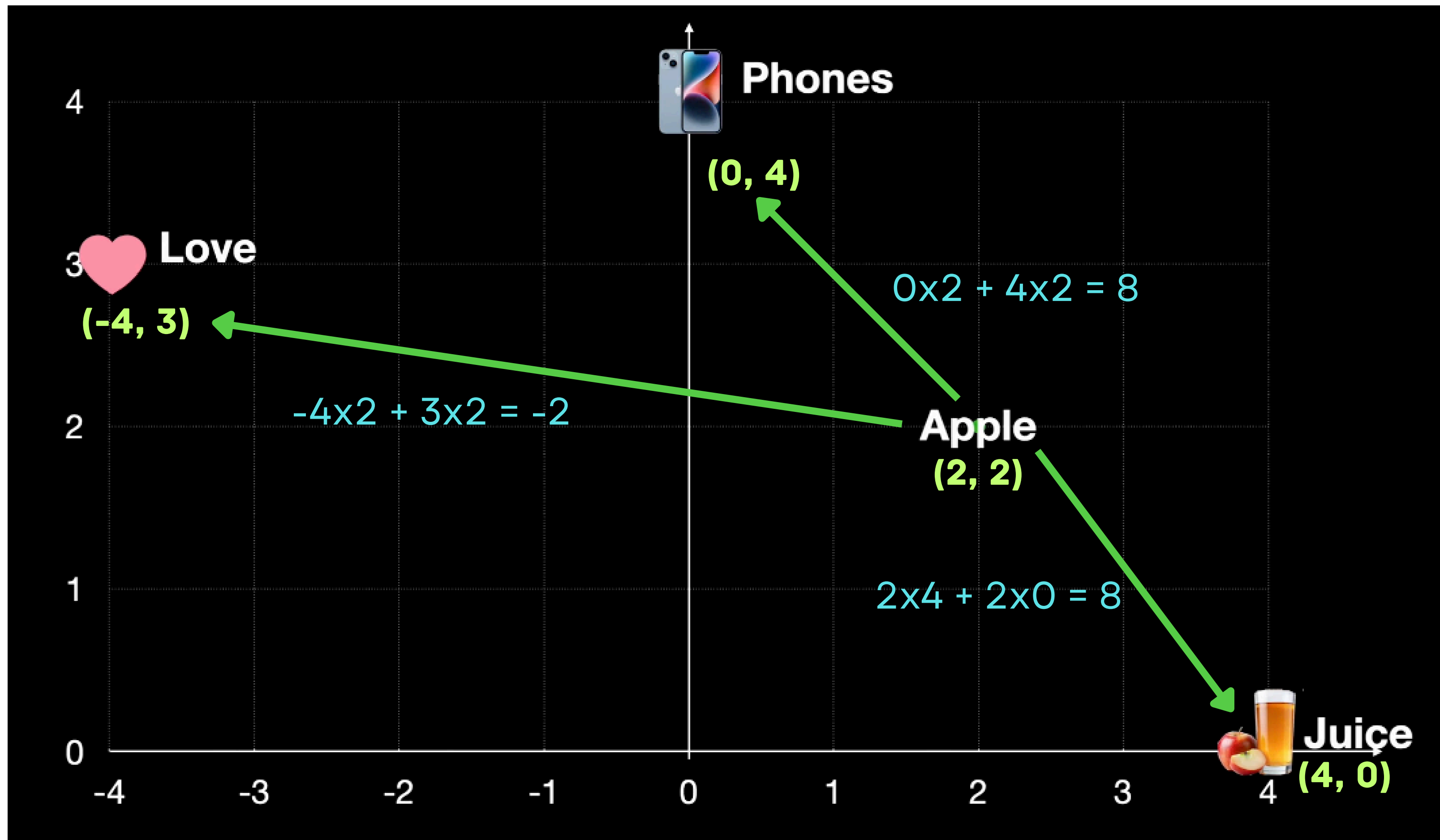


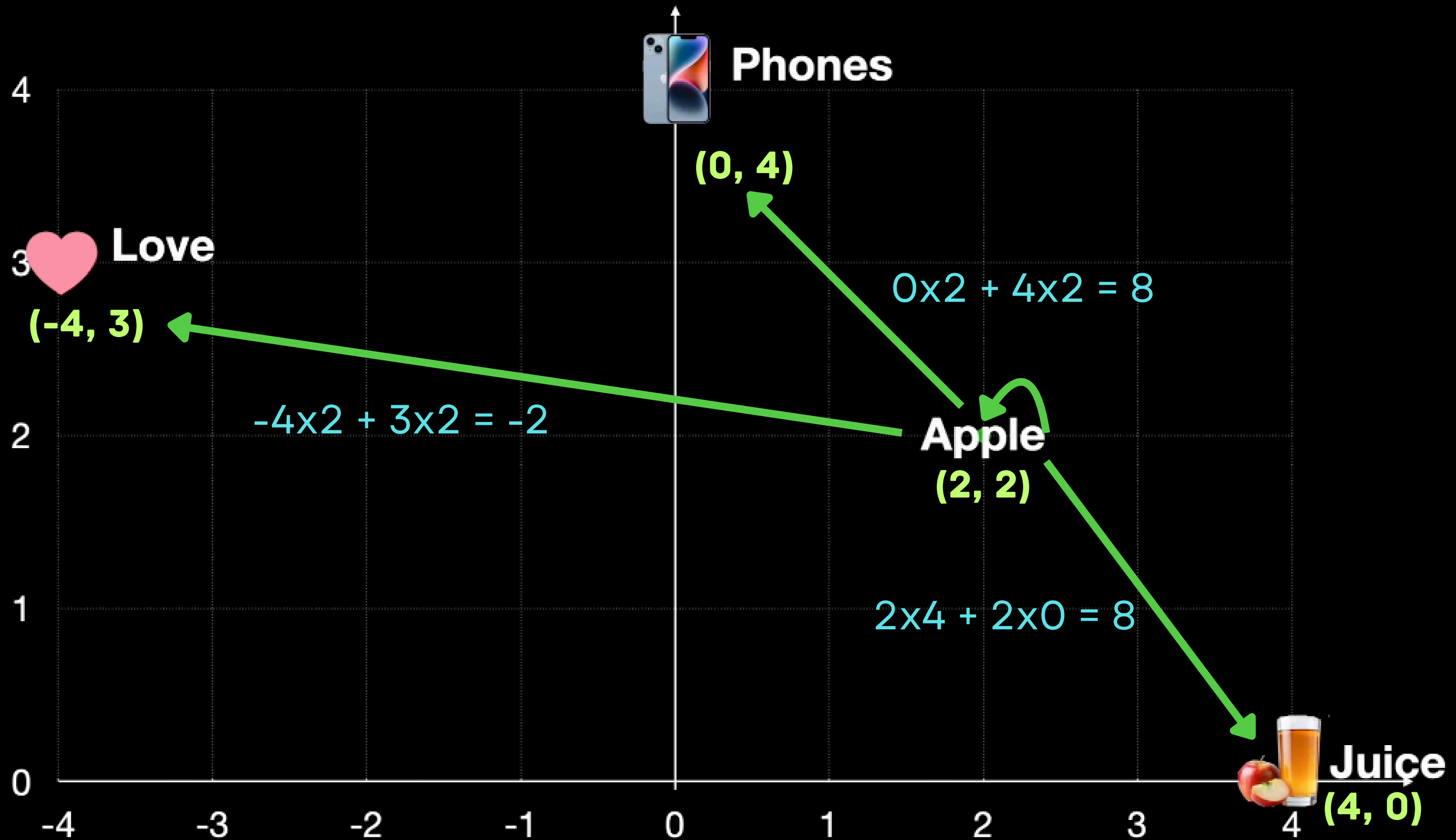


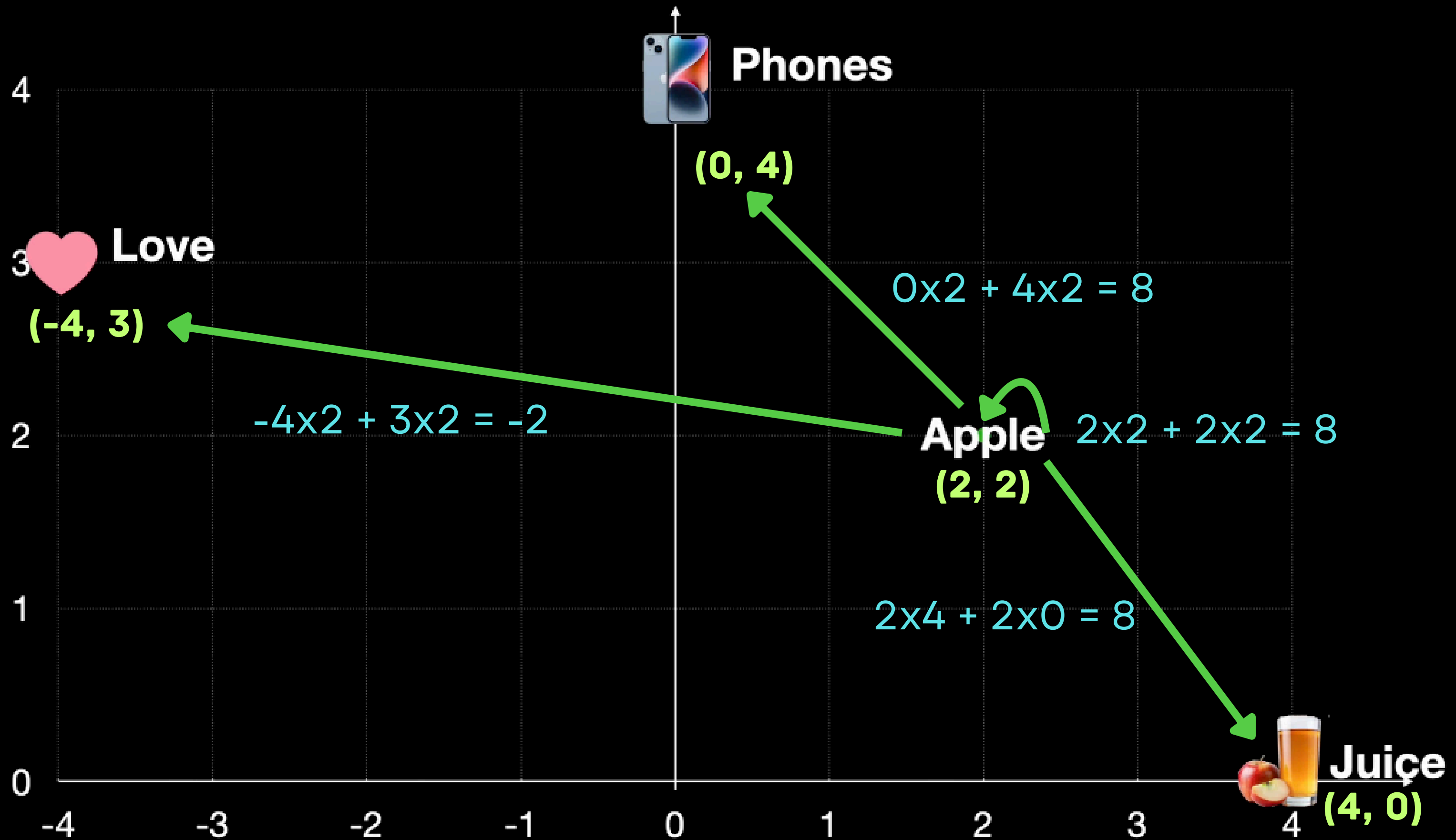












LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

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E_{apple} x E_{phone}

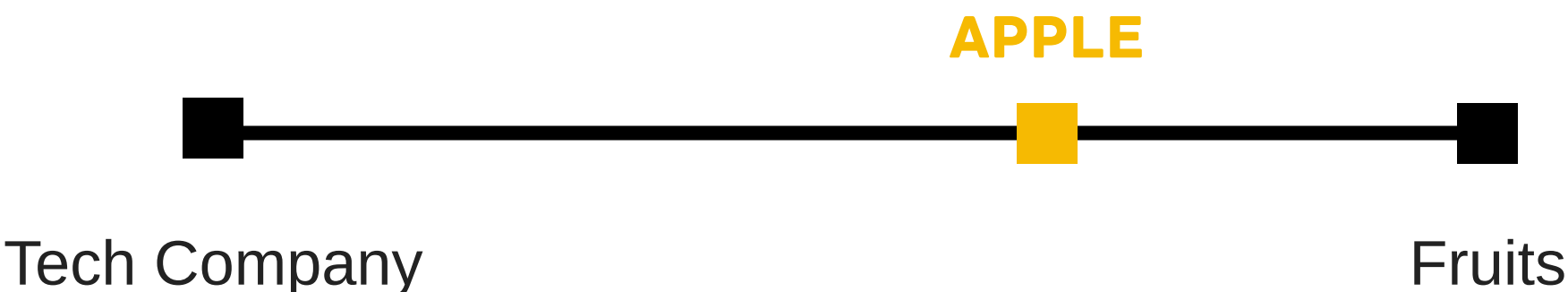
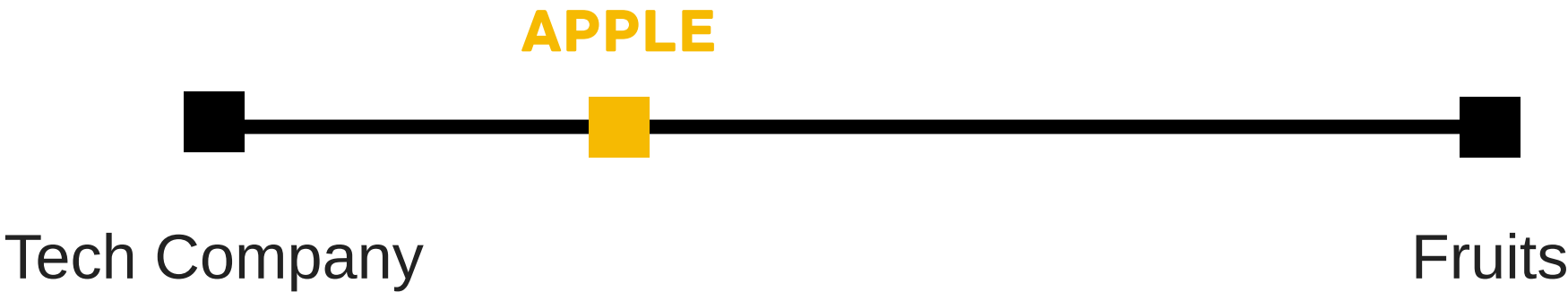
$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$



LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$(1, 2) \times (2, 1)$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E_{\text{apple}} \times E_{\text{phone}}^T$$

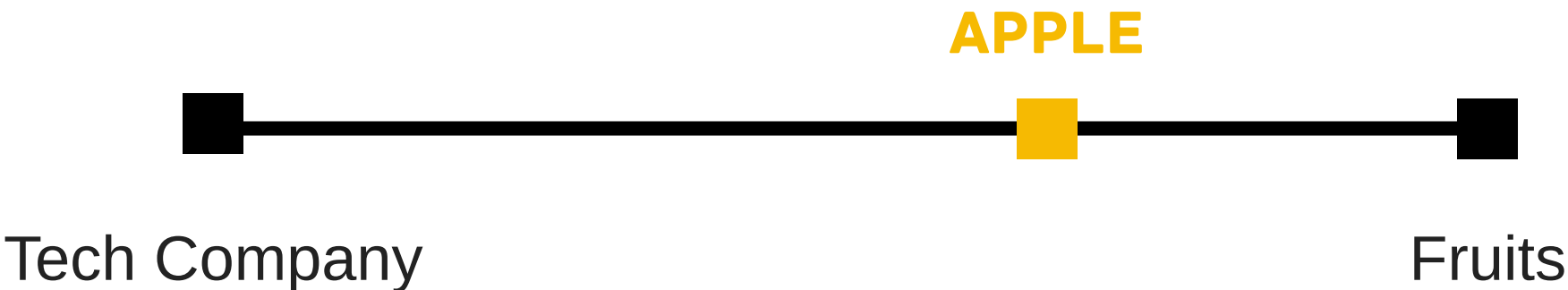
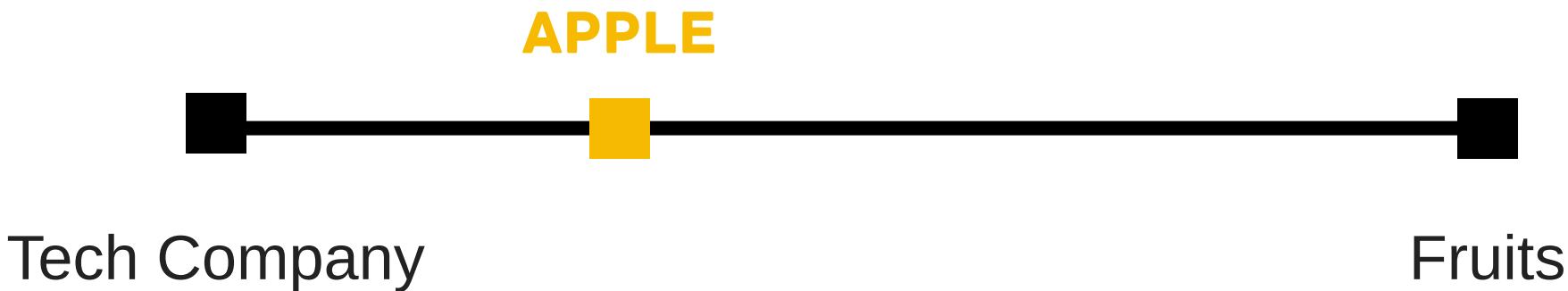
$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

LOVE APPLE JUICE

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$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$



LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(1, 1)
(1, 2) x (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

E_{apple} x E_{phone}

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

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$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

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(1, 1)
(1, 2) x (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$E_{\text{apple}} \times E_{\text{apple}}^T$ $E_{\text{apple}} \times E_{\text{phone}}^T$

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

LOVE APPLE JUICE

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$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(1, 1)
(1, 2) × (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$E_{\text{apple}} \times E_{\text{apple}}^T$ $E_{\text{apple}} \times E_{\text{phone}}^T$

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

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APPLE



Tech Company

Fruits

APPLE



Tech Company

Fruits

LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(1, 1)

(1, 2) × (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

E_{apple} × E_{love}^T

E_{apple} × E_{apple}^T

E_{apple} × E_{phone}^T

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

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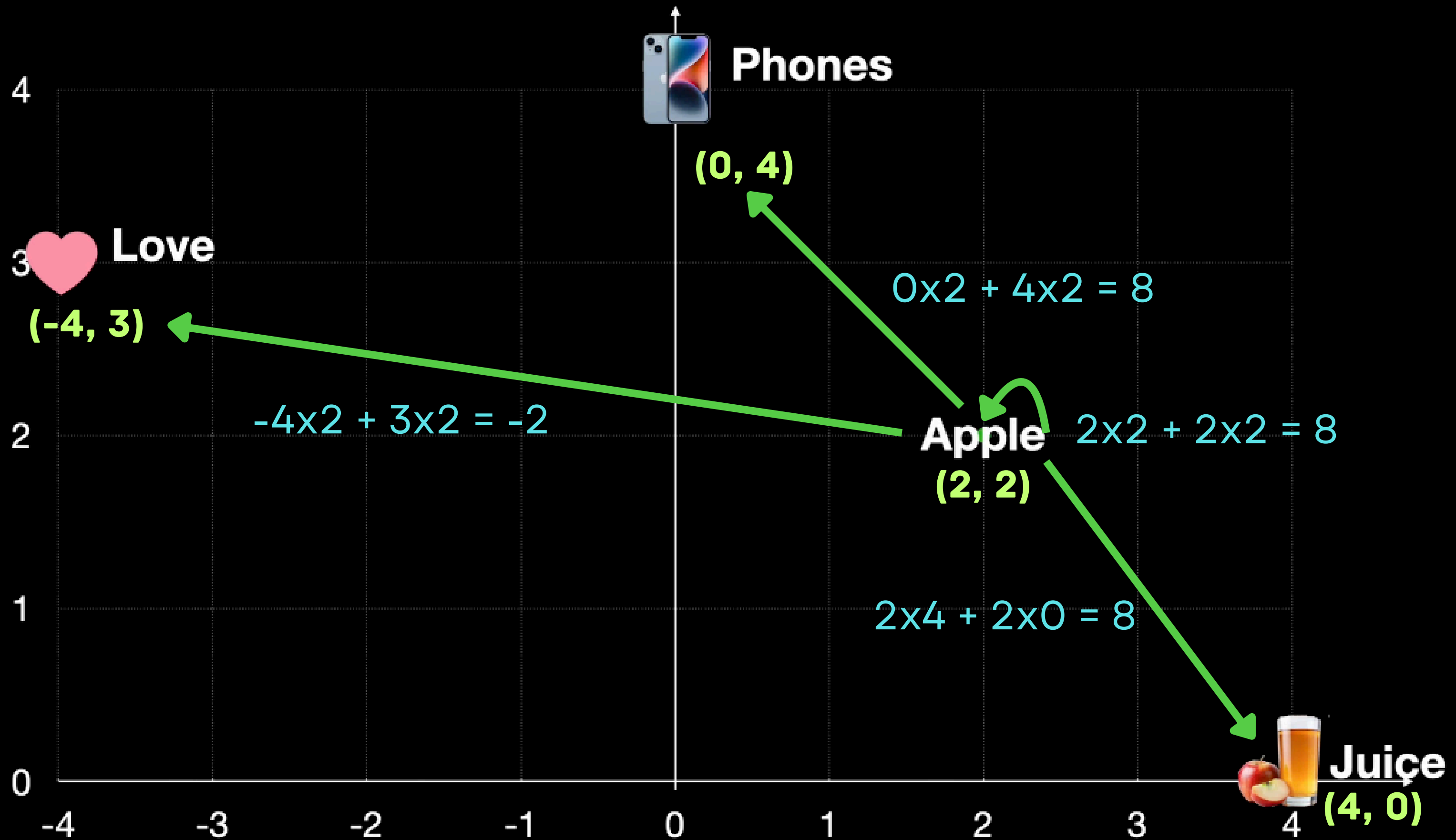
$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits



LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(1, 1)

(1, 2) × (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

E_{apple} × E_{love}^T

E_{apple} × E_{apple}^T

E_{apple} × E_{phone}^T

-2

8

8

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

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$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

Q: How to convert to probability?

LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(1, 1)

(1, 2) × (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$E_{\text{apple}} \times E_{\text{love}}^T$
 $E_{\text{apple}} \times E_{\text{apple}}^T$
 $E_{\text{apple}} \times E_{\text{phone}}^T$

-2

8

8

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

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$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE



Tech Company

Fruits

Q: How to convert to probability? Softmax

LOVE APPLE PHONE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

(1, 1)

(1, 2) × (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$E_{\text{apple}} \times E_{\text{love}}^T$
 $E_{\text{apple}} \times E_{\text{apple}}^T$
 $E_{\text{apple}} \times E_{\text{phone}}^T$

-2

8

8

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

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APPLE



Tech Company

Fruits

Q: How to convert to probability? Softmax

LOVE APPLE PHONE

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(1, 1)

(1, 2) × (2, 1)

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

E_{\text{apple}} \times E_{\text{love}}^T
E_{\text{apple}} \times E_{\text{apple}}^T
E_{\text{apple}} \times E_{\text{phone}}^T

-2 0.0

8 0.5

8 0.5

$$E'_{\text{phone}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{phone}}$$

APPLE



Tech Company

Fruits

LOVE APPLE JUICE

$$E'_{\text{love}} = 0.8 \times E_{\text{love}} + 0.1 \times E_{\text{apple}} + 0.1 \times E_{\text{phone}}$$

$$E'_{\text{apple}} = 0.1 \times E_{\text{love}} + 0.5 \times E_{\text{apple}} + 0.4 \times E_{\text{phone}}$$

$$E'_{\text{juice}} = 0.1 \times E_{\text{love}} + 0.4 \times E_{\text{apple}} + 0.5 \times E_{\text{juice}}$$

APPLE

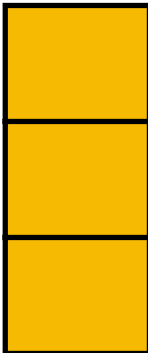


Tech Company

Fruits



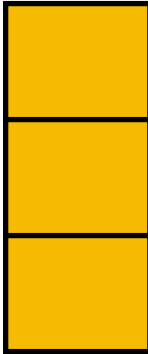
E_{apple}



E_{love}



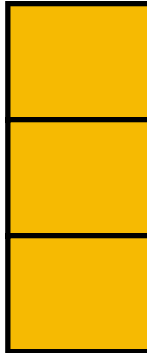
E_{apple}



E_{apple}



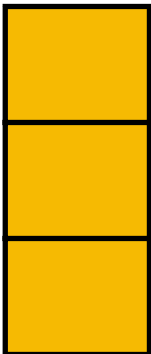
E_{apple}



E_{phones}



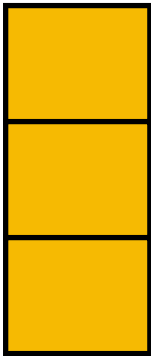
E_{apple}



E_{love}



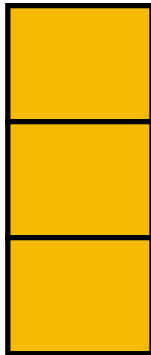
E_{apple}



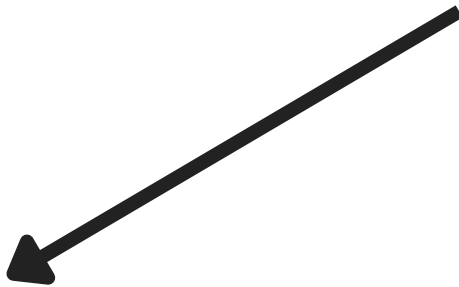
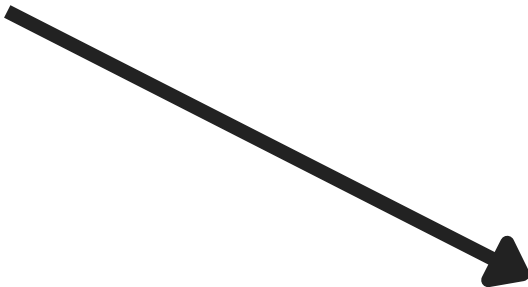
E_{apple}



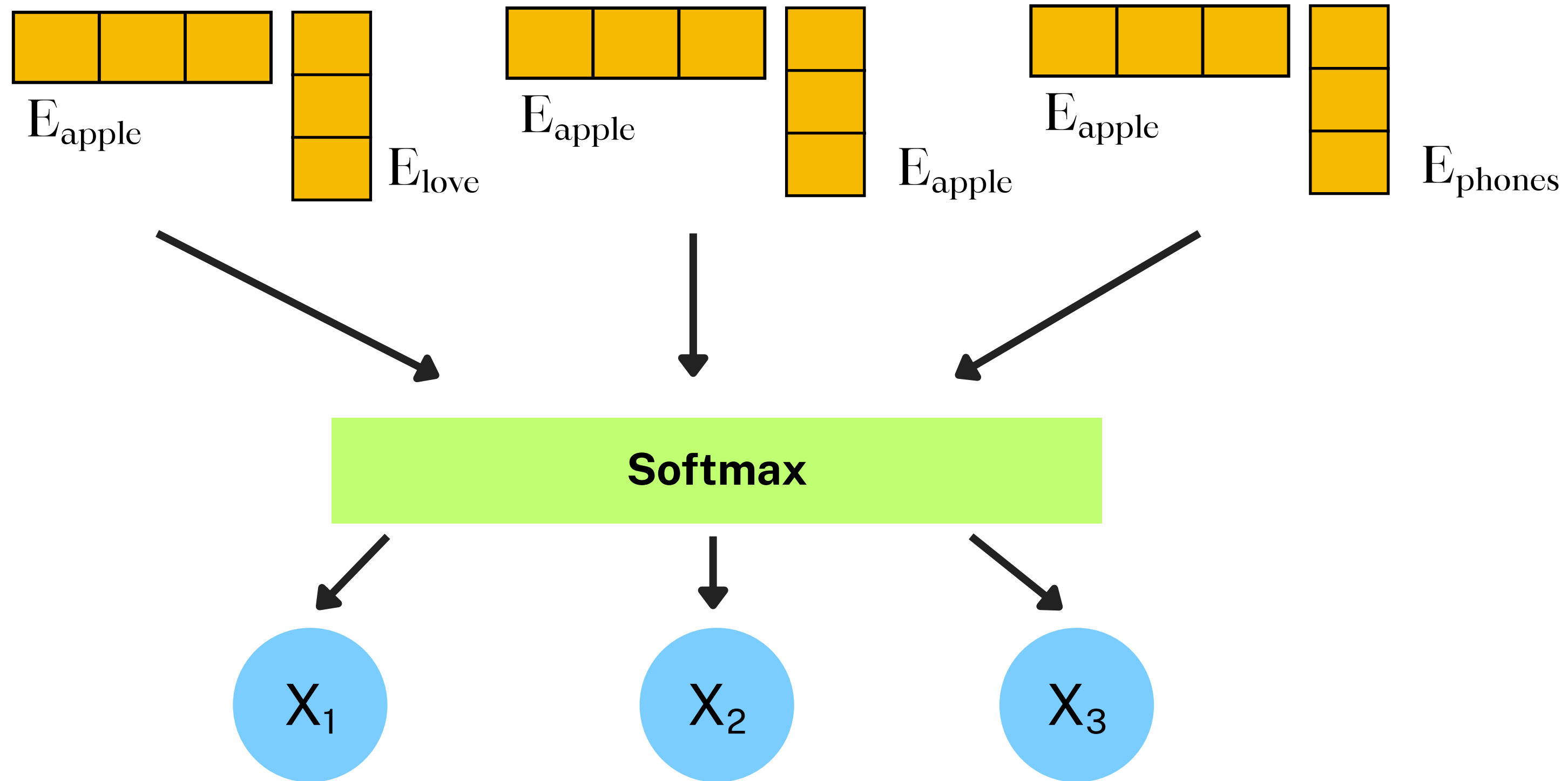
E_{apple}

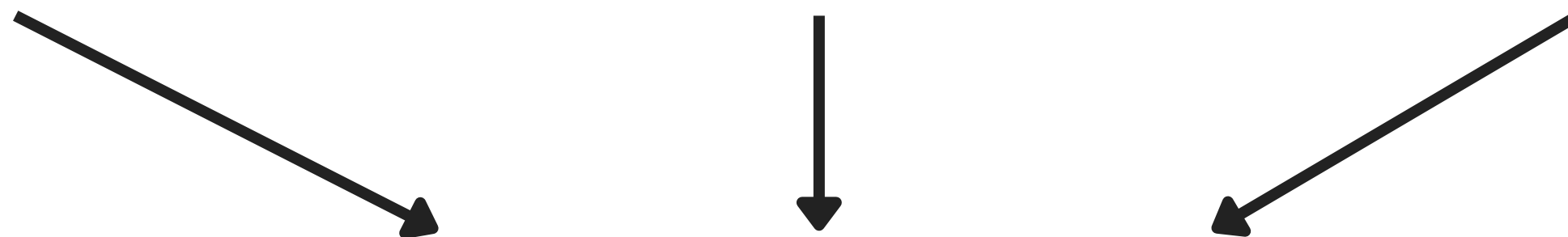
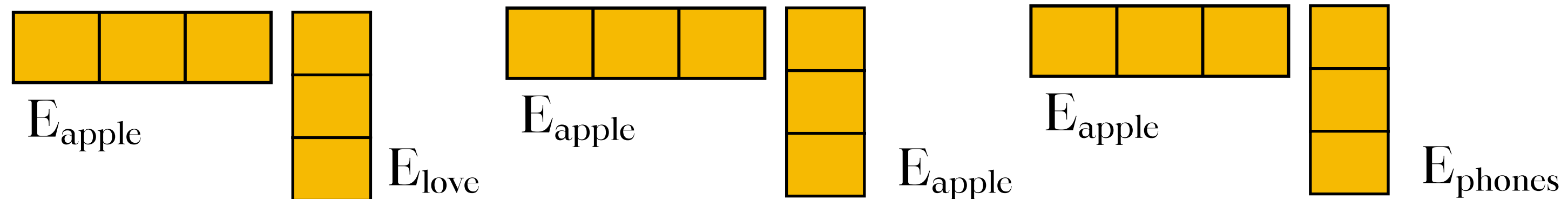


E_{phones}

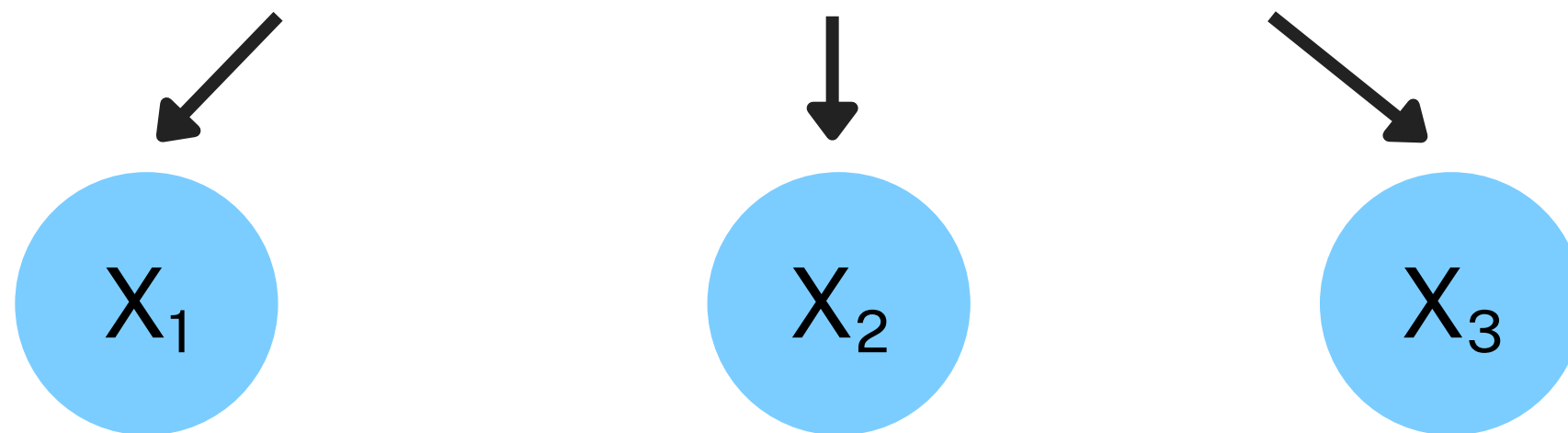


Softmax

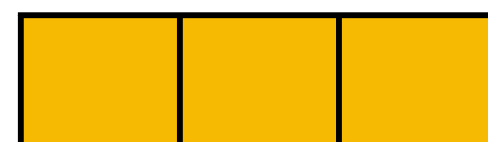




Softmax



*



E_{love}

*

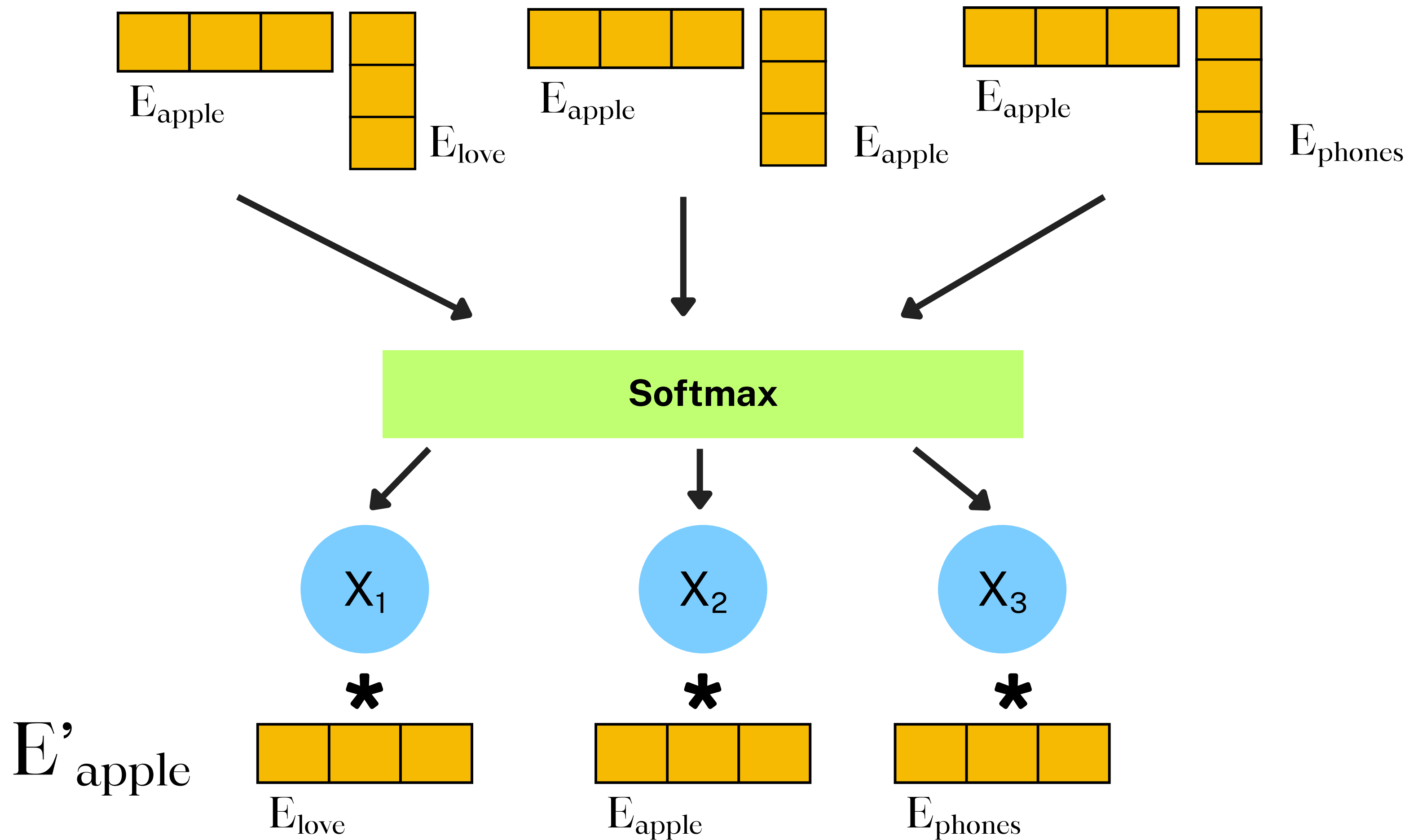


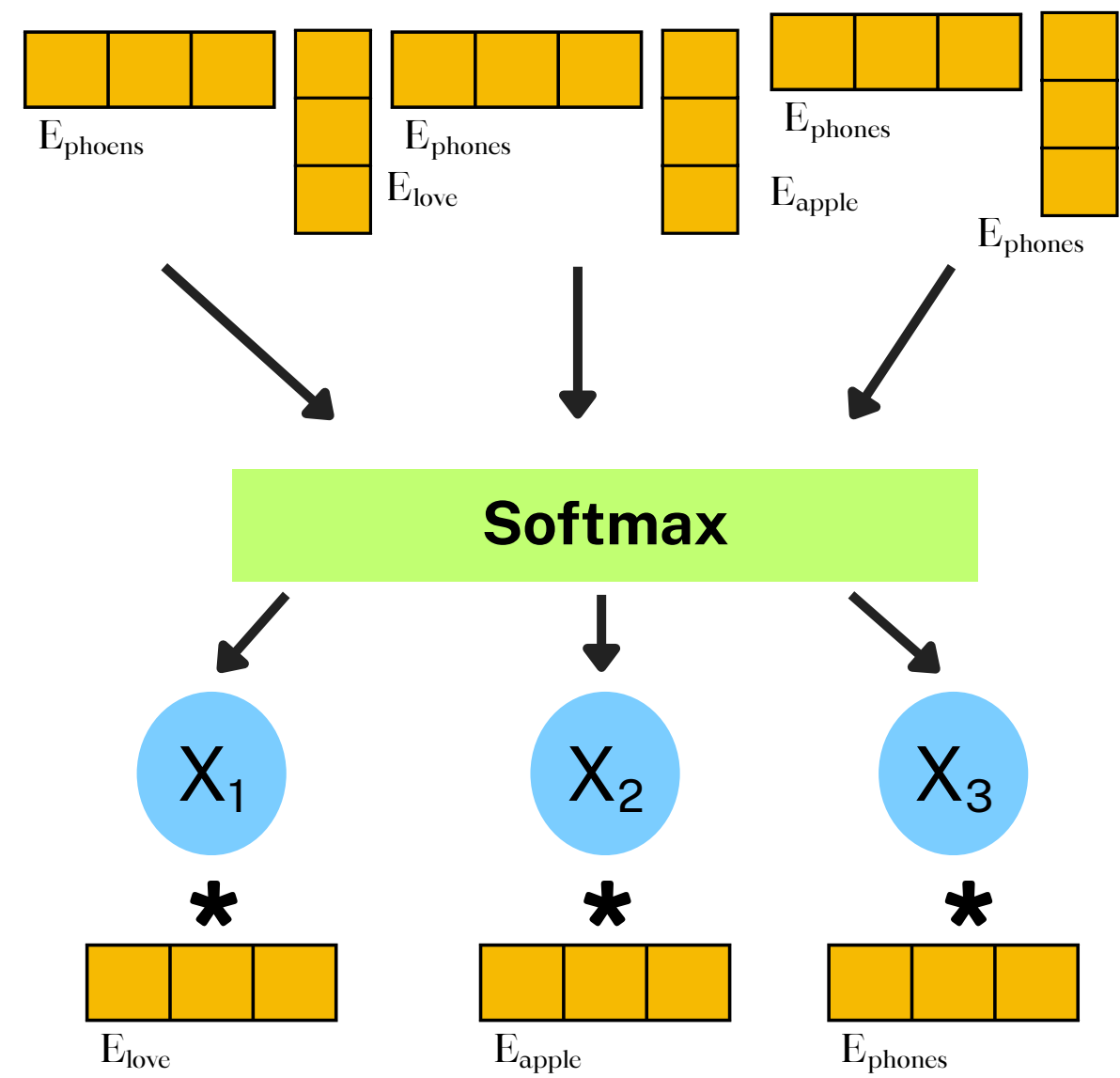
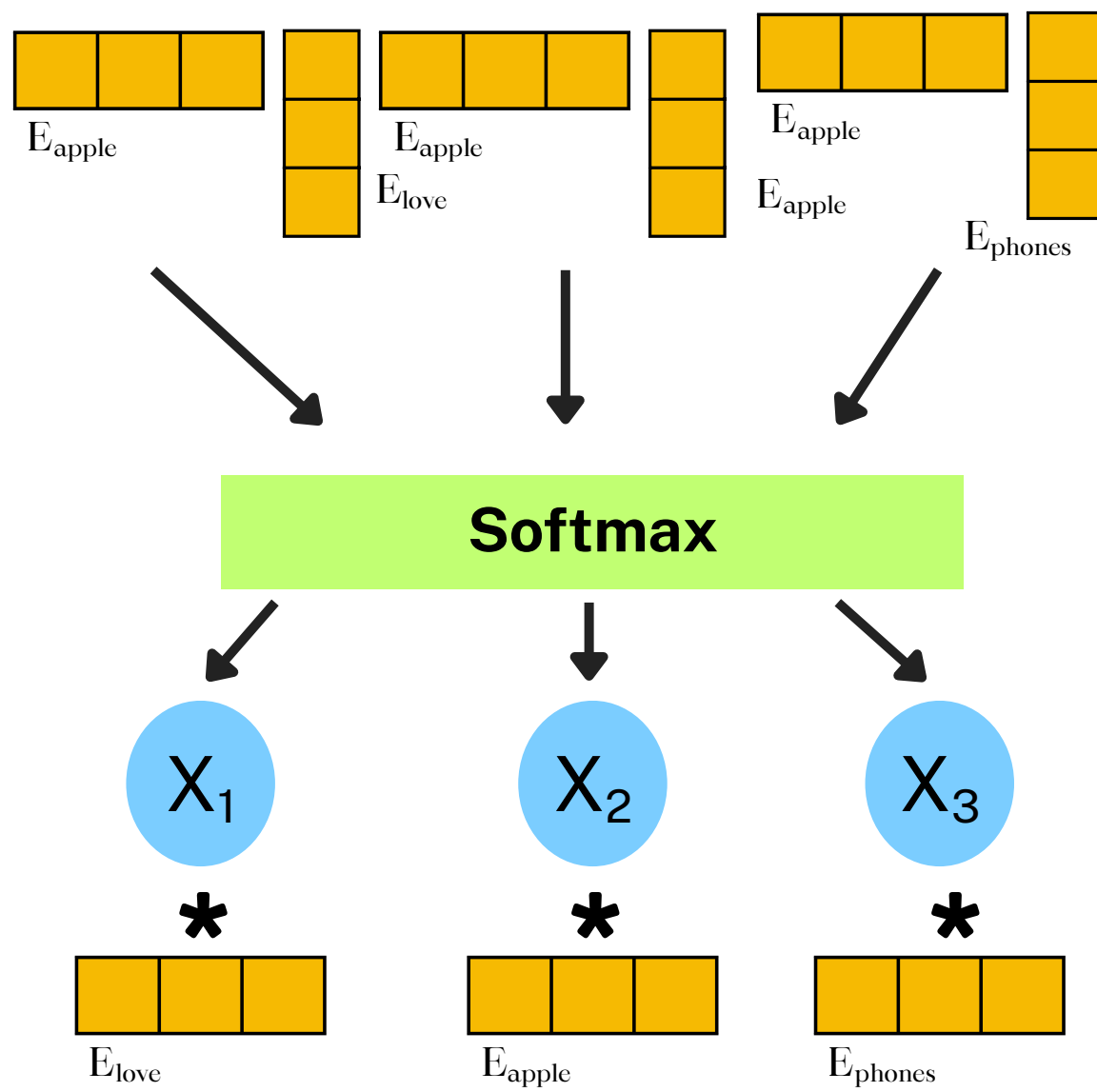
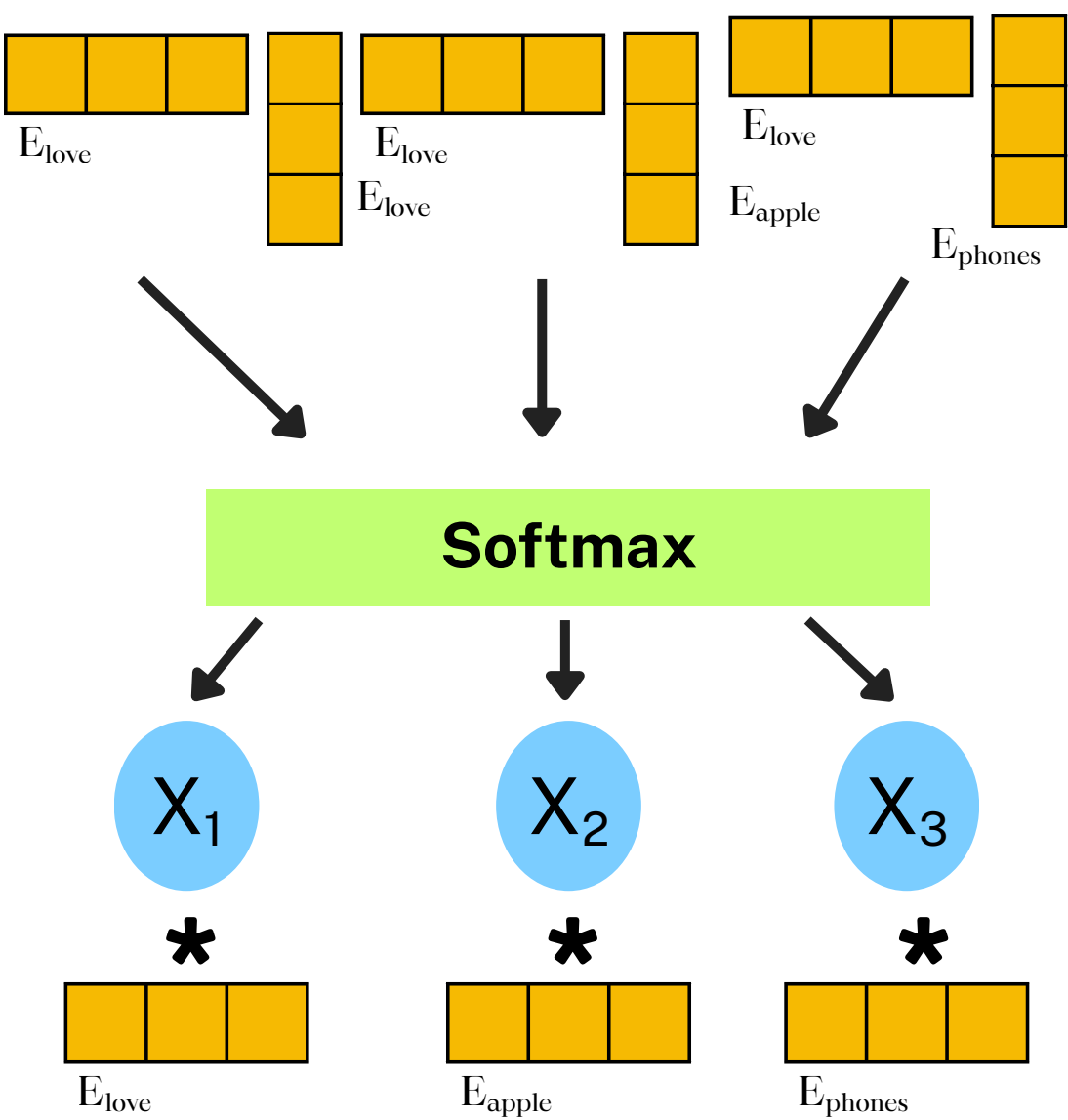
E_{apple}

*

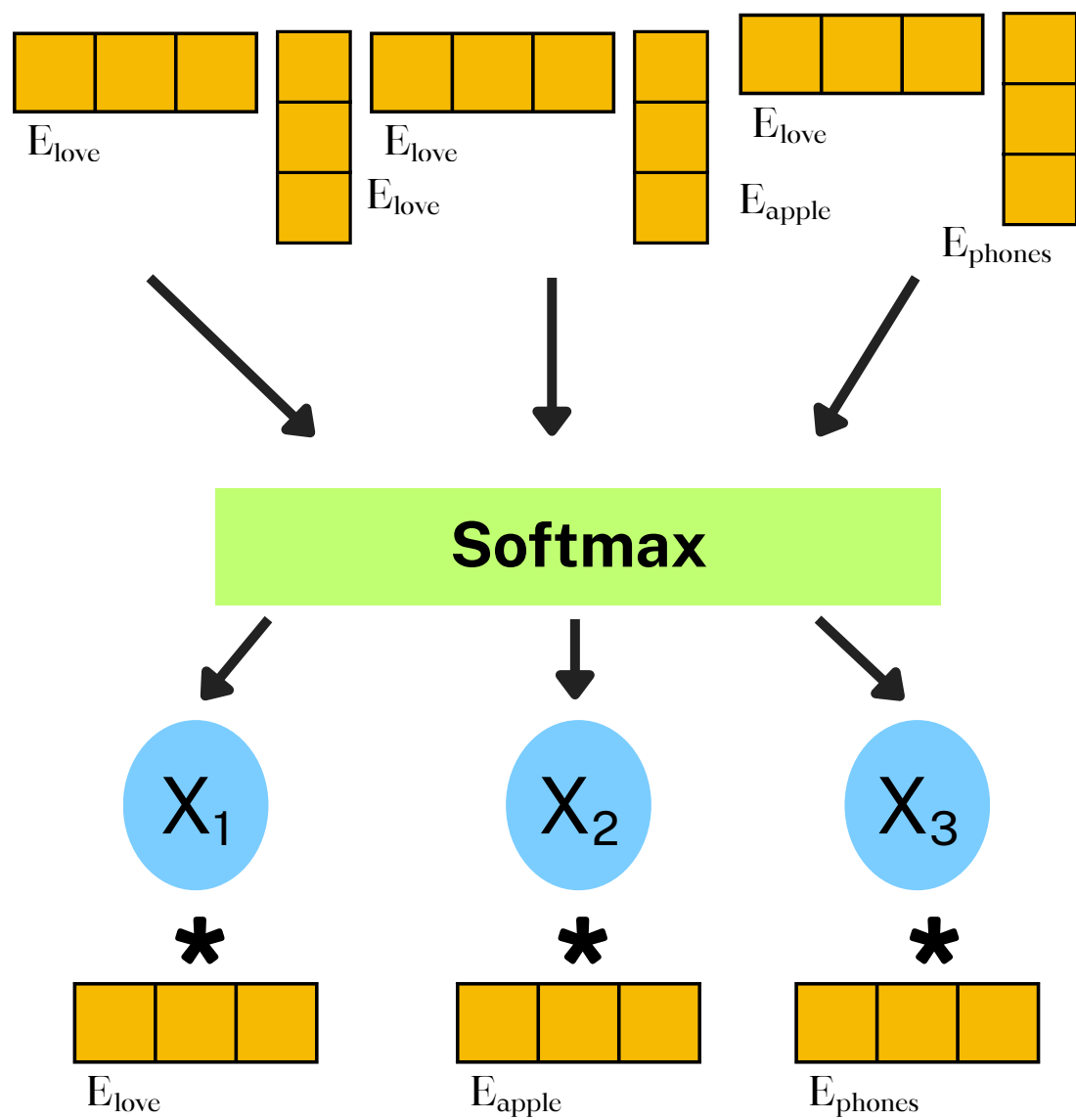


E_{phones}

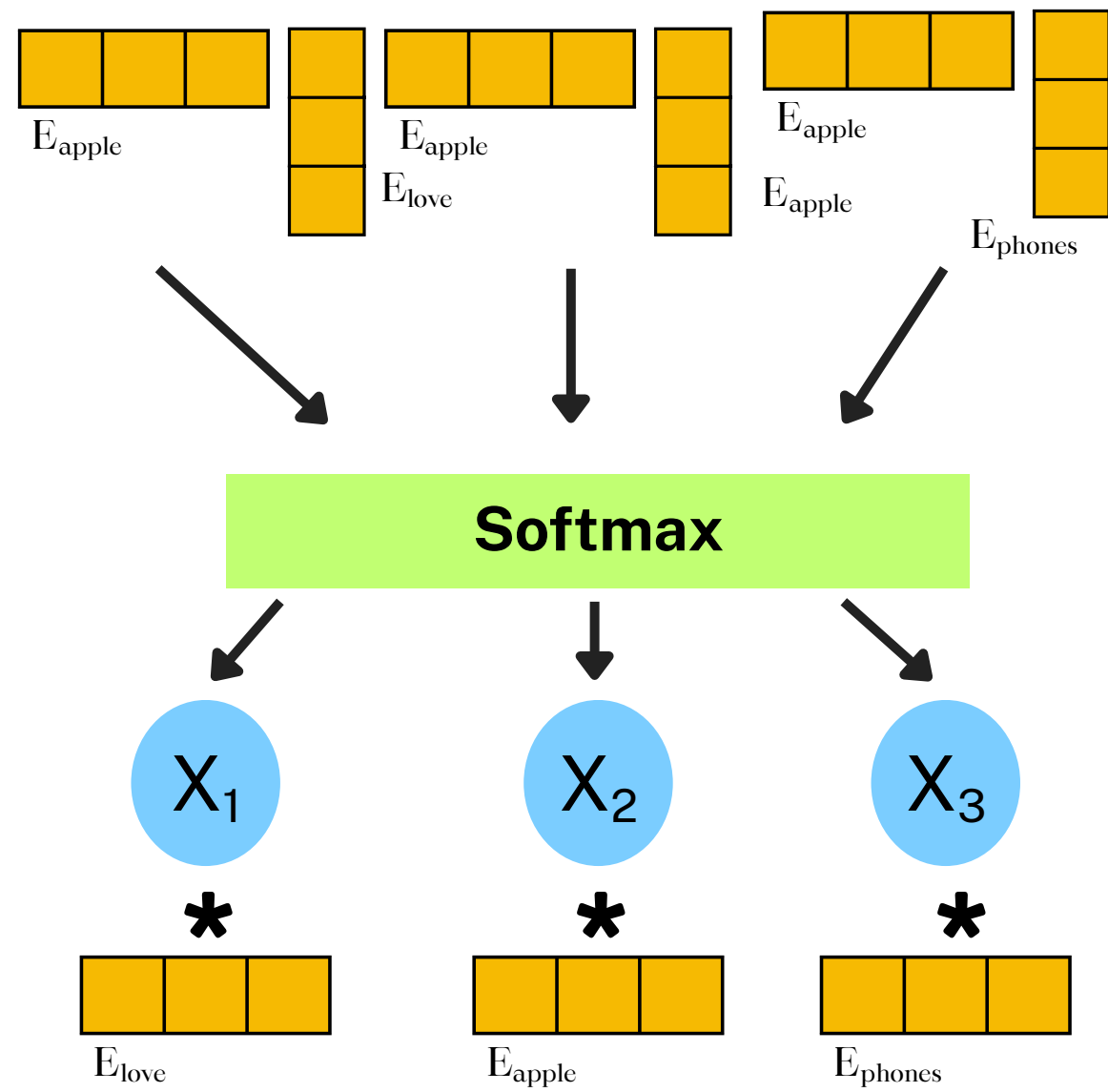




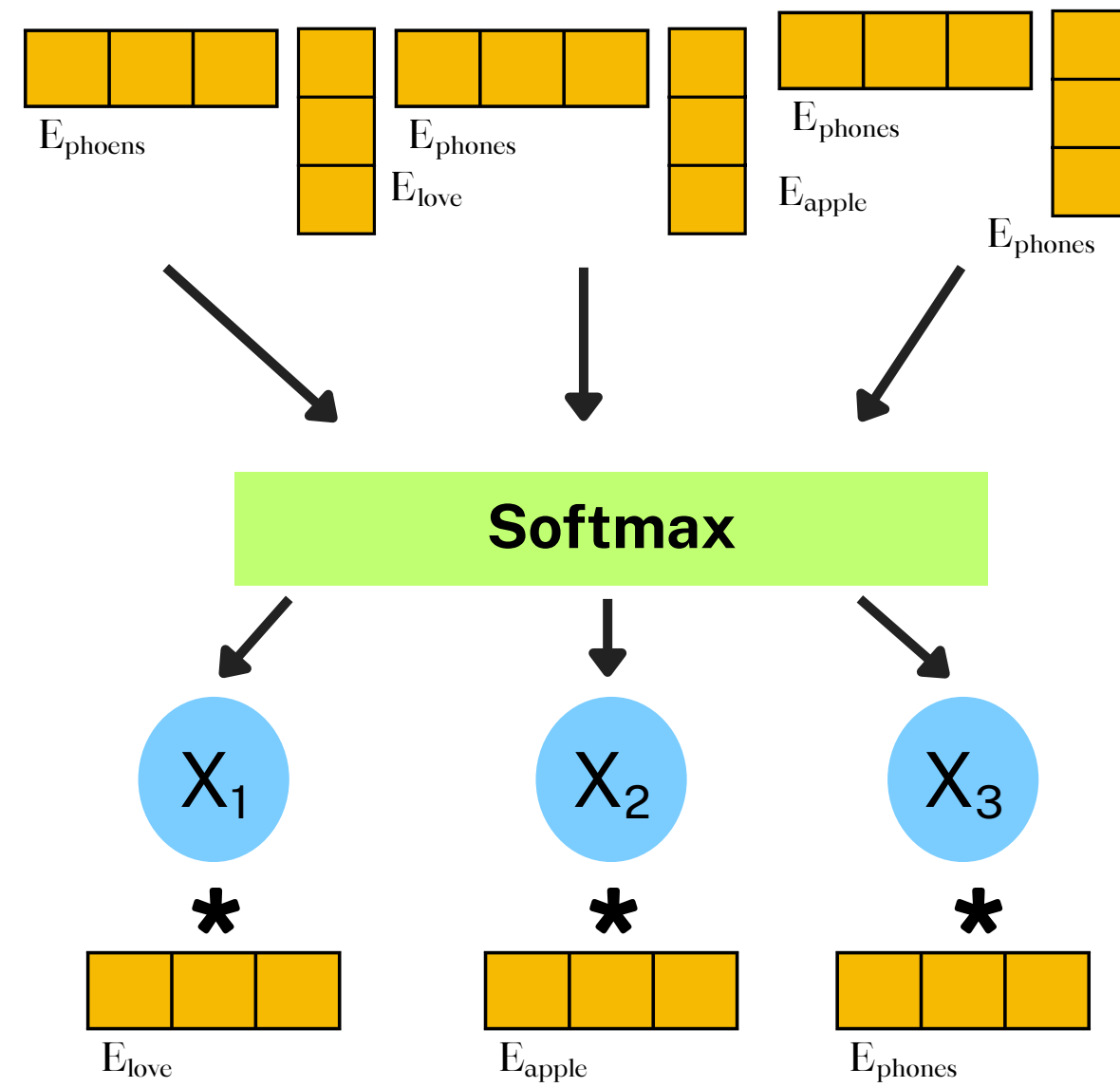
E'_{love}

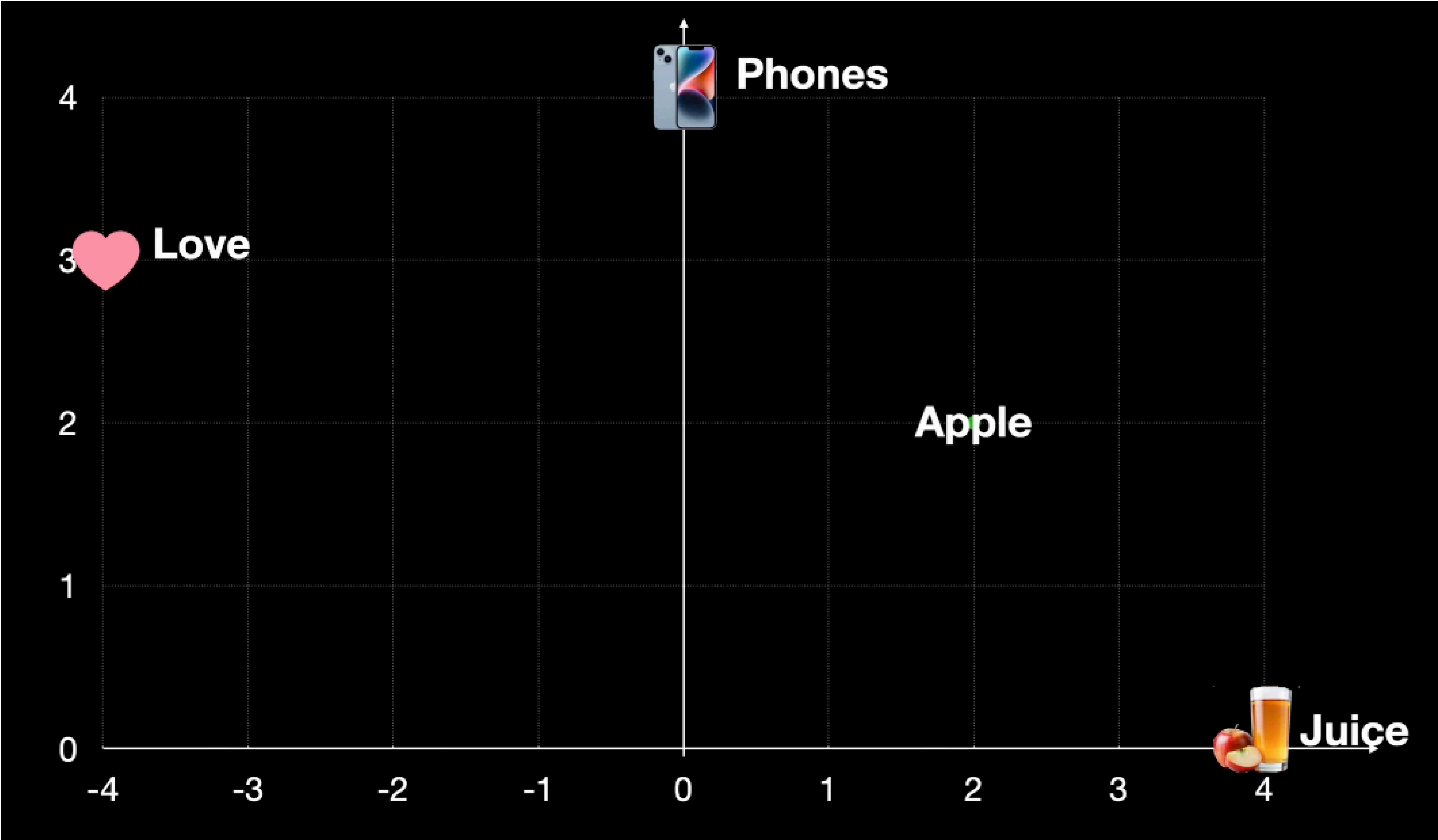


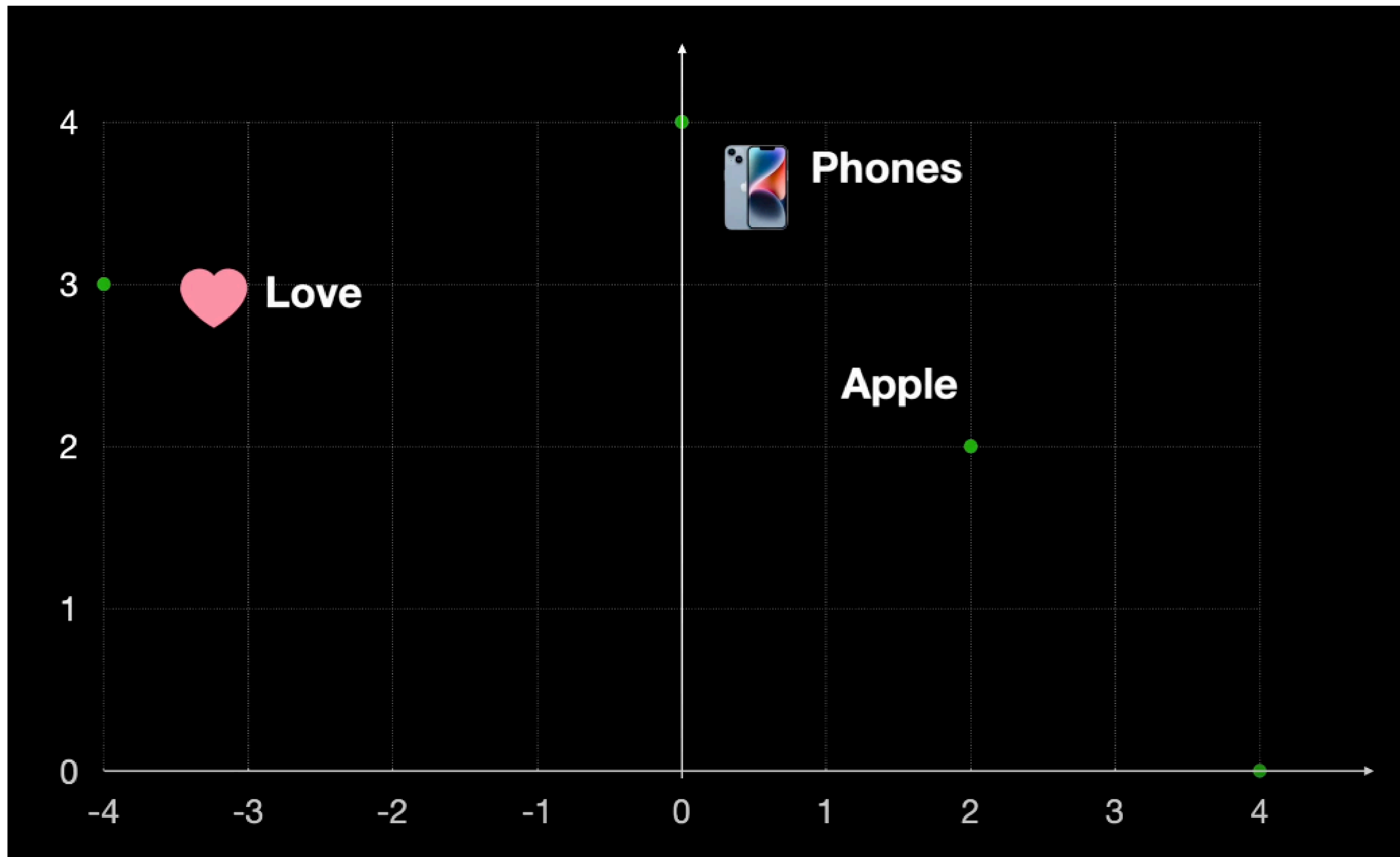
E'_{apple}

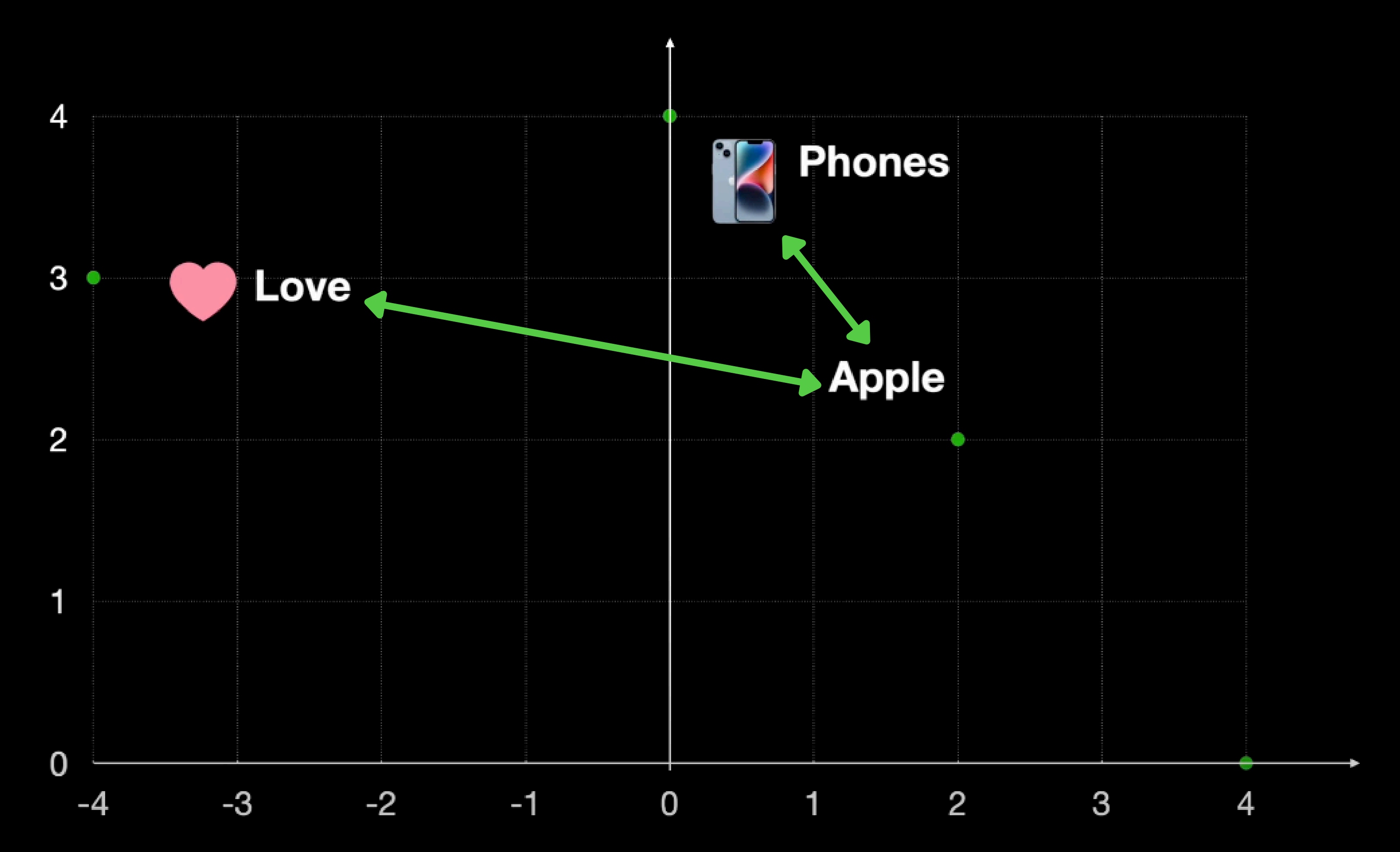


E'_{phones}











PRIOR PROBLEMS IN SELF ATTENTION

Problems in Self Attention

- **PROBLEM 01:** No Learnable Parameters
 - No Task Specific Learning

Juice
Orange
Fruit

Apple

Smart
Technology
Gadgets

See
Observe

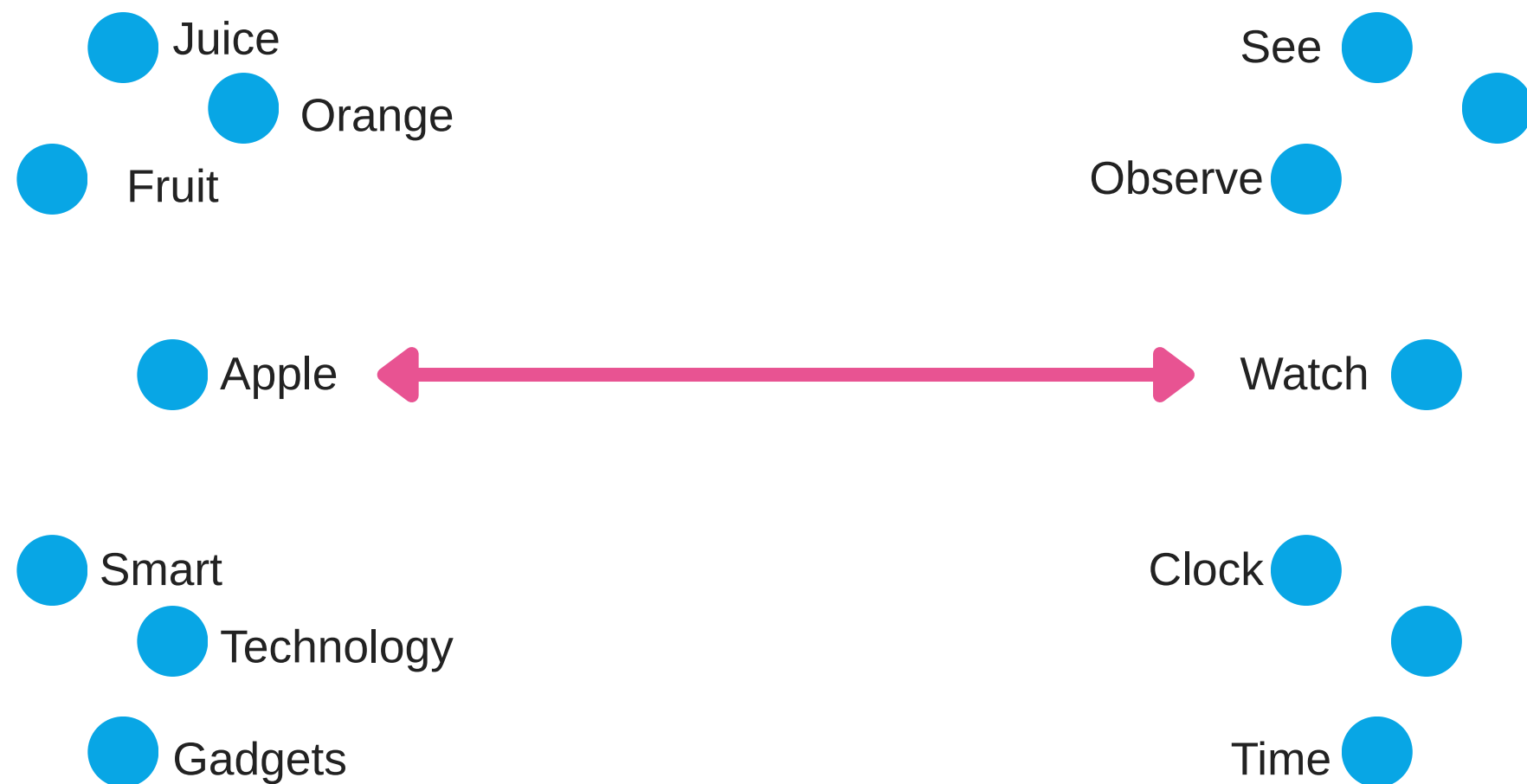
Watch

Clock
Time

I love Apple Watch

Problems in Self Attention

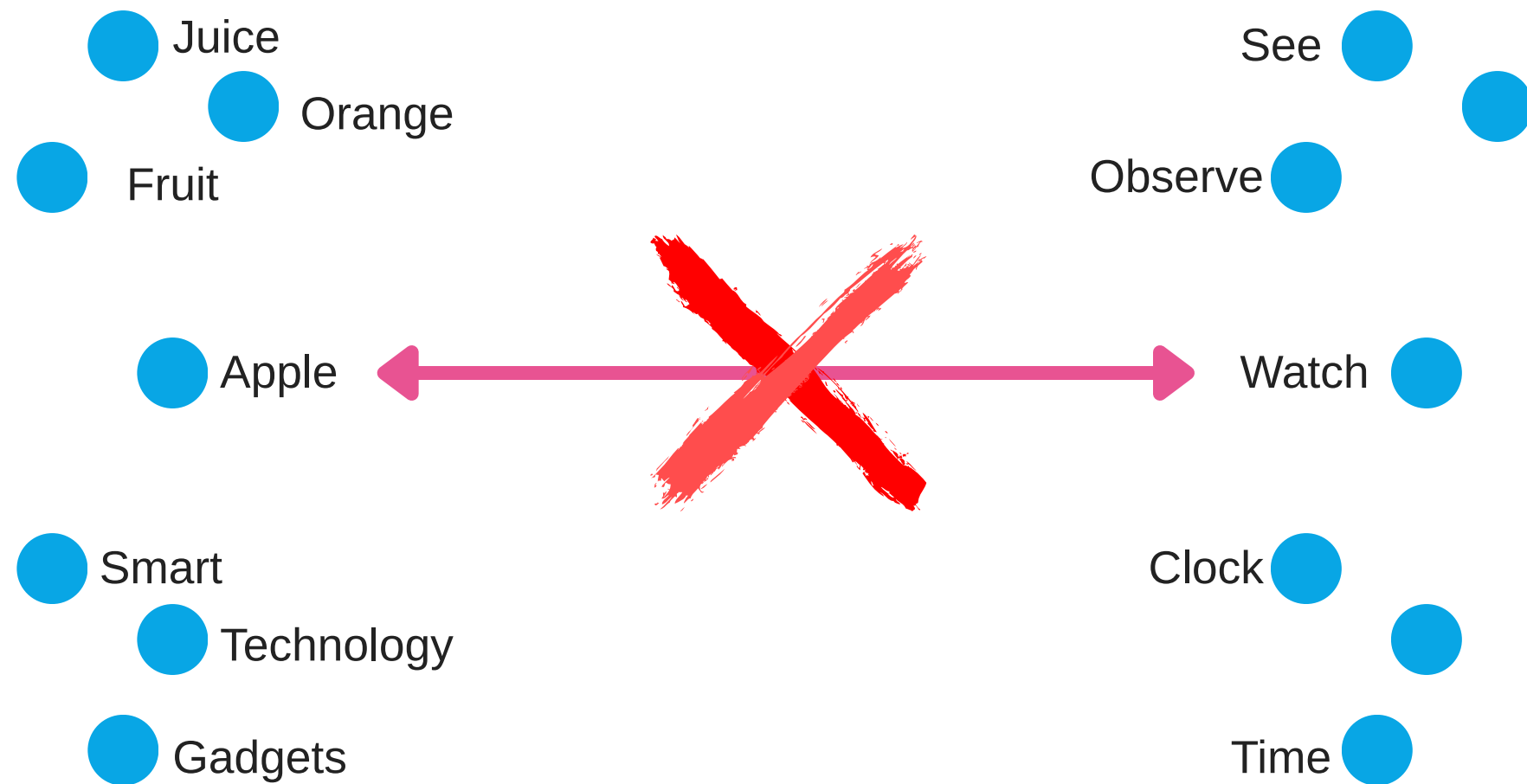
- **PROBLEM 01:** No Learnable Parameters
 - No Task Specific Learning



I love Apple Watch

Problems in Self Attention

- **PROBLEM 01:** No Learnable Parameters
 - No Task Specific Learning



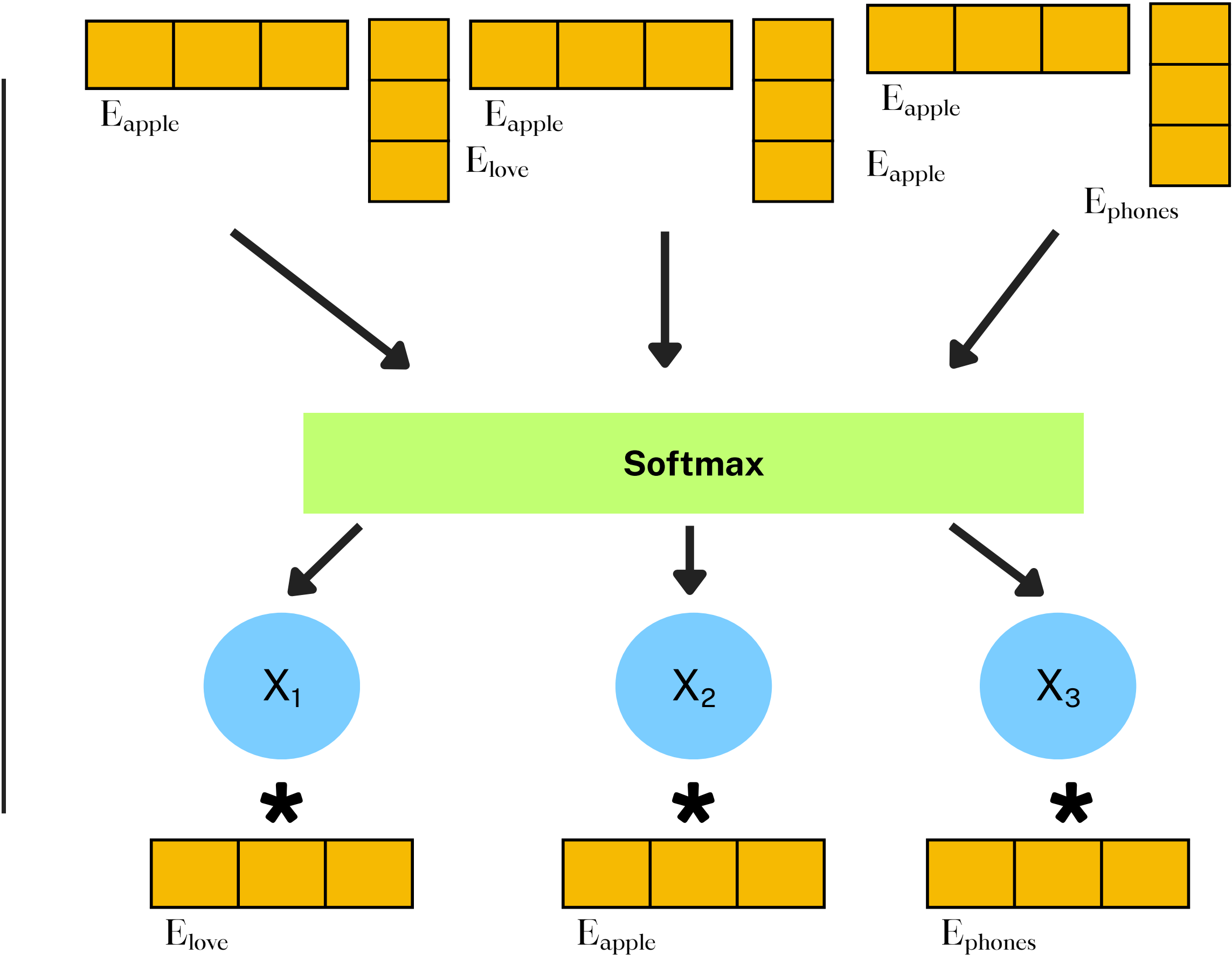
I love Apple Watch

PROBLEM 01:

No Learnable Parameters

PROBLEM 02:

Use of Same Vector

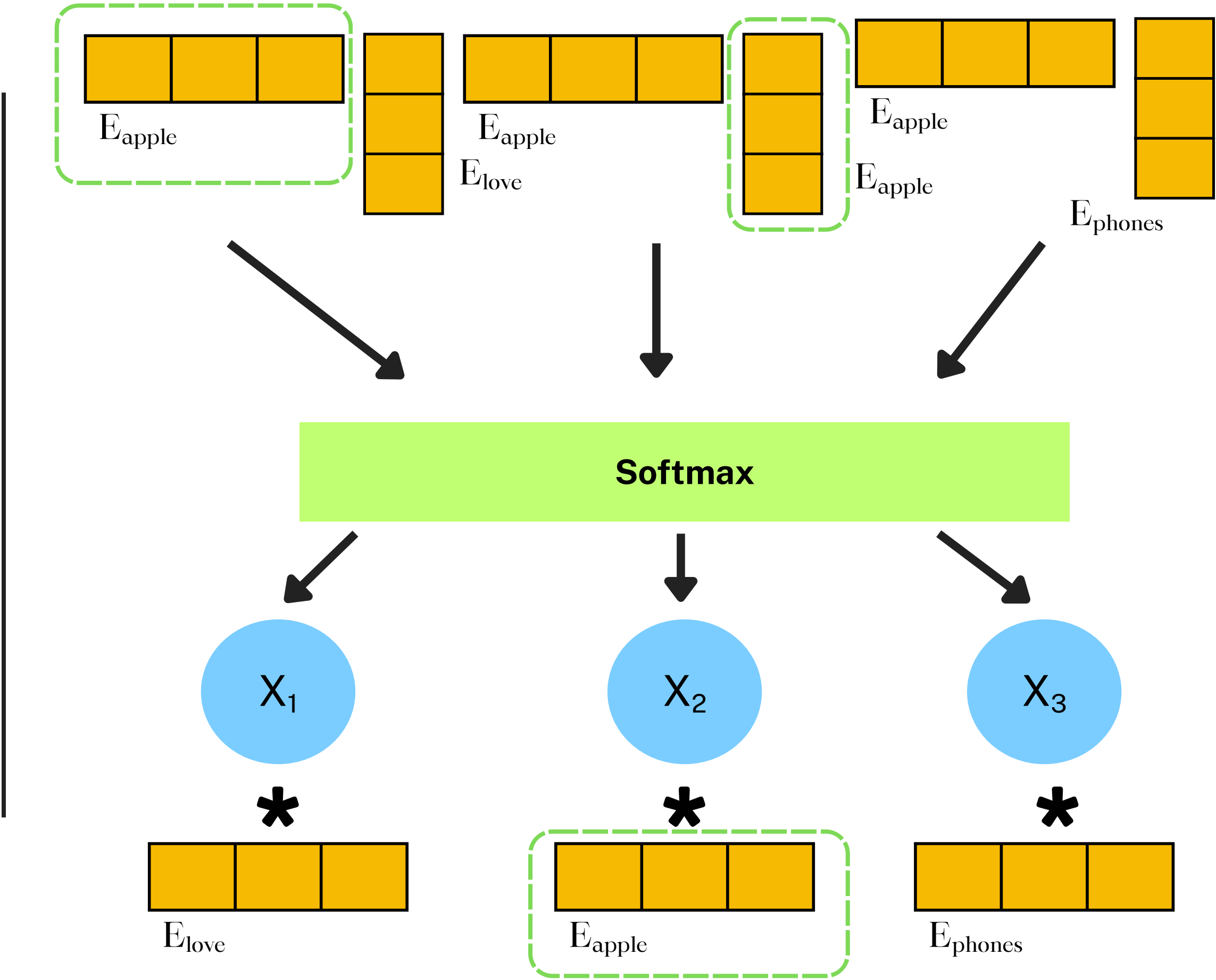


PROBLEM 01:

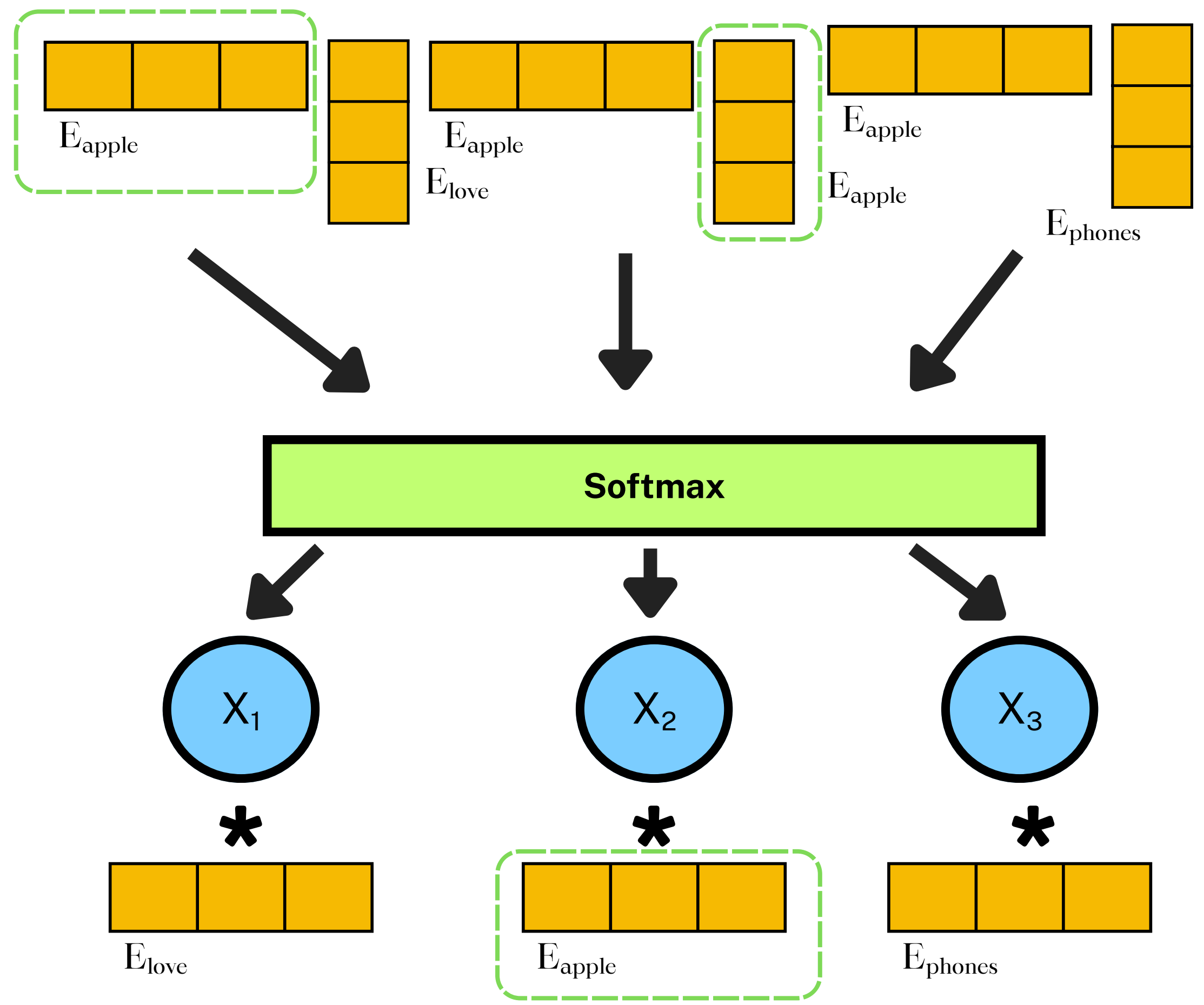
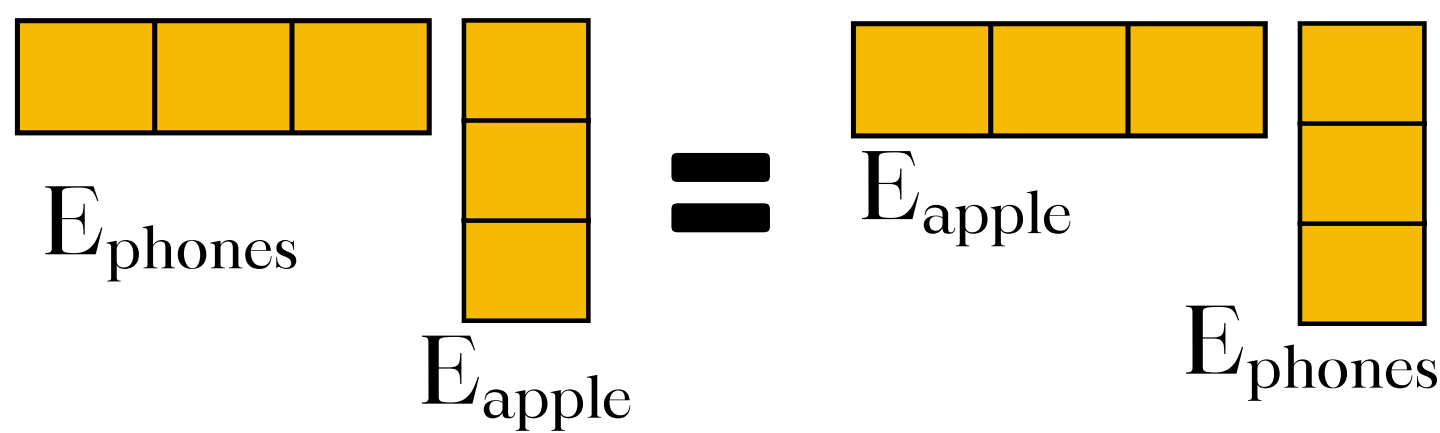
No Learnable Parameters

PROBLEM 02:

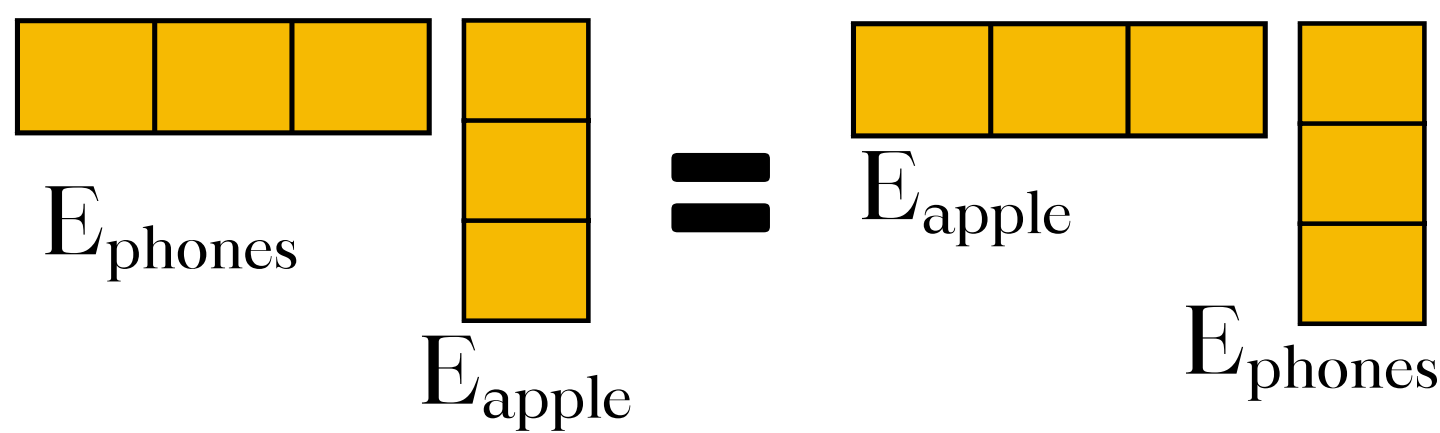
Use of Same Vector



PROBLEM 02:
Use of Same Vector

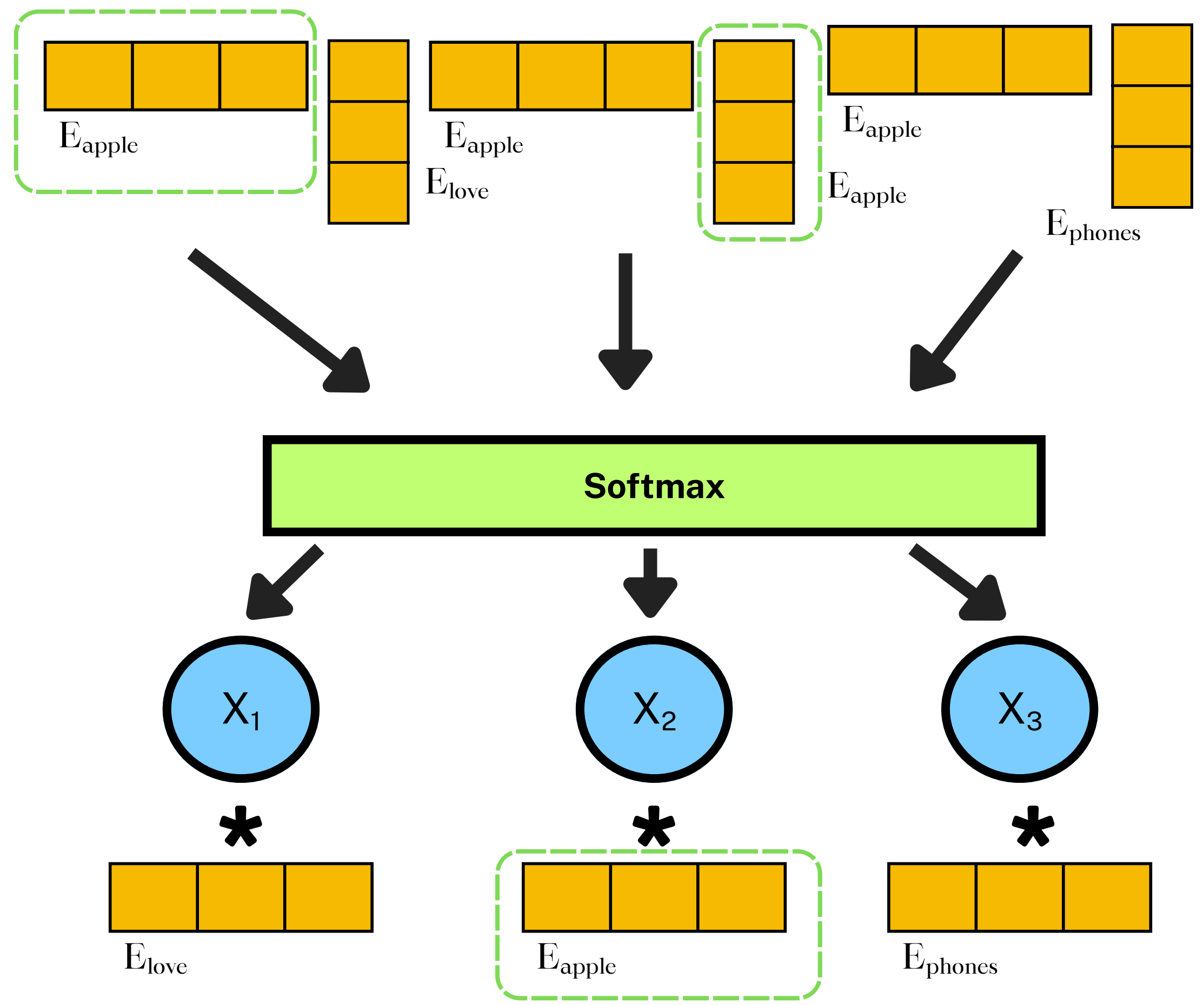


PROBLEM 02:
Use of Same Vector



APPLE $\longleftrightarrow$ **PHONES**

*pull each other equally

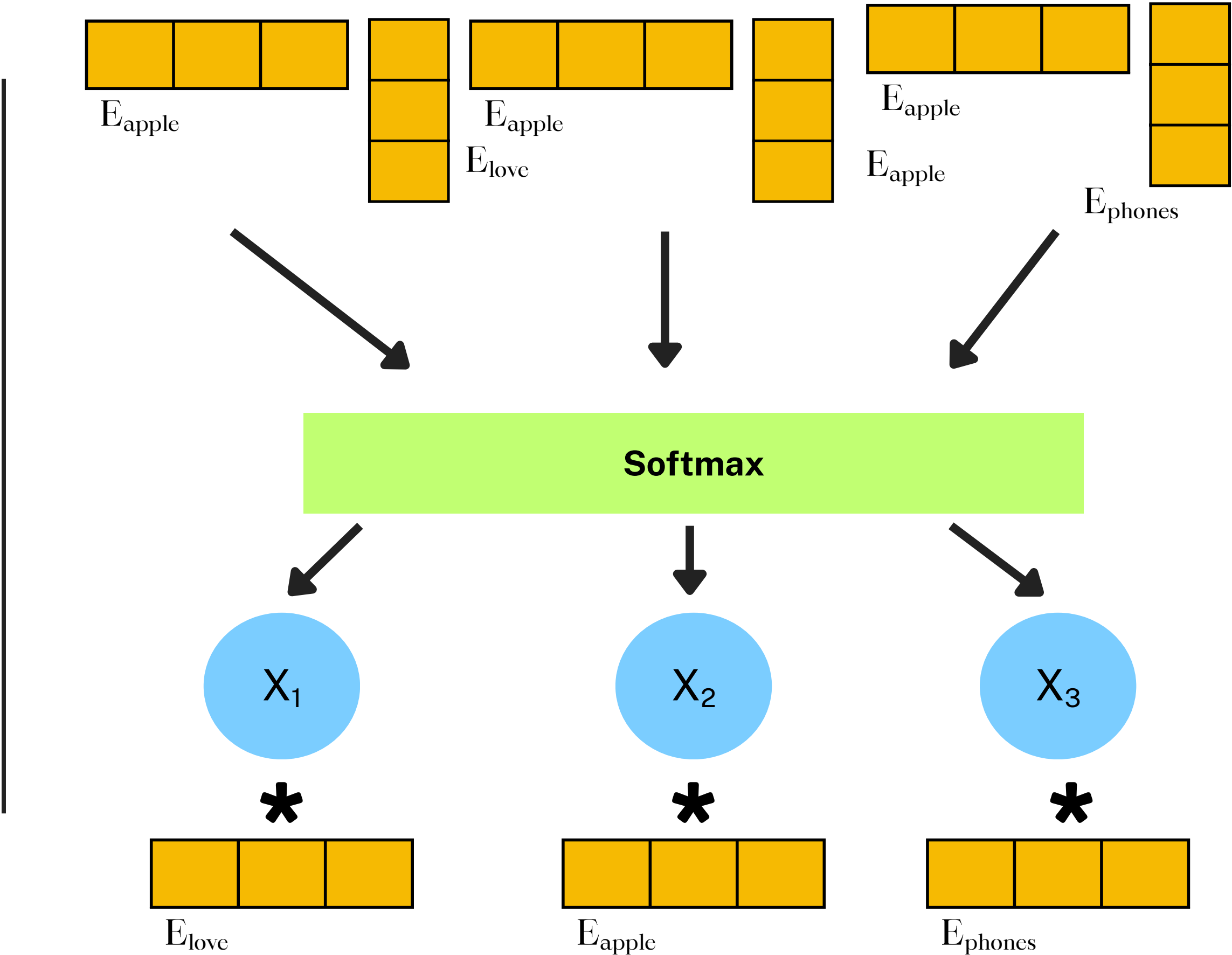


PROBLEM 01:

No Learnable Parameters

PROBLEM 02:

Use of Same Vector

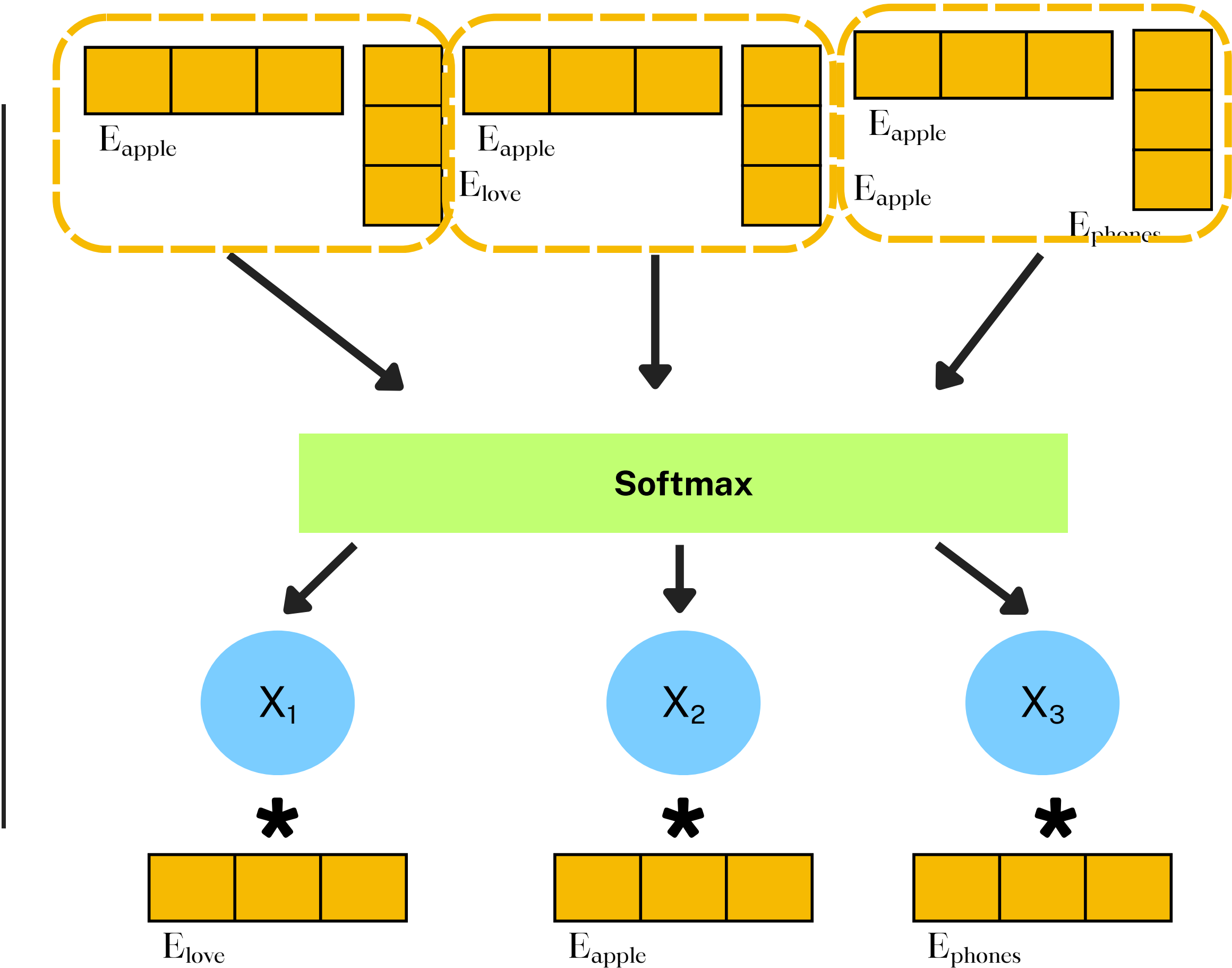


PROBLEM 01:

No Learnable Parameters

PROBLEM 02:

Use of Same Vector

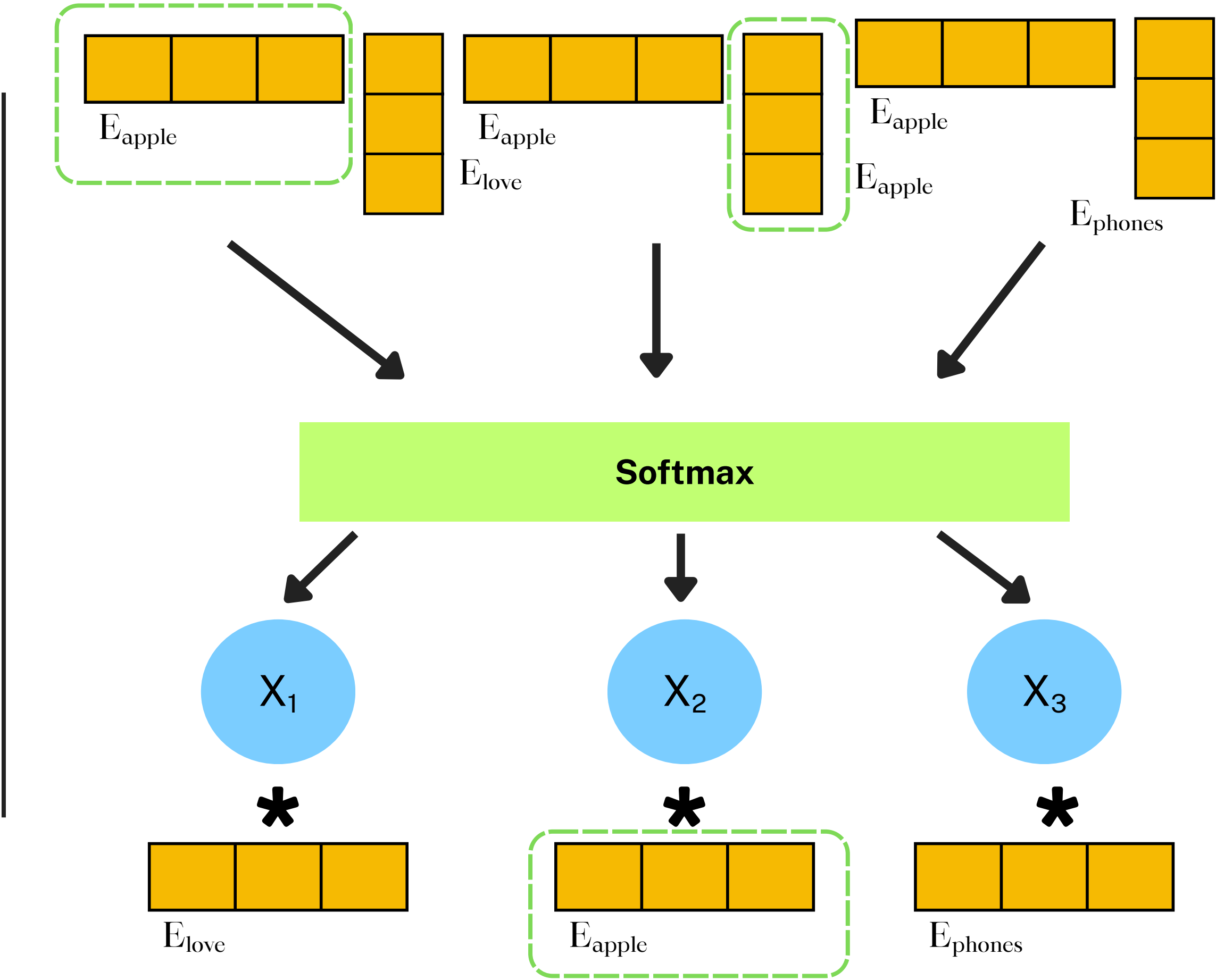


PROBLEM 01:

No Learnable Parameters

PROBLEM 02:

Use of Same Vector



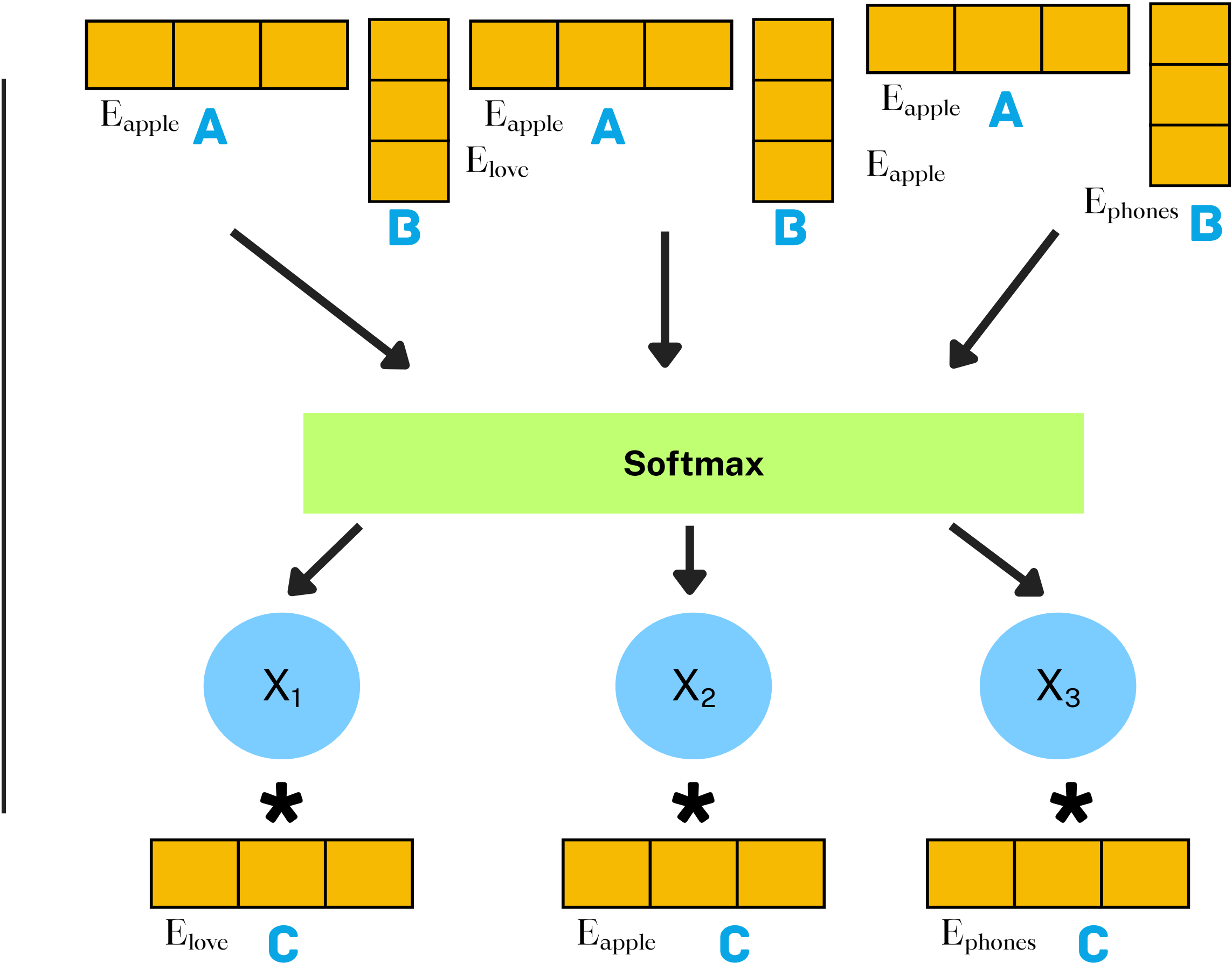
*Use different variation of the vectors at every place

PROBLEM 01:

No Learnable Parameters

PROBLEM 02:

Use of Same Vector



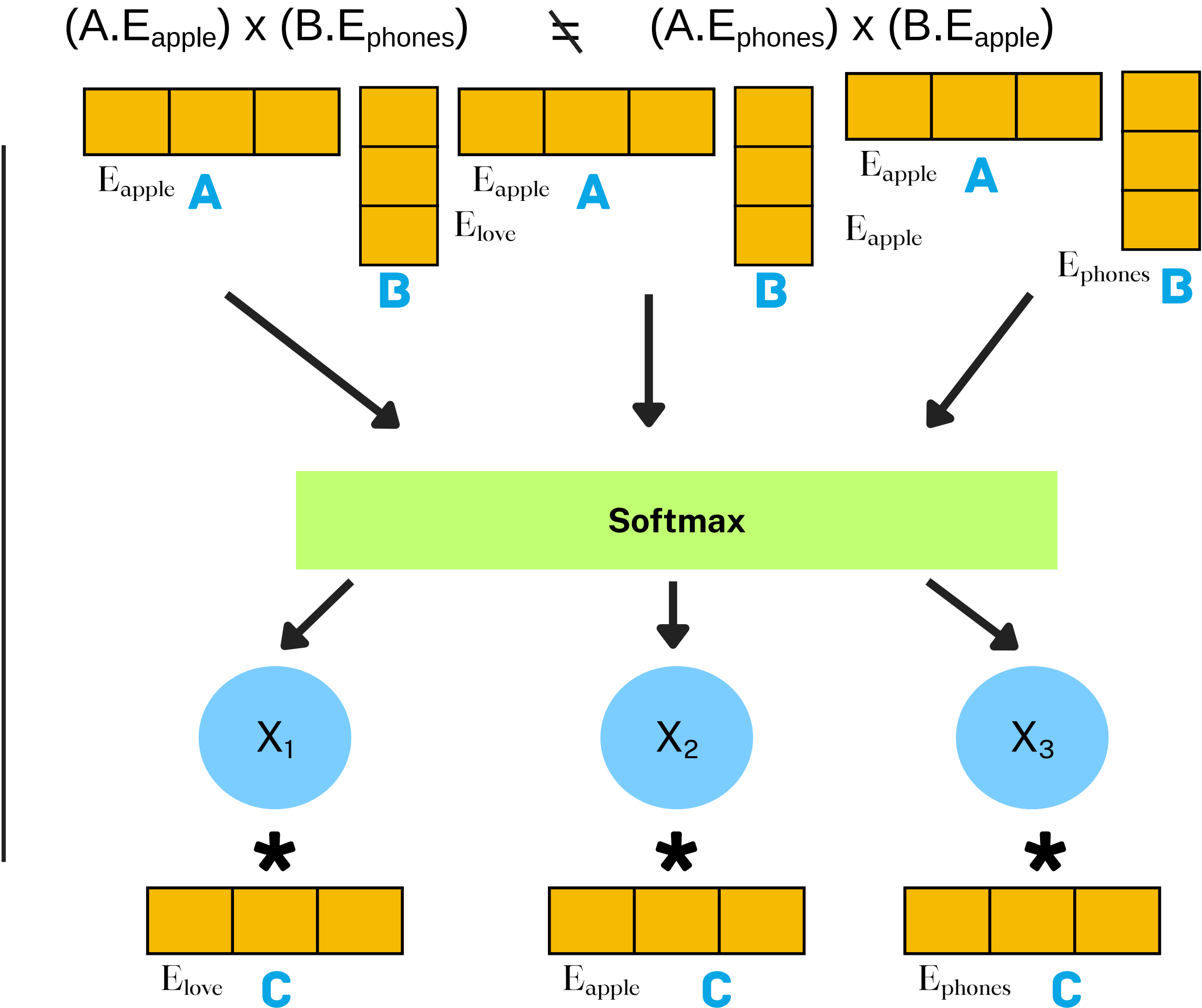
*Use different variation of the vectors at every place

PROBLEM 01:

No Learnable Parameters

PROBLEM 02:

Use of Same Vector



*Use different variation of the vectors at every place

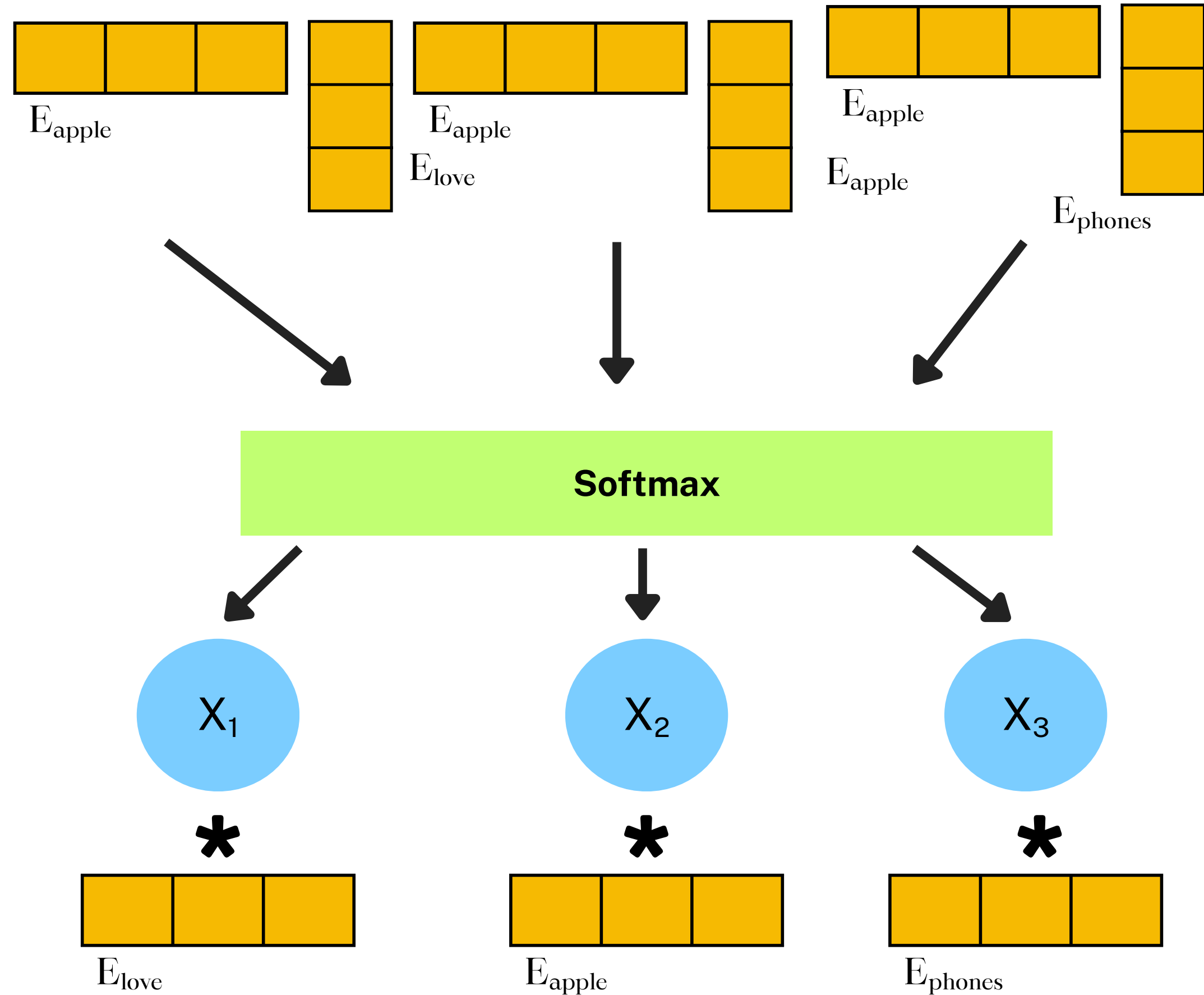
PROBLEM 01:

No Learnable Parameters

$\mathbf{W}$ = weight parameters

PROBLEM 02:

Use of Same Vector



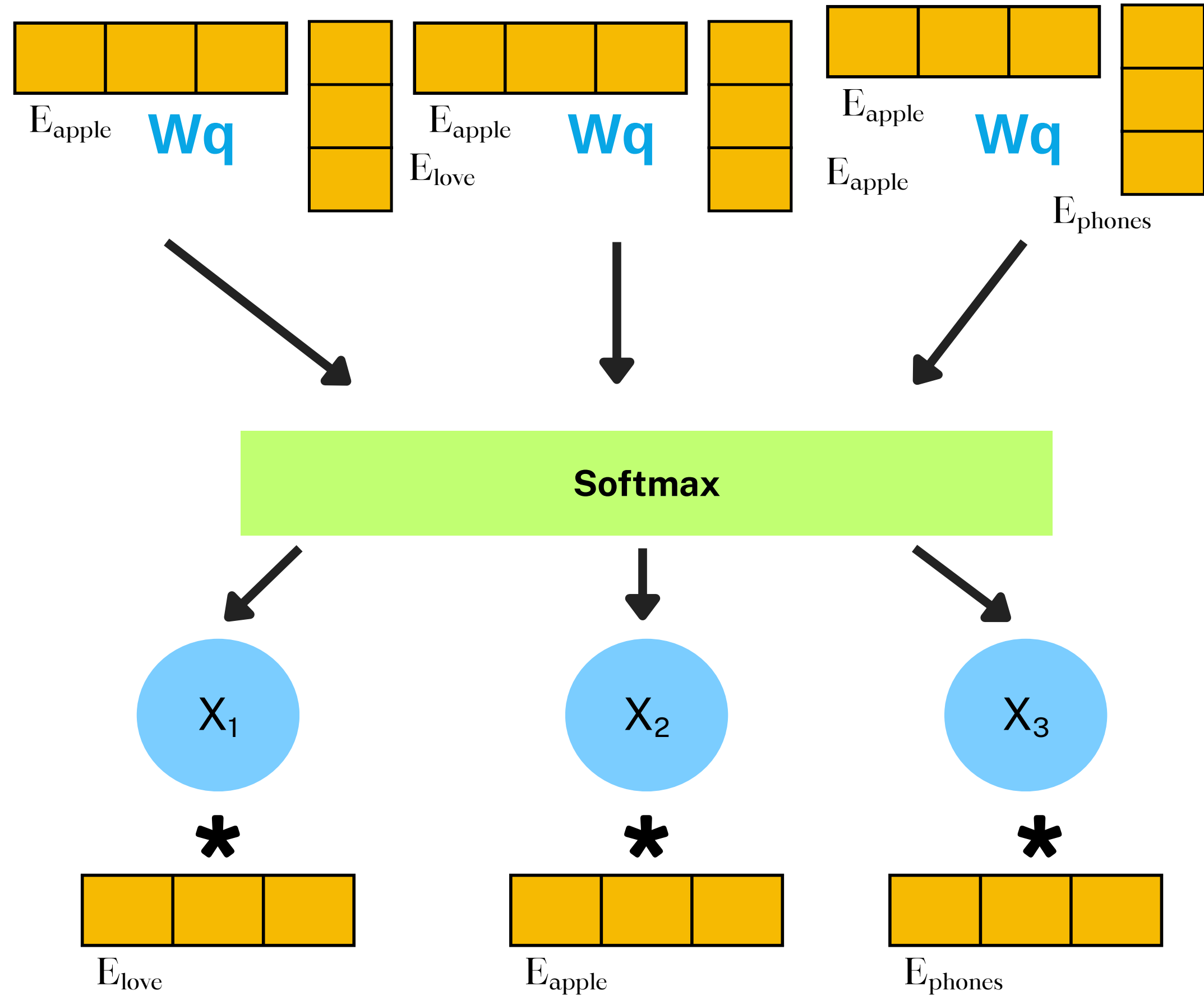
PROBLEM 01:

No Learnable Parameters

$\mathbf{W}$ = weight parameters

PROBLEM 02:

Use of Same Vector



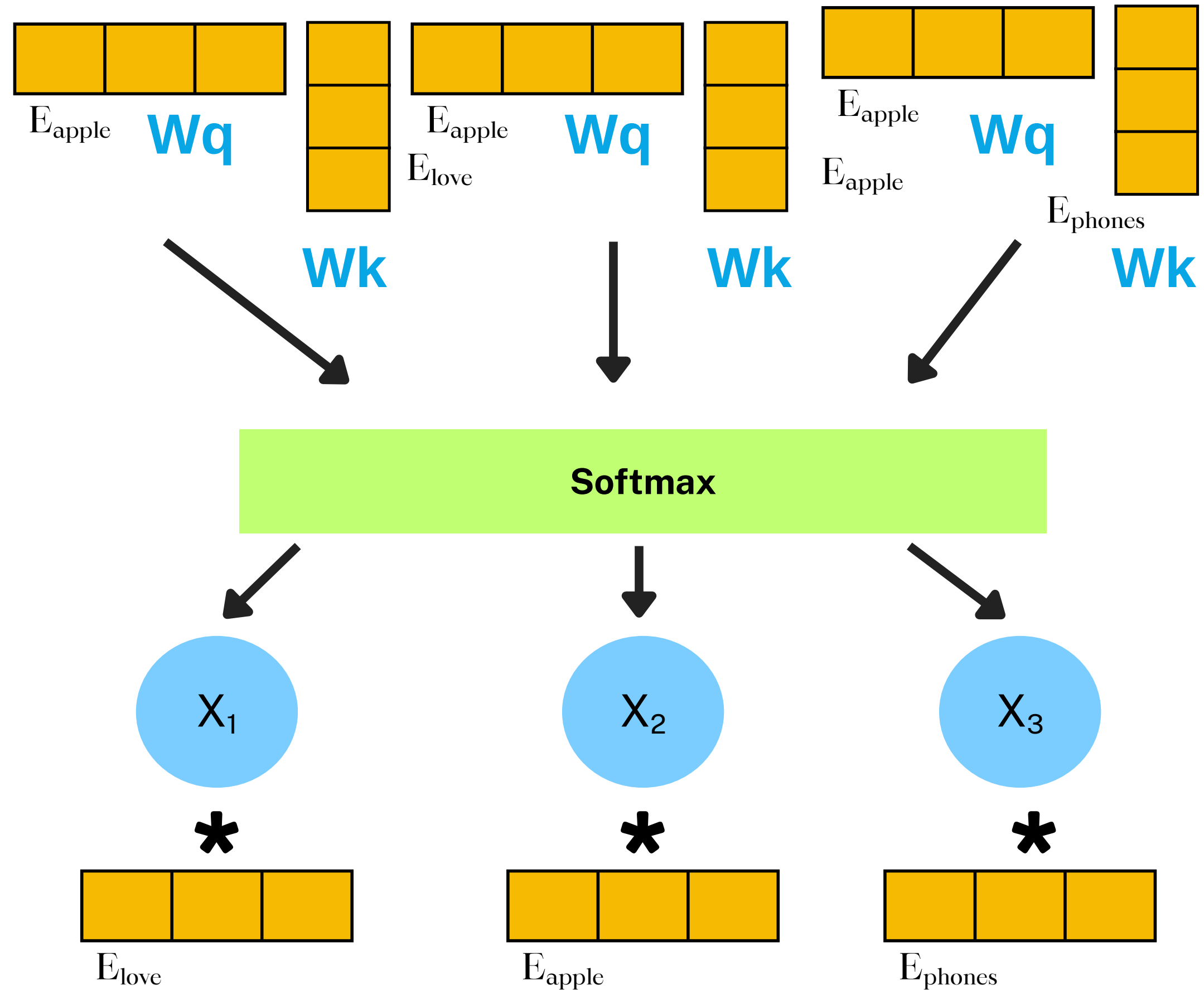
PROBLEM 01:

No Learnable Parameters

$\mathbf{W}$ = weight parameters

PROBLEM 02:

Use of Same Vector



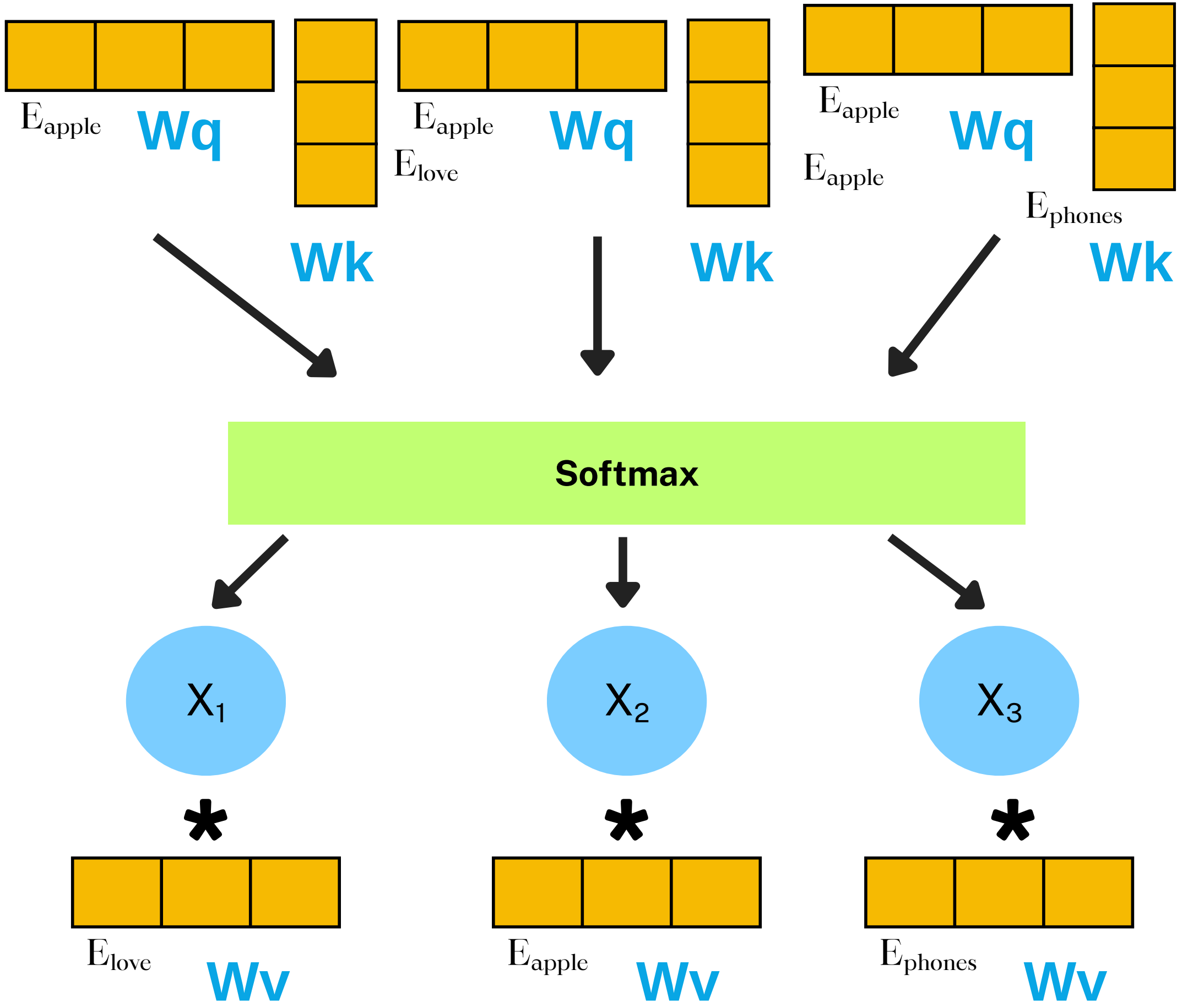
PROBLEM 01:

No Learnable Parameters

W = weight parameters

PROBLEM 02:

Use of Same Vector



*Self Attention should now have some common sense

PROBLEM 01:

No Learnable Parameters

W = weight parameters

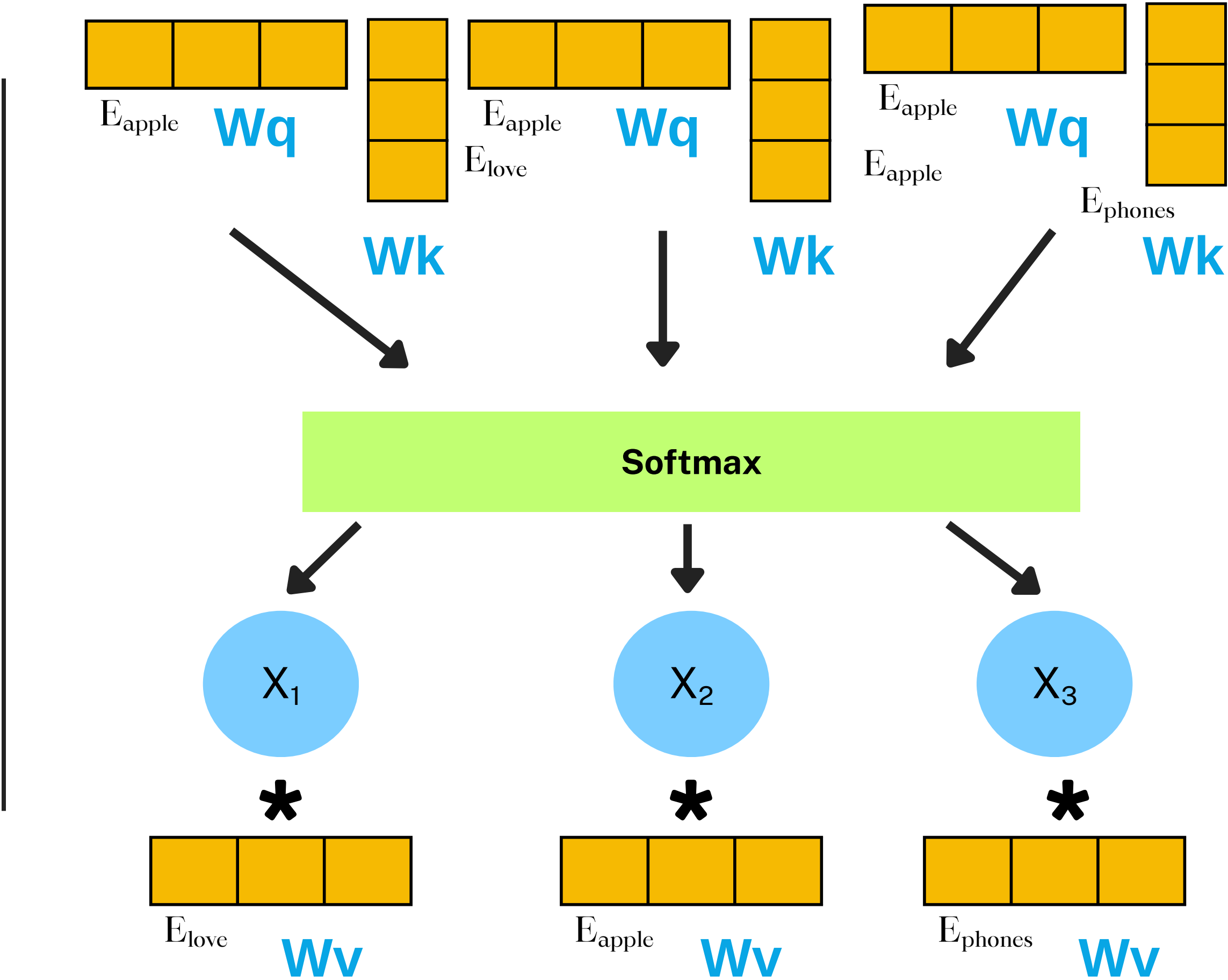
PROBLEM 02:

Use of Same Vector

APPLE → PHONES

PHONES ~~→~~ APPLE

*Self Attention should now have some common sense



SELF ATTENTION
FULL PICTURE

E_{love}

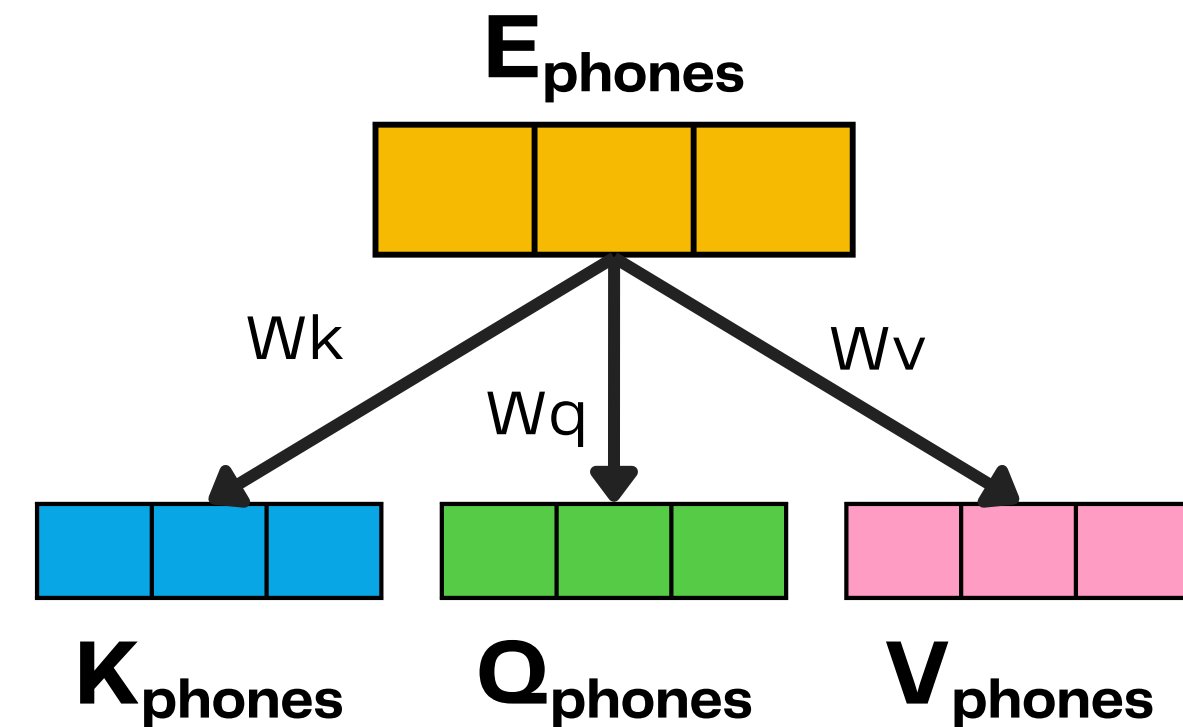
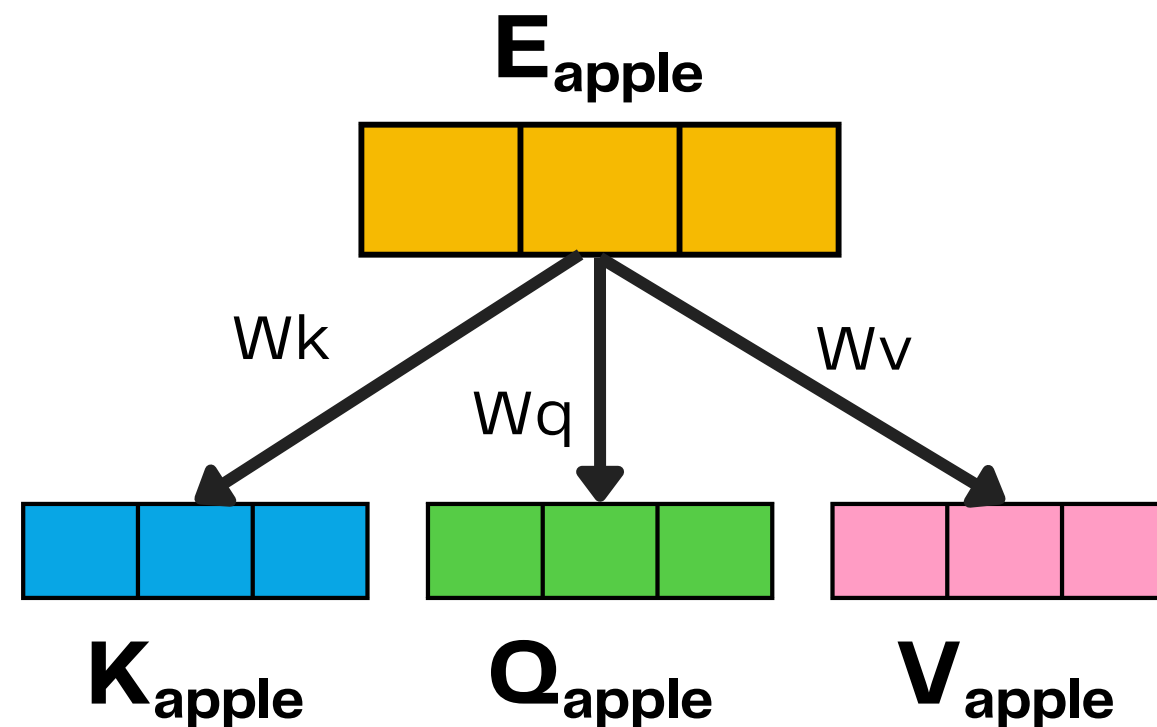
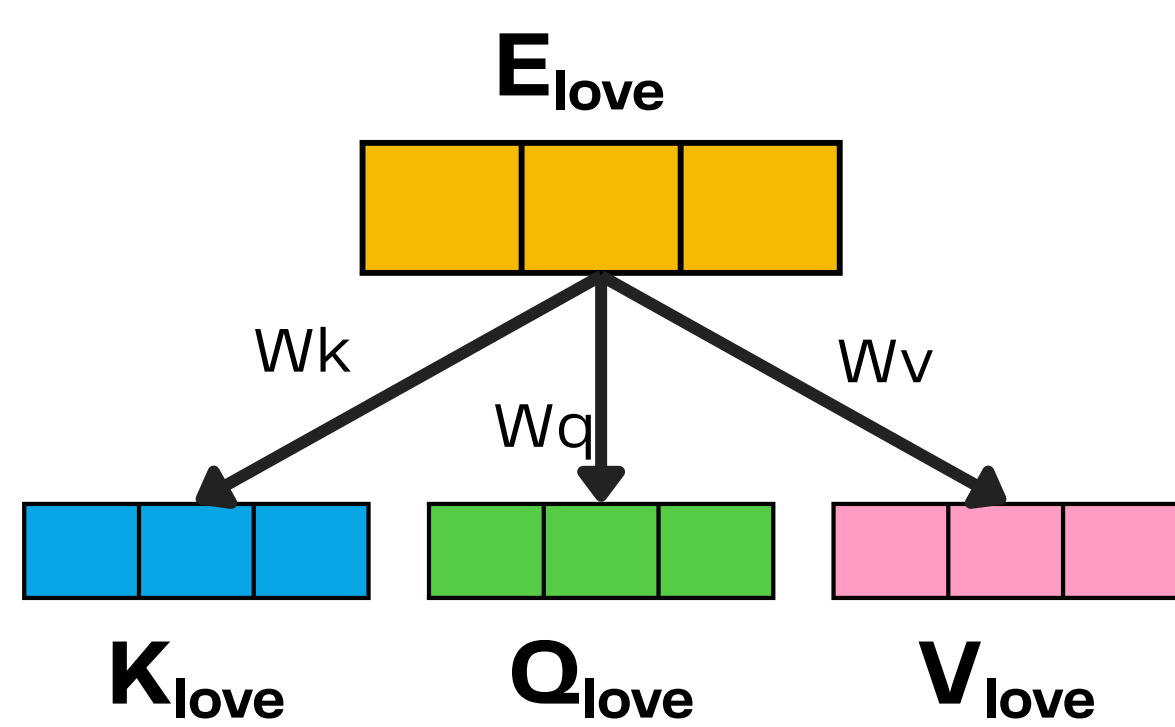


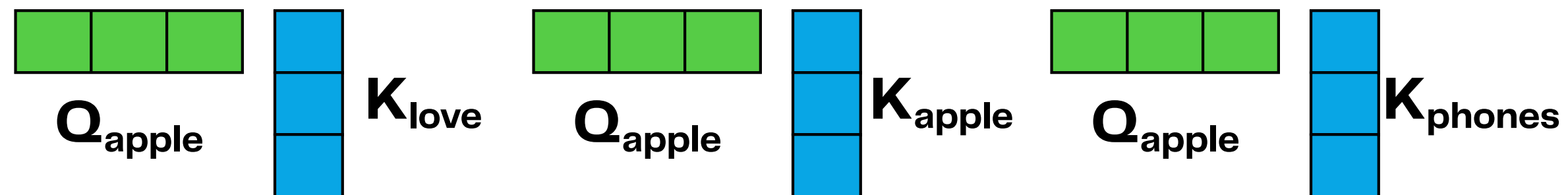
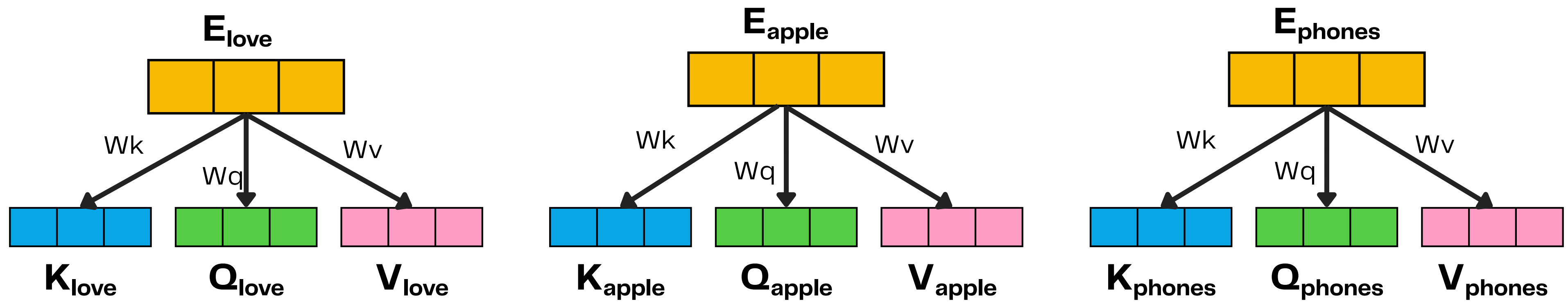
E_{apple}

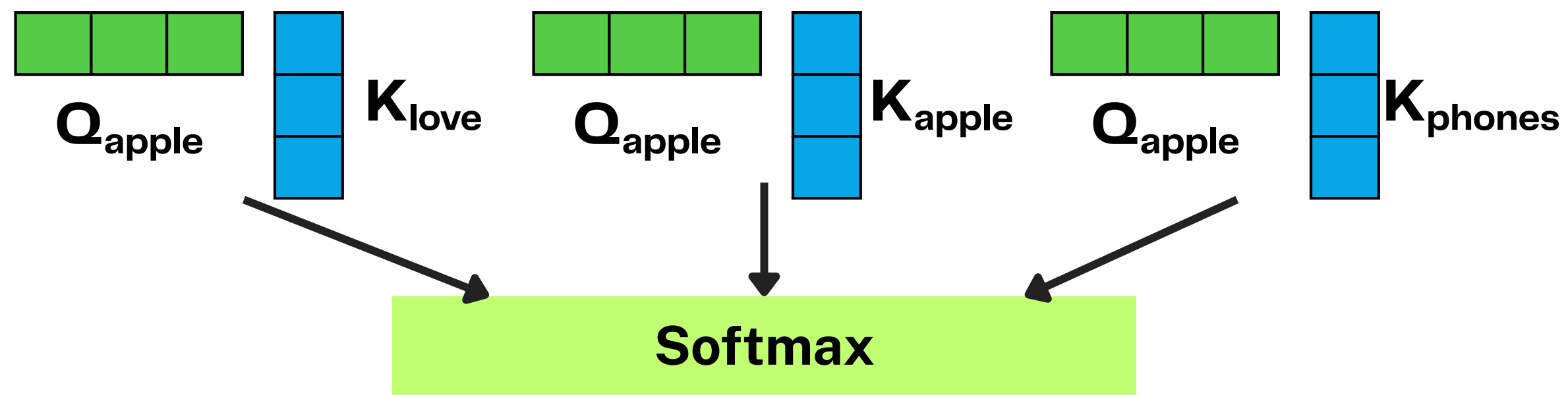
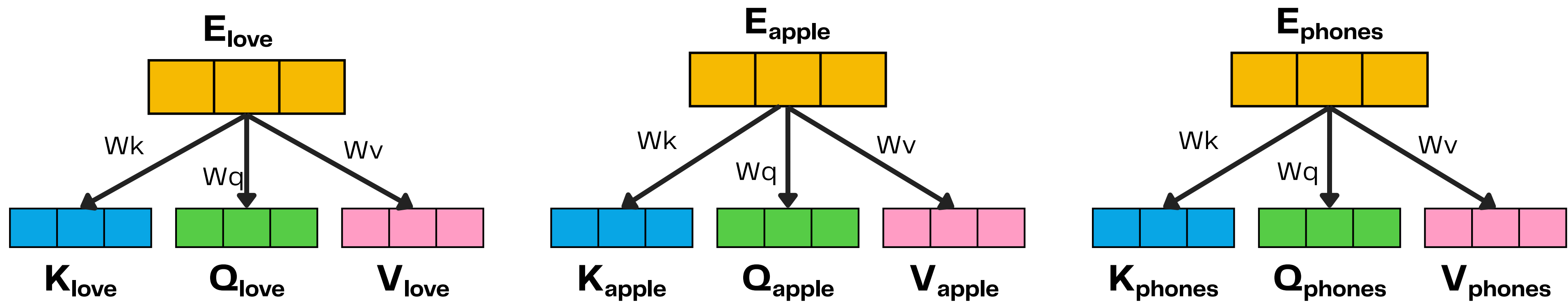


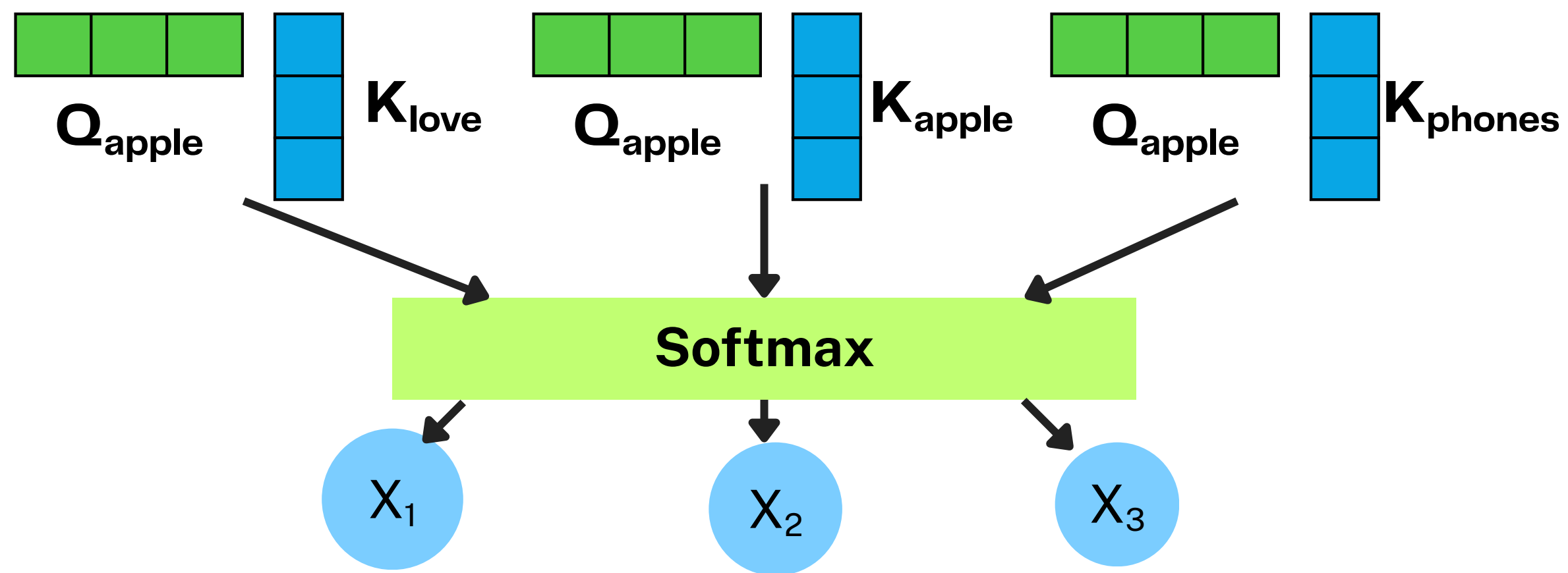
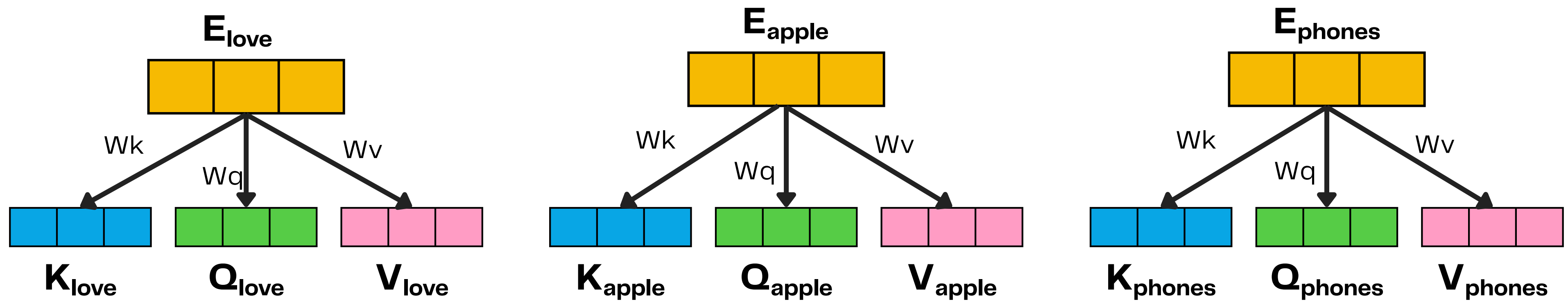
E_{phones}

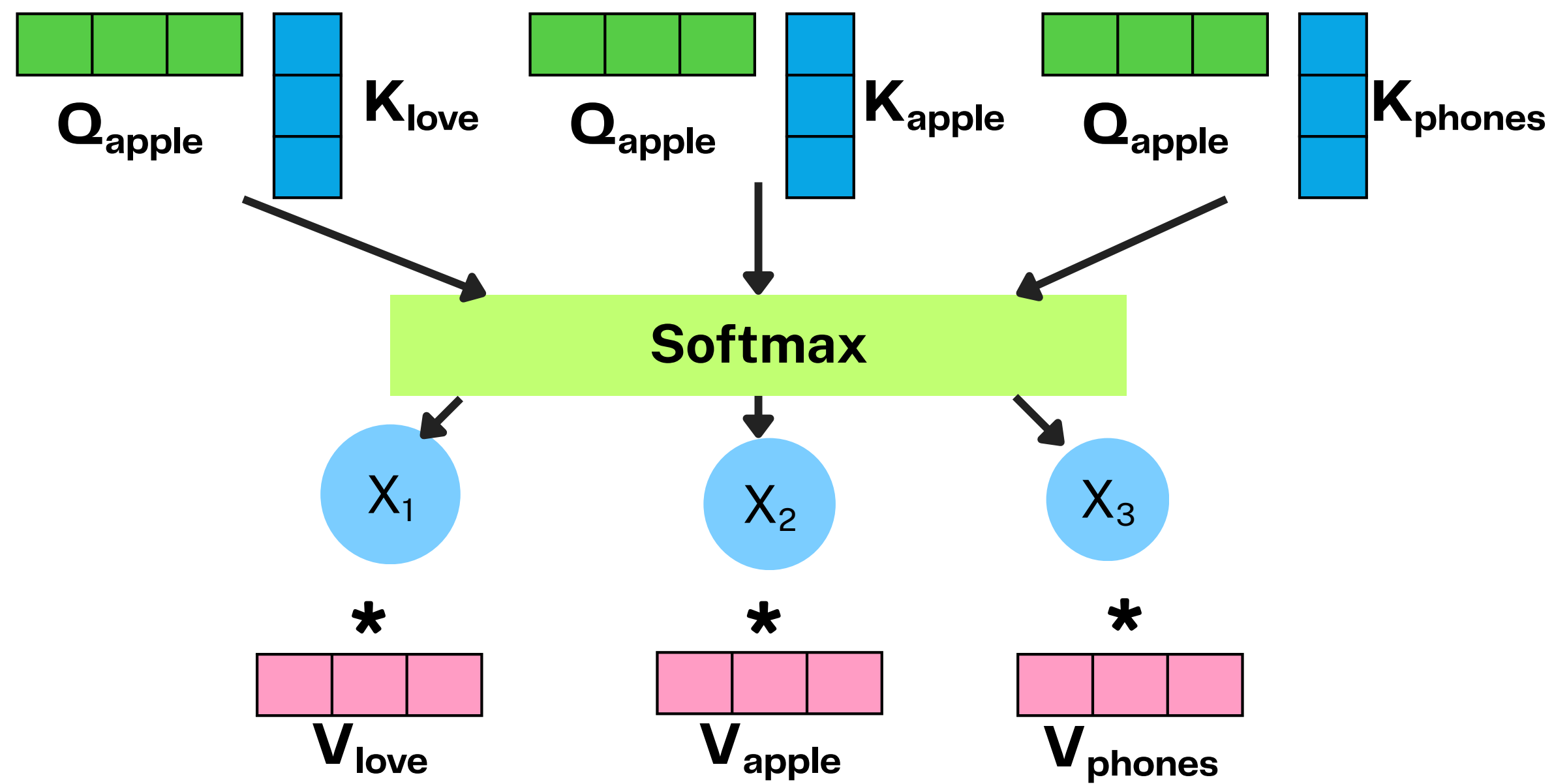
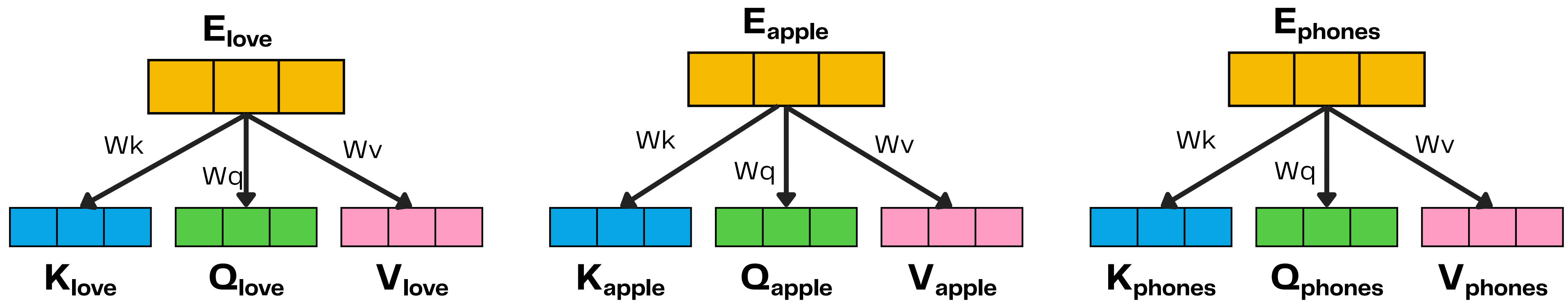


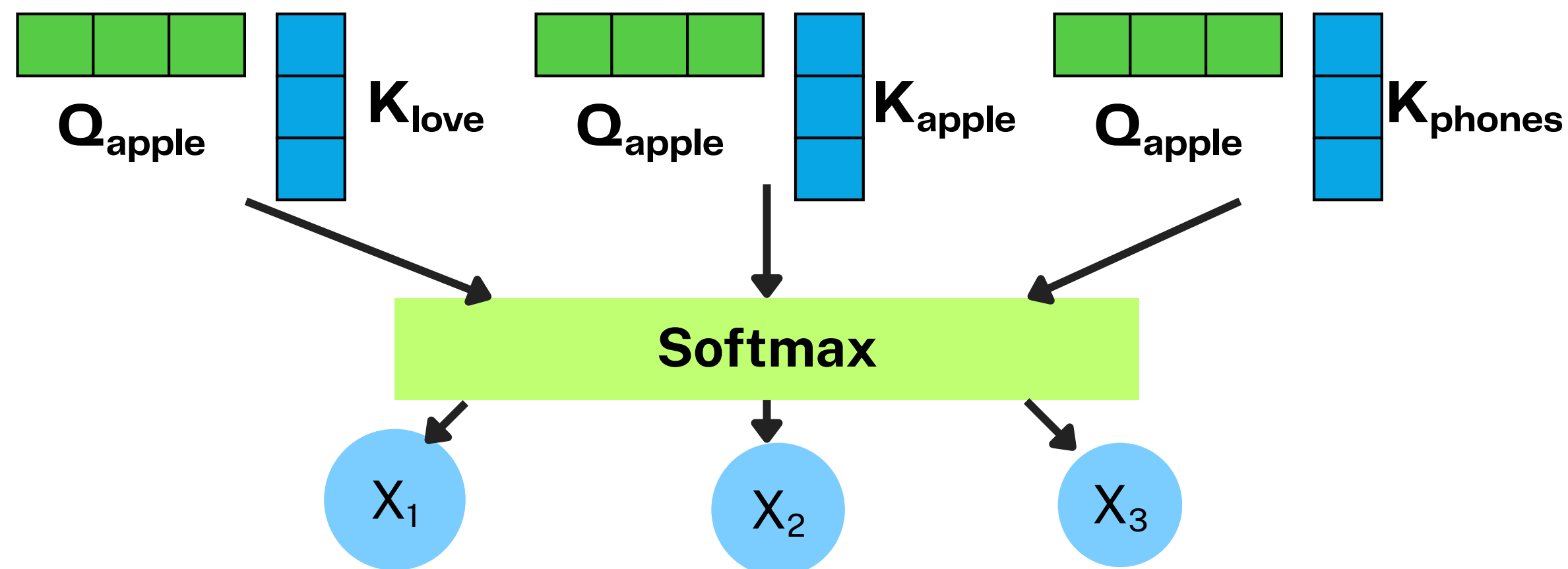
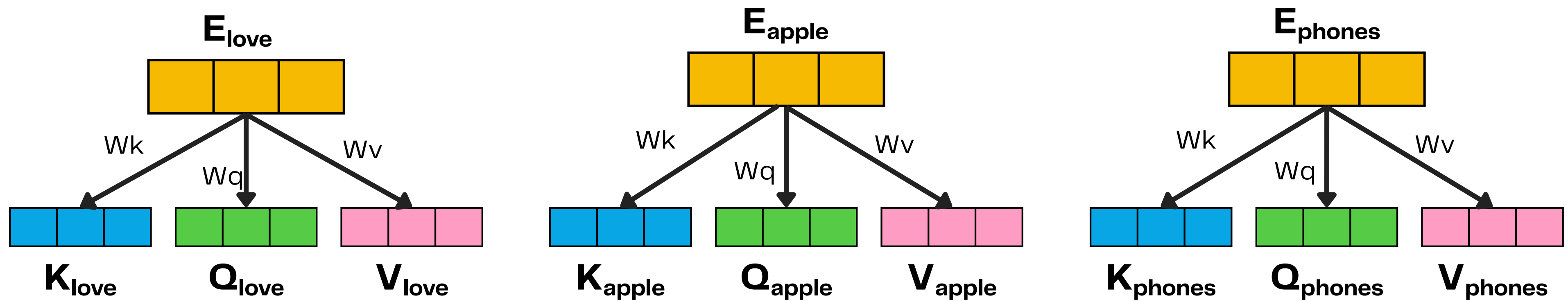












$$E'_{\text{apple}} = V_{\text{love}} * X_1 + V_{\text{apple}} * X_2 + V_{\text{phones}} * X_3$$

The equation shows the weighted sum of Value vectors. Each V vector (pink, 1x3) is multiplied (indicated by $*$) by its corresponding score (X_1, X_2, X_3), and the results are summed (indicated by $+$) to produce the final output E'_{apple} .